

TABLE OF CONTENTS

ABBREVIATIONS USED	7
INTRODUCTION.....	8
1. STATE OF THE ART	11
1.1. Methodology of literature search	11
1.1.1. Databases, keywords, Boolean logic	11
1.1.2. Inclusion/exclusion criteria	12
1.1.3. Screening summary and PRISMA flow diagram	13
1.1.4. Bibliometric analysis.....	14
1.2. Forest Inventory Data Quality.....	18
1.3. Remote Sensing in Forestry	21
1.3.1. Sentinel-2 characteristics.....	21
1.3.2. Sentinel-1 SAR characteristic	22
1.3.3. Satellite-derived features for forest structure estimation.....	23
1.4. Machine Learning for Forest Attribute Modelling.....	25
1.4.1. Supervised learning approaches for forest attribute estimation	25
1.4.2. Feature engineering, feature importance, and uncertainty	26
1.4.3. Summary of existing studies	28
1.5. Identified Knowledge Gap and Justification for Chosen Approach.....	29
2. Research methodology	31
2.1. Introduction - Data Sources.....	32
2.2. Forest Inventory Dataset Description.....	33
2.3. Sentinel Satellite Imagery Selection and Acquisition	35
2.4. Selection of Forest Inventory Attributes for Modelling	36
2.5. Data Preprocessing	37
2.5.1. NFI dataset preprocessing	37
2.5.2. Satellite data cloud masking and temporal filtering	38
2.5.3. Satellite data harmonisation and alignment.....	38
2.5.4. Satellite data polygon-based spectral extraction	39
2.6. Sentinel-derived Feature Description.....	39
2.7. Feature Engineering	40
2.7.1 Vegetation index calculation and selection justification	41
2.7.2 Correlation analysis and feature selection.....	42
2.8. Machine Learning Modelling.....	43
2.8.1. Selected algorithms	43
2.8.2. Training and validation strategy.....	44

2.8.3. Reproducibility considerations.....	45
2.9. Accuracy Assessment and Evaluation Metrics	46
2.10. Ethical and Legal Considerations.....	46
3. RESULTS	48
3.1. Dataset Characteristics	48
3.1.1. Forest inventory attribute statistics.....	48
3.1.2. Correlation analysis of Sentinel-derived predictors and feature selection	52
3.1.3. Predictor variable characteristics.....	53
3.2. Model Performance	54
3.2.1. Regression results.....	54
3.2.2. Classification results	56
3.2.3. Model comparison.....	59
3.3. Feature Importance and Model Interpretation.....	59
3.3.1. Regression models.....	59
3.3.2. Classification models	61
3.4. Regression Model Residual Diagnostics and Pattern Analysis	63
3.4.1. Regression residual diagnostics	64
3.4.2. Residual patterns across stand characteristics	65
3.4.3. Identification of high-deviation stands.....	68
3.5. Classification Error Analysis	71
4. DISCUSSION	75
4.1. Interpretation of results	75
4.2. Practical relevance for forest management	76
4.3. Limitations	77
4.4. Opportunities for future work	78
CONCLUSIONS.....	79
REFERENCES.....	81
APPENDICES.....	87

ABBREVIATIONS USED

AI	– Artificial Intelligence
DBH	- diameter at breast height
GB	- Gradient Boosting
GBC	- Gradient Boosting Classifier
GBR	- Gradient Boosting Regressor
GEE	- Google Earth Engine
GSV	- Growing Stock Volume
LiDAR	- Light Detection and Ranging
ML	- Machine Learning
NDRE	- Red Edge Normalised Difference Vegetation Index
NDVI	- Normalised Difference Vegetation Index
NIR	- Near-infrared
NFI	- National Forest Inventory
PRISMA	- Preferred Reporting Items for Systematic Reviews and Meta-Analyses
RF	- Random Forest
SAR	- Synthetic Aperture Radar
SVM	- Support Vector Machine
SVR	- Support Vector Regression
SWIR	- Short-Wave Infrared
UAV	- Unmanned Aerial Vehicle
VH	- vertical transmit – horizontal receive
VV	- vertical transmit – vertical receive

INTRODUCTION

Background (Actuality of research)

Forests represent a vital component of the global environment, offering key ecosystem services such as carbon sequestration, biodiversity conservation, soil protection, and regulation of hydrological processes. As highlighted by Puliti et al. (2020) and Silveira et al. (2023), reliable information on forest structure, species composition, and Growing Stock Volume (GSV) is essential for sustainable forest management, climate change mitigation, and international reporting systems. In many countries, such data is largely obtained from National Forest Inventory (NFI), which aims to provide statistically representative estimates of forest resources at national and regional scales.

Field-based data collection and the maintenance of forest inventory data are very complex and require significant resources. Inventory datasets are often affected by uncertainties in sampling design, plot configuration, and field measurements, as well as inconsistencies across inventory cycles. At the same time, increasing demand for timely and spatially detailed forest information has exposed the limitations of relying primarily on traditional inventory methods.

Recent developments in remote sensing and data-driven modelling have created new opportunities to complement forest inventories. Amuyou et al. (2022) and Khan et al. (2024) showed that combining satellite imagery with Machine Learning (ML) can support spatially continuous estimation of forest attributes. For NFI purposes, these estimates also need to meet data quality requirements to be useful for validation and decision-making.

Problem statement

Accurate forest inventory data are crucial for sustainable forest management, biodiversity conservation and climate change mitigation. These data are used to estimate GSV, biomass, carbon reserves, forest health and disturbance patterns, which are important for national reporting, forest policy and international climate commitments. Large scale studies on forest structure and carbon mapping show that uncertainty in inventory-based variables can affect the spatial distribution of estimated biomass and GSV. Signs of forest degradation may be overlooked, or forest recovery may be overestimated (Silveira et al., 2023; Chen et al., 2019). Such inaccuracies can make forest disturbances more difficult to detect, reduce the reliability of biodiversity assessments and weaken the evidence base for climate-related forest policy.

In recent studies, remote sensing and ML were used for forest disturbance monitoring and biomass estimation. The results were found to depend strongly on the quality and consistency of the reference inventory data (Molnár and Király, 2024; Yadehari et al., 2025). In validation-focused research, uncertainties related to sampling design, plot layout, time differences and measurement errors in NFI data were shown to affect spatial predictions when they were not addressed (Kovárník and Janová, 2025; Suleymanov et al., 2024). Investigations into data homogeneity and reference data consistency demonstrated that inconsistent inventory data can reduce model reliability more significantly than the selection of the ML algorithm (D'Amico et al., 2024).

As a result, even though satellite-based forest monitoring methods have improved, inventory data quality has still not been given enough attention. This remained an important limitation. The issue needed to be addressed so that forest attribute estimates produced by ML could be accurate, understandable and useful for operational forest inventories, environmental assessments and decision-making.

The goal of this thesis is to develop an approach to improve forest inventory data quality using satellite data and supervised ML.

Object of the research: forest inventory data quality.

Subject of the research: a forest inventory data quality improvement procedure based on supervised ML techniques.

The tasks/objectives of the research

1. To review existing scientific literature on forest inventory data quality, remote sensing in forestry, and ML methods for forest attribute estimation.
2. To prepare data for analysis, including preprocessing and derivation of relevant spectral features:
 - a. explore forest inventory data
 - b. extract Sentinel satellite data.
3. To choose and apply supervised ML methods to estimate selected forest inventory attributes using satellite-based predictors.
4. To evaluate the accuracy and reliability of the model results through comparison with forest inventory reference data.
5. To analyse feature importance and model residuals to identify possible uncertainty or inconsistencies in forest inventory data.
6. To synthesise and generalise the research findings, including documentation of the developed approach and presentation of the thesis results.

Research questions

RQ1: Which factors influence the quality and internal consistency of NFI data?

RQ2: Which Sentinel-1- and Sentinel-2-derived features contribute most to the prediction of selected forest inventory attributes?

RQ3: Which supervised ML methods provide the most accurate and reliable estimates of selected forest inventory attributes and support the assessment of NFI data quality?

Research methods and software tools

This research combined a systematic review of scientific literature with practical data analysis using remote sensing and ML methods. Forest inventory data were merged with satellite imagery to estimate specific forest attributes and to evaluate how data quality and uncertainty influence modelling results.

The study used forest inventory data in polygon format together with Sentinel-1 and Sentinel-2 satellite data. Cloud-based platforms and open-source software facilitated data preprocessing, feature extraction, model development, validation, and visualisation. Specialised tools supported literature management, bibliometric analysis, and thesis development documentation.

The main stages of the research process, data sources, methods, and software tools applied in this study are summarised in Appendix VI.

Structure of the thesis

Chapter 1 presents the theoretical background, including forest inventory data quality, the use of remote sensing in forestry, ML approaches for forest attribute modelling, and the identified research gap.

Chapter 2 describes the materials and methods applied in the study.

Chapter 3 presents the results of the analysis.

Chapter 4 discusses the results in relation to existing research and their practical implications.

Finally, the Conclusions section summarises the main findings and provides recommendations for future research.

1. STATE OF THE ART

1.1. Methodology of literature search

1.1.1. Databases, keywords, Boolean logic

For the literature search, three major databases were selected: Scopus, ScienceDirect, and IEEE Xplore. These sources include a wide range of peer-reviewed literature in environmental sciences, forestry and ML applications. The use of multiple databases reduced the publication bias and ensured a wider coverage of relevant studies.

Zotero was used for reference management, duplicate detection and tagging of studies by topic and methodology. Artificial Intelligence (AI) based tools, such as ChatGPT, were used to support the development of synonyms and Boolean search strings. References were exported in RIS format for merging and screening in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) process.

The search strategy was designed around the main research question: “How can Sentinel-2 imagery and ML be used to improve the quality of forest inventory data by predicting stand attributes and correcting inaccurate records?” The literature search was structured around four thematic keyword groups (Table 1.1), which were combined using Boolean operators to capture relevant studies across the domains of forestry, remote sensing, and ML.

table 1.1

Keyword groups

Keyword group	Search terms
Forest inventory	Forest inventory; national forest inventory; NFI; timber inventory; woodland inventory
Sentinel-2 and remote sensing	Sentinel-2; sentinel; optical remote sensing; multispectral imagery; satellite imagery
Machine learning	Machine learning; predictive modelling; random forest; XGBoost; support vector regression
Data quality and accuracy	Enhancement; accuracy; precision; validation; assessment

The Boolean logic, using AND and OR operators and truncations (e.g., Sentinel*, enhanc*), was applied to expand the search terms to capture all relevant literature. The search was limited to English-language publications, and the period covered 2015 to 2025, as Sentinel-2 was launched in 2015. In the Scopus and ScienceDirect databases, an additional filter was applied to include

only open-access articles. Figure 1.1. shows the Boolean search used in the Scopus literature search.

Enter query string

```
TITLE-ABS-KEY( ("forest inventory" OR "timber inventory" OR "woodland inventory" OR "national forest inventory" OR "NFI") AND ("Sentinel*" OR "multispectral imag*" OR "optical remote sensing") AND ("machine learning" OR "predictive modelling" OR "random forest" OR "XGBoost" OR "support vector" OR "SVM") AND ("quality enhanc*" OR "accuracy enhanc*" OR "quality improv*" OR "quality control" OR "accuracy assessment" OR "accuracy" OR "validation" OR "data quality" OR "quality control" OR "uncertainty" OR "precision" OR "performance" OR "estimation" OR "assessment" OR "mapping")) AND PUBYEAR > 2014 AND PUBYEAR < 2026 AND ( LIMIT-TO ( LANGUAGE,"English" ) )
```

Fig. 1.1. Scopus search boolean

1.1.2. Inclusion/exclusion criteria

The inclusion and exclusion criteria were defined to ensure that the systematic review included only relevant, methodologically sound studies. The inclusion criteria targeted peer-reviewed papers in English, published from 2015 to 2025, that utilised Sentinel-2 optical imagery and ML techniques for estimating, validating, or enhancing forest inventory attributes. Studies addressing data quality issues such as accuracy, validation, uncertainty, or correction of inventory records were also included. The exclusion criteria removed studies unrelated to forestry, articles focused on carbon without direct forest inventory links, those relying only on non-Sentinel data sources, and publications lacking full-text access. A detailed overview of the inclusion and exclusion criteria is provided in Table 1.2.

table 1.2

Inclusion/ exclusion criteria

Inclusion/ Exclusion	Category	Criteria
Inclusion criteria	Language	Publications written in English
	Publication period	Studies published between 2015 and 2025 (Sentinel-2 was launched in 2015).
	Document type	Peer-reviewed journal articles and conference papers
	Data source	Use of Sentinel-2 optical satellite imagery
	Methodology	Application of machine learning or predictive modelling methods
	Research focus	Estimation, validation, or improvement of forest inventory attributes (e.g., growing stock, biomass, stand density, canopy structure)

Inclusion/ Exclusion	Category	Criteria
Inclusion criteria	Data quality focus	Studies addressing accuracy, validation, uncertainty, or correction of forest inventory data
Exclusion criteria	Non-forestry focus	Studies focused on agriculture, grasslands, wetlands, or non-forest ecosystems.
	Carbon-only studies	Studies addressing carbon stocks or climate modelling without a direct linkage to forest inventory attributes
	Non-Sentinel data	Studies relying exclusively on Light Detection and Ranging (LiDAR), Synthetic Aperture Radar (SAR), hyperspectral, or Unmanned Aerial Vehicle (UAV) data without Sentinel-2 imagery
	Accessibility	Publications without full-text availability (non-open-access)

1.1.3. Screening summary and PRISMA flow diagram

An initial database search identified 434 records, including 137 articles from Scopus, 287 from ScienceDirect, and 10 from IEEE Xplore. After these records were imported into Zotero, 27 duplicate entries were identified and removed, leaving 407 unique publications. The remaining records were screened based on their titles and abstracts. In this stage, 339 records were excluded: 137 studies were excluded for not being relevant to forestry, and 202 were excluded because their methodologies did not align with the study's goals, such as focusing only on carbon analyses without forest inventory attributes or using data sources other than Sentinel-2.

The entire selection process was documented using a PRISMA flow diagram (Figure 1.2.), illustrating the stages of identification, screening, eligibility, and inclusion, thereby ensuring transparency and reproducibility of the review process.

Following the screening stage, 68 full-text articles were assessed for eligibility. An additional 11 studies were excluded due to the unavailability of full text. As a result, 57 studies met all inclusion criteria and were included in the systematic review.

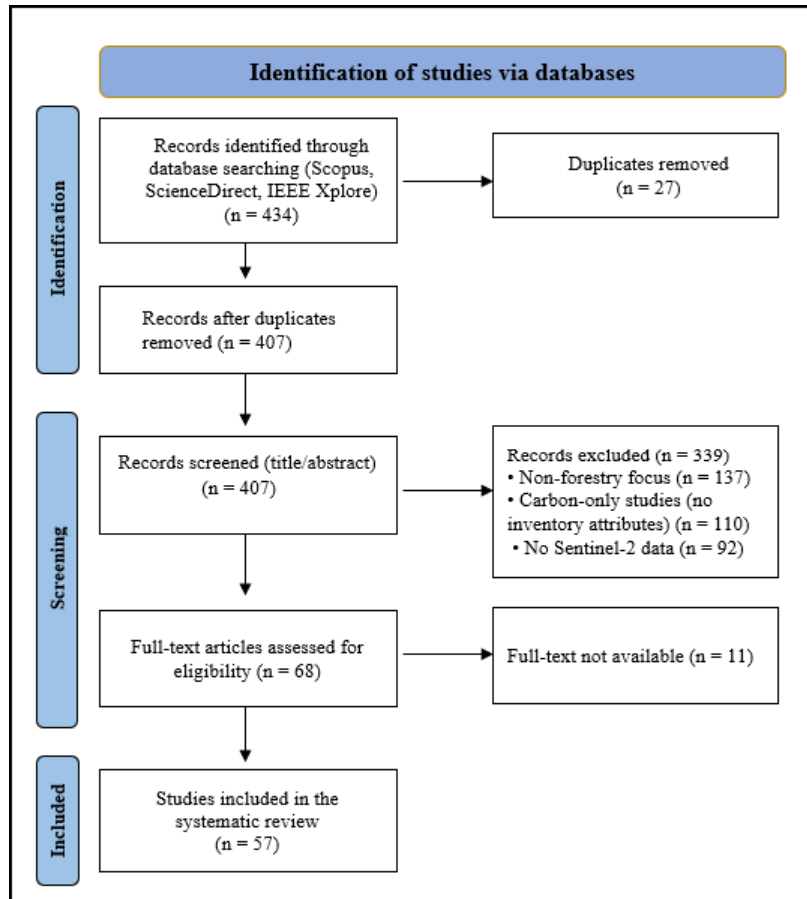


Fig.1.2. PRISMA flow diagram

1.1.4. Bibliometric analysis

The final collection of publications was organised and managed in Zotero. The records were grouped into main themes, such as remote sensing, ML and NFI applications. The final reference database was exported in RIS format for bibliometric analysis and visualisation. Additional analyses were carried out using Python tools and the Litmaps platform to examine publication trends, citation patterns, journal distribution, keyword co-occurrence and the citation network structure in the selected literature

Annual publication output showed a clear increase in research activity, as shown in Figure 1.3. Between 2016 and 2018, only a small number of studies were published, which suggested that academic interest in Sentinel-2 imagery and ML applications in forestry was still limited. From 2019 onwards, the number of publications increased and reached 16 publications per year in both 2024 and 2025. This trend indicated growing research interest and progress in combining remote sensing and ML for forest monitoring.

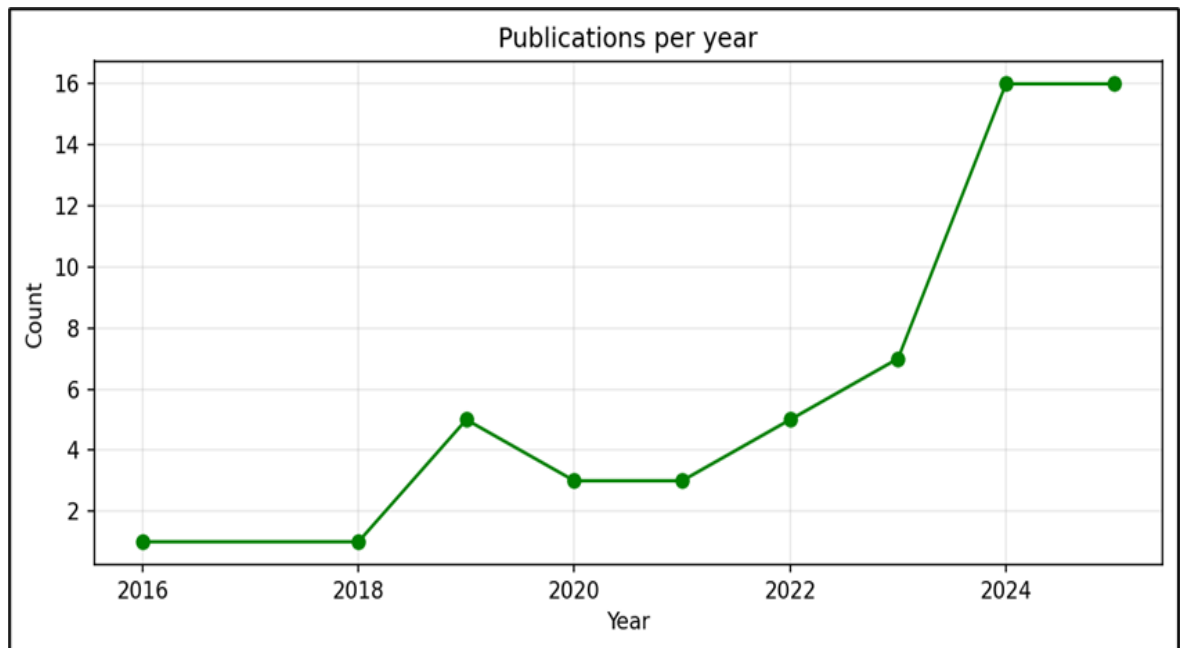


Fig. 1.3. Publications per year

Of the 57 articles in the filtered database, 17 received citation counts above the dataset average of 22, while 12 articles had not yet been cited at the time of analysis (Appendix I). The three most cited papers (Tolan et al., 2024; Laurin et al., 2016; Wessel et al., 2018) showed a strong influence in the research field, with 160, 157 and 141 citations. These works underscore the promising future of combining Sentinel-2 imagery with ML for forest monitoring.

The database included 25 journals contributing to the reviewed literature (Fig.1.4). The *Remote Sensing of Environment* and the *International Journal of Applied Earth Observation and Geoinformation* were the most productive sources, each with nine publications. They are followed by *Remote Sensing* and *Forests* (five publications each) and *Ecological Informatics* (four). The remaining journals included only one or two papers each. Publication activity was concentrated in a small number of leading remote sensing and environmental science journals. This reflected their important role in research on Sentinel-1/2 and forest monitoring applications (Puliti et al., 2020; Silveira et al., 2023).

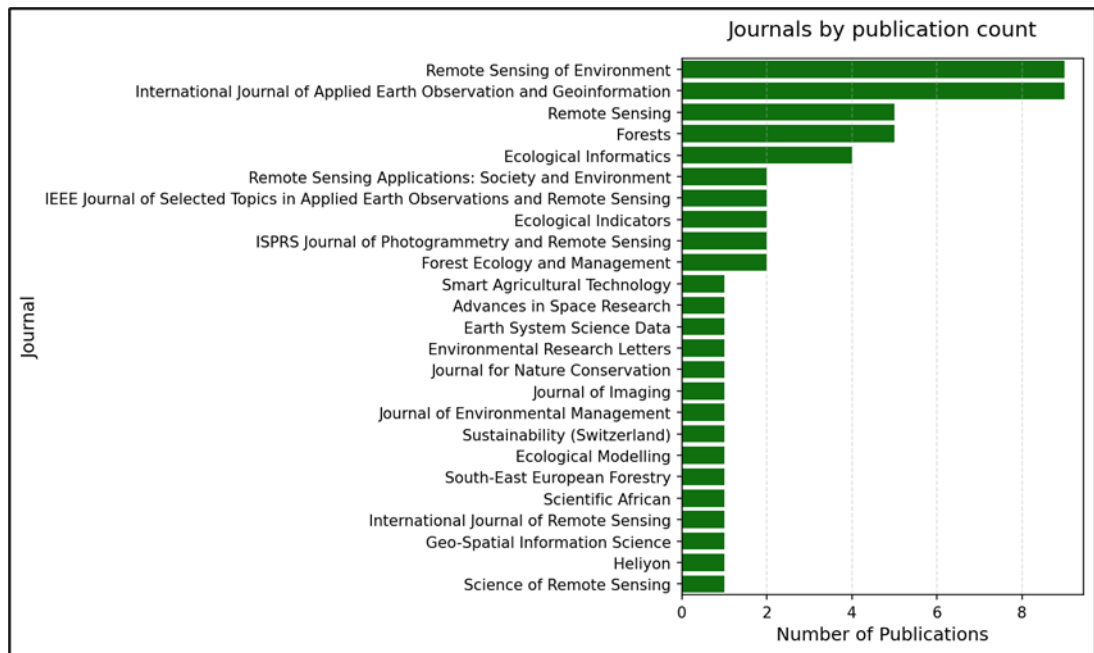


Fig. 1.4. Journals by publication count

Figure 1.5 presents the five most highly cited journals within the analysed dataset, with bubble size representing the H-index. *Sustainability (Switzerland)* and *Remote Sensing* had the highest total citation counts, with more than 220,000 and 90,000 citations, respectively. This showed that these journals played an important role in publishing research on Sentinel-2 and forest monitoring. The H-index values ranged from 103 to 236, which suggested that the leading journals had strong long-term academic influence. *Sustainability (Switzerland)* had the highest H-index at 236, followed by *Remote Sensing* at 216 and the *Journal of Environmental Management* at 210. Overall, the results suggested that research on Sentinel-2, remote sensing and forest monitoring was mainly published in a small number of well-established journals.

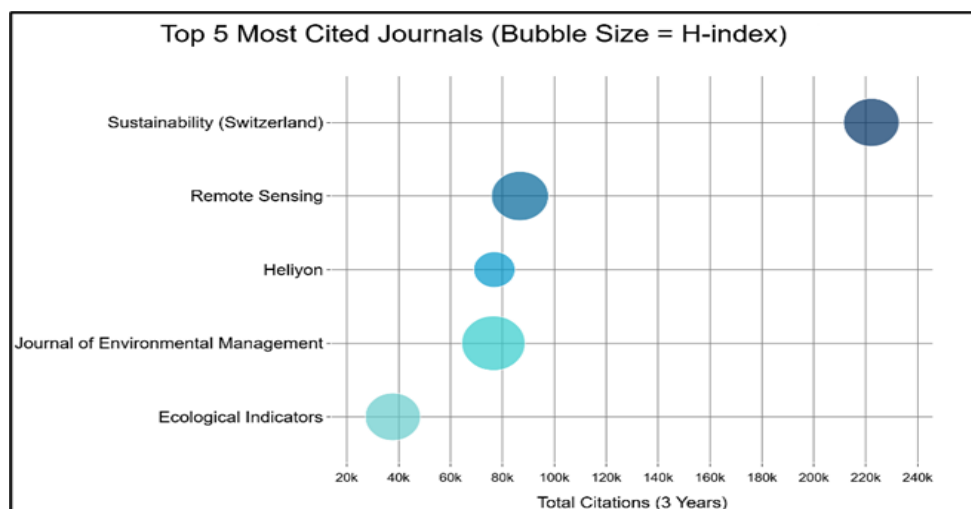


Fig. 1.5. Most cited journals

The keyword cooccurrence analysis revealed a well-connected research network with consistent thematic patterns across the analysed studies (Figure 1.6). Core terms such as Sentinel-2, remote sensing, machine learning, random forest, and forest inventory formed a dominant cluster, indicating a strong research focus on integrating satellite-derived data with ML approaches for forest monitoring and inventory attribute estimation. The strong links between these keywords suggested that they were often used together in the reviewed studies, especially in research focused on improving classification accuracy and prediction performance. Overall, the results indicated that the combination of ML techniques and remote sensing data constituted a central methodological framework within the reviewed research field.

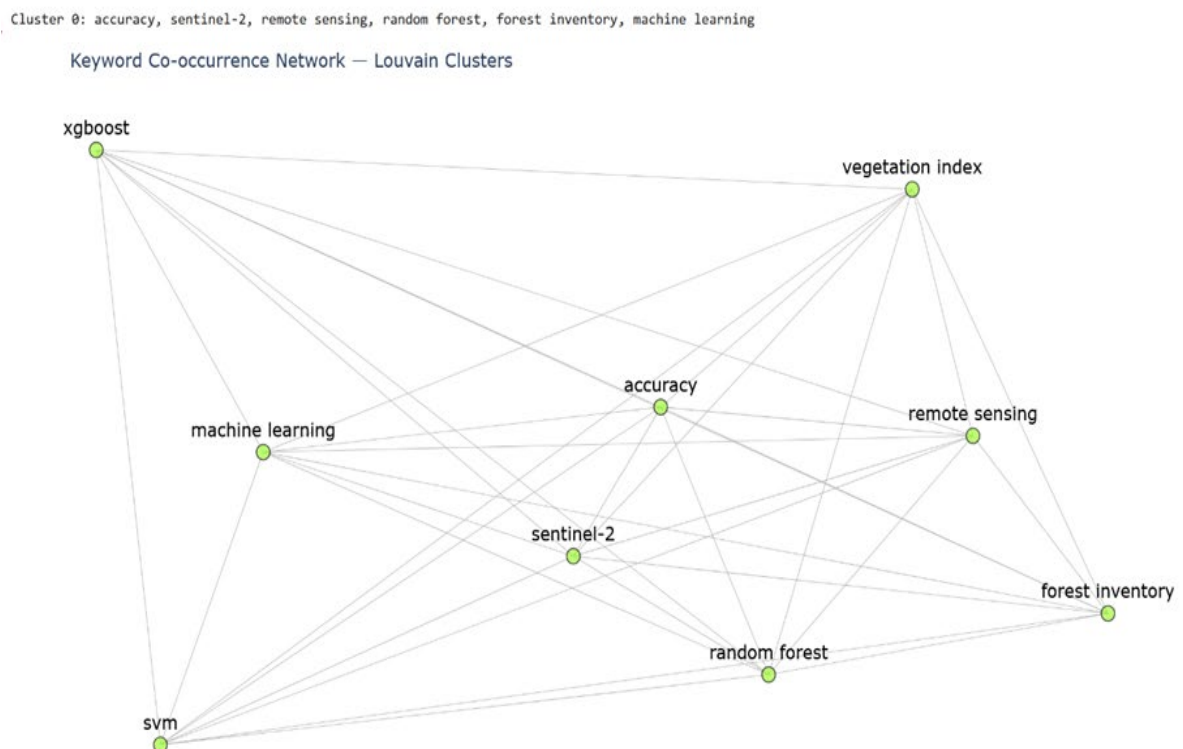


Fig. 1.6. Keyword Co-occurrence network

The citation network showed the relationships between key publications and how the research field developed over time (Figure 1.7). Foundational studies, including Breiman (2001, 2004), Gitelson (1996) and Lu (2016), were highly cited and placed in central positions in the network. This indicated their importance in shaping the main methods and concepts used in the field. More recent studies, such as Grabska (2019) and Breidenbach (2021), also appeared clearly in the network. These studies had high citation impact and helped connect earlier methodological research with later applications. Overall, the network showed how early methodological studies

supported later research on remote sensing and ML applications in forest inventory and monitoring.

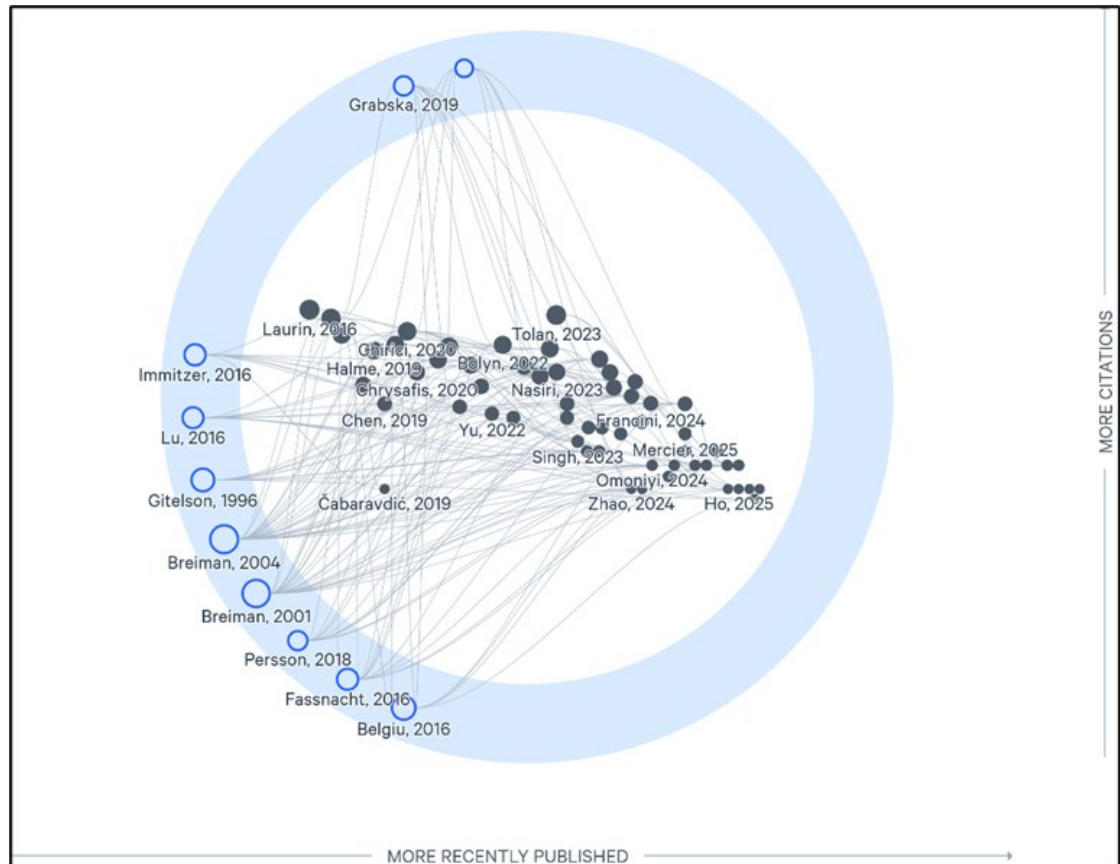


Fig. 1.7. Citation network

The bibliometric results showed that research on Sentinel-2 imagery and ML methods in forest monitoring and inventory became more active after 2020. More papers were published in leading remote sensing and environmental science journals, which suggested that this topic had become more established in the field. The keyword and citation network analysis also showed clear links between remote sensing, ML and forest attribute estimation. This indicated that these methods were often used together in the reviewed studies. Overall, the results showed steady progress in the field, supported by better access to satellite data, improved analysis methods and wider collaboration between researchers..

1.2. Forest Inventory Data Quality

National forest inventories were used as the main framework for collecting information on forest resources at national and regional levels. Studies based on Norwegian and Argentine inventory datasets showed that NFIs were designed to provide statistically representative estimates for large areas, rather than exact measurements at stand level (Puliti et al., 2020; Silveira et al.,

2023). Therefore, some uncertainty was expected when these data were used for more detailed stand-level analysis.

In data management, data quality was commonly assessed through dimensions such as accuracy, completeness, consistency, currency, integrity and uniqueness. These dimensions helped to determine whether data were suitable for their intended purpose (Yatskiv, 2025). When this framework was applied to NFIs, several dimensions became especially important, because inventory systems were shaped by statistical design, operational limits and changing protocols, rather than by the need for exact stand-level measurements.

Accuracy in NFI data was strongly affected by sampling design and spatial representativeness. Francini et al. (2024) showed that sparse and systematic NFI plot placement limited the ability to detect small-scale variation in species composition and forest structure. Similar results were reported in national-scale studies, where plot networks were designed to provide unbiased estimates for large areas, but were less precise at local scale (Puliti et al., 2020). As a result, NFI estimates were reliable at national level, while their accuracy was lower at sub-regional and stand levels.

Plot geometry and scaling were also closely linked to accuracy and consistency. Čabaravdić and Balić (2019) showed that differences in plot size and expansion rules affected the estimation of stand-level variables. Smaller plots could increase local variability, while larger plots could hide fine-scale structural differences. Biomass inventory studies also showed that differences in plot configuration made it more difficult to combine plot-level measurements into regional estimates, even when field protocols were consistent (Khan et al., 2024; Amuyou et al., 2022).

Measurement errors and observer related uncertainty also reduced the accuracy and reliability of NFI datasets. Field measurements of diameter at breast height (DBH), tree height and species identification were affected by observer experience, terrain conditions and instrument limitations (Kovárník and Janová, 2025). Small errors at individual tree level could build up when the data were combined into stand-level variables, such as GSV. This caused uncertainty at both plot and regional scales (Omoniyi and Sims, 2024; Novo-Fernández et al., 2024).

The currency and internal consistency of NFI data were impacted by the timing of inventory cycles. Plots in the Czech NFI were measured throughout a number of years rather than all at once, as demonstrated by Kovárník and Janová (2025). This meant that forest conditions from many years could be included in a single dataset. National canopy cover evaluations revealed similar variations between reporting and measuring periods (Heyer et al., 2025). Temporal inconsistency was also increased by forest growth, harvesting, and disturbance events between measurement years (Puliti et al., 2020).

Consistency between inventory cycles was impacted by modifications to inventory processes. To increase productivity or satisfy policy criteria, field definitions, classification schemes, and mapping guidelines were frequently revised. For instance, Punalekar et al. (2021) demonstrated how remote sensing may provide updated forest type information for management purposes by using Sentinel-2 time-series data and machine learning for national-scale larch mapping in Wales. The direct comparability of data from several inventory periods was diminished by similar methodological modifications in national mapping studies from Argentina and Poland (Grabska-Szwagrzyk et al., 2024; Silveira et al., 2023). The data was broken as a result of these modifications, making long-term trend analysis more challenging.

Limited documentation and metadata availability also affected completeness, reliability and transparency. Kovárník and Janová (2025) noted that incomplete documentation of measurement conditions or protocol changes made it harder to trace possible sources of error. Suleymanov et al. (2024) also showed that limited uncertainty reporting made inventory-derived products more difficult to interpret and reuse.

Figure 1.8 summarises the main structural sources of NFI data quality limitations identified in the literature, including sampling design constraints, plot geometry and scaling effects, measurement and observer uncertainty, temporal inventory cycles, protocol updates, and limited documentation. These factors relate to recognised data quality dimensions such as accuracy, consistency, currency, completeness, integrity, and uniqueness, and they originate from the initial stages of inventory planning and field data collection. These limitations remain regardless of the analytical or modelling methods used later.

Recognising NFI data quality as multidimensional is essential when integrating inventory attributes with satellite-derived predictors. Rather than assuming inventory variables represent absolute ground truth, they must be interpreted within the context of their design-based and operational constraints. This perspective provides the conceptual foundation for the present study and directly informs the formulation of the research questions concerning the quality and internal consistency of NFI data.

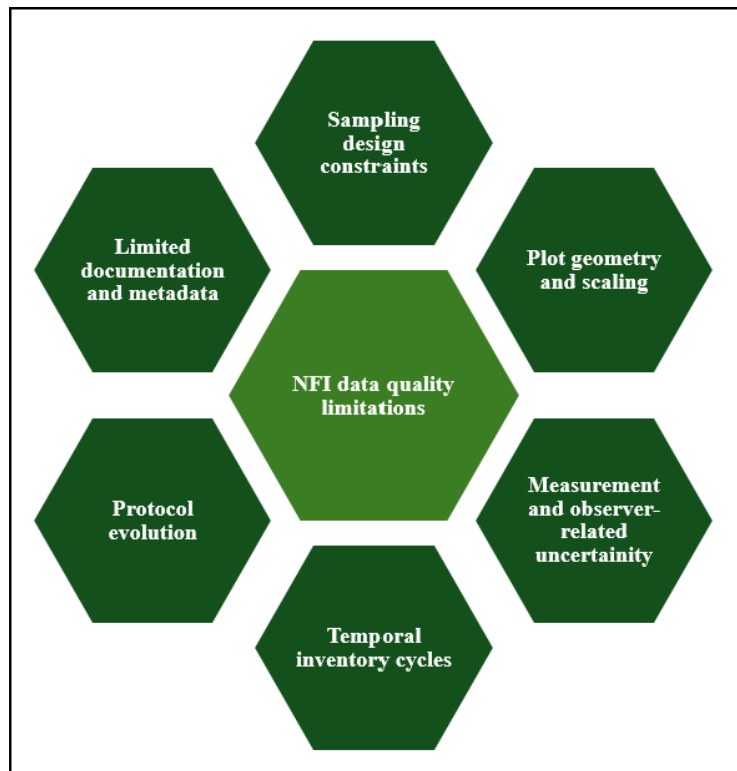


Fig. 1.8. NFI data quality limitations

1.3. Remote Sensing in Forestry

Remote sensing includes different methods for observing forest ecosystems, including optical, radar and Light Detection and Ranging (LiDAR) based approaches. Although all of these methods are relevant in forestry, this study focuses mainly on Sentinel-based satellite data, with particular attention given to Sentinel-2 multispectral imagery. Sentinel-1 SAR data are also considered as a potential additional source of information, depending on their suitability for the selected forest inventory attributes and modelling tasks.

1.3.1. Sentinel-2 characteristics

Sentinel-2 has been widely adopted for forestry applications because its spatial resolution enables canopy-level analysis, its spectral configuration captures vegetation-related properties, and its frequent revisit time enables repeated observation of forest changes. Grabska et al. (2019) showed that the (10–20) m spatial resolution of Sentinel-2 imagery is sufficient to capture canopy-level variability in both managed and semi-natural forests while remaining suitable for large-area analyses. Similar conclusions were reported by Čabaravdić and Balić (2019), who demonstrated that this spatial resolution supported stand-level forest structure modelling, although fine-scale within-stand variability could remain unresolved.

Temporal resolution was another feature that was consistently highlighted in the reviewed literature. Puliti et al. (2020) demonstrated that the short revisit cycle of Sentinel-2 supported the use of multi-temporal observations, thereby improving the robustness of forest attribute estimation by reducing the influence of single-date anomalies, including cloud contamination and transient phenological effects. In national-scale applications, Punalekar et al. (2021) and Heyer et al. (2025) further showed that consistent temporal coverage facilitated harmonised forest monitoring across reporting periods, enabling more reliable comparison of forest conditions over time.

A key feature of Sentinel-2, often mentioned in the reviewed literature, was its red-edge spectral bands. Grabska et al. (2019) showed that these bands were useful for detecting vegetation properties related to canopy condition and structure. This gave Sentinel-2 an advantage over the use of visible and near-infrared bands alone. Similar results were reported in studies estimating forest variables across different biomes, where red-edge information helped to improve separation in dense or structurally complex forest stands (Novo-Fernández et al., 2024).

Although Sentinel-2 had clear advantages, several studies also pointed out its limitations for forest structure analysis. Puliti et al. (2020) noted that Sentinel-2 became less sensitive in high-biomass conditions, especially in boreal forests, where optical saturation limited the detection of structural differences. The reviewed studies showed that Sentinel-2 was more useful for forest monitoring and inventory analysis when it was used in multi-temporal approaches or combined with other data sources.

Overall, the analysed studies showed that Sentinel-2 was well suited to large-scale forestry applications. It provided a useful balance of spatial detail, regular image availability and spectral information for forest monitoring and inventory analysis. These properties explained the widespread adoption of Sentinel-2 across forest structure estimation, species mapping, and inventory support studies included in this review.

1.3.2. Sentinel-1 SAR characteristic

Sentinel-1 is an active microwave satellite mission that delivers C-band Synthetic Aperture Radar (SAR) data, allowing for consistent Earth observation regardless of cloud cover or sunlight. This feature is frequently emphasised as a significant benefit for forest monitoring, especially in areas where optical data is hindered by persistent clouds or restricted acquisition periods (Puliti et al., 2020; Suleymanov et al., 2024). Since SAR is an active sensor, it can provide regular observations across different seasons and acquisition conditions. This makes it useful for large-area and operational forest monitoring.

Unlike optical sensors, which mainly captured canopy reflectance and vegetation condition, Sentinel-1 SAR data were sensitive to forest structure. Radar backscatter was linked to forest properties such as canopy density, woody biomass and differences between stands. Large-scale forest mapping studies showed that this made Sentinel-1 useful for estimating structural forest attributes, including growing stock, biomass and height (Becker et al., 2023; Puliti et al., 2020). This structural sensitivity makes SAR particularly useful in mature or dense forests, where optical signals tend to saturate and lose discriminatory power.

Sentinel-1 data are usually collected in dual polarisation modes: vertical transmit-vertical receive (VV) and vertical transmit-horizontal receive (VH), which are commonly used as predictors in ML models for forest attribute estimation. Puliti et al. (2020) found that cross-polarised backscatter (VH) is especially valuable for indicating vegetation structure. They also reported that combining VV and VH, along with derived metrics, enhances model robustness in inventory-focused studies. Similar findings were presented by Silveira et al. (2023), who showed that SAR-derived features, when combined with inventory data, significantly improved nationwide mapping of forest structure and biomass.

Recent research highlights the advantages of combining Sentinel-1 SAR data with optical imagery. Nationwide forest structure retrieval using both optical and SAR satellite data demonstrated that multi-sensor methods decrease spatial uncertainty and enhance consistency across various forest types. Furthermore, multimodal deep learning techniques that integrate SAR and optical data allow for more precise forest height mapping, especially in structurally complex forests and regions with diverse reference data coverage (Chen et al., 2025).

Overall, the reviewed literature showed that Sentinel-1 SAR data provided additional structural information that optical sensors alone could not capture. When Sentinel-1 was combined with Sentinel-2 imagery and analysed using supervised ML methods, forest attribute estimates became more reliable and spatially consistent. This was especially useful when uncertainty was present in the reference forest inventory data (Silveira et al., 2023; Suleymanov et al., 2024).

1.3.3. Satellite-derived features for forest structure estimation

In the reviewed literature, forest attribute estimation for NFI was mainly based on satellite-derived features, especially vegetation indices from Sentinel-2 imagery. These features were often used as indicators of forest attributes such as biomass, growing stock volume and canopy structure, because they reflected changes in canopy greenness, chlorophyll content and moisture conditions. The Normalised Difference Vegetation Index (NDVI) was commonly used to describe canopy condition and vegetation cover. Amuyou et al. (2022) and Omoniyi and Sims (2024) showed that

NDVI supported the estimation of biomass and growing stock volume by capturing differences in canopy density between forest stands.

Multiple studies also noted the limitations of NDVI in dense, mature forests. Puliti et al. (2020) found that NDVI became less sensitive in high-biomass areas due to saturation, thereby reducing its ability to distinguish more complex forest structures. These results led to a greater reliance on alternative optical indicators that better capture forest structural differences.

To address the limitations of traditional vegetation indices, recent studies increasingly relied on red-edge-based indices, especially the Red Edge Normalised Difference Vegetation Index (NDRE). Grabska et al. (2019) demonstrated that red-edge spectral data enhanced sensitivity to vegetation properties related to canopy condition and structure. Additional studies by Amuyou et al. (2022) and Omoniyi and Sims (2024) indicated that NDRE and similar red-edge indices offered better estimates of biomass and growing stock volume compared to NDVI alone. Novo-Fernández et al. (2024) supported these findings, demonstrating that red-edge indices excelled in distinguishing dense and structurally complex forest stands where traditional indices often saturated.

Beyond NDVI and NDRE, multiple studies have highlighted the complementary role of features that include shortwave infrared (SWIR) data. Puliti et al. (2020) found that SWIR metrics enhance sensitivity to variations in forest structure by capturing canopy moisture and shadow effects, particularly in boreal forests. Likewise, Suleymanov et al. (2024) demonstrated that SWIR-based indices improve model stability and support uncertainty assessment when used alongside vegetation indices targeting canopy greenness and chlorophyll content.

The reviewed literature showed that forest attribute estimation for NFI was improved when different satellite-derived features were combined, rather than relying on one index alone. NDVI was mainly used to describe canopy condition, while NDRE was more sensitive to structural characteristics. SWIR-based indices added useful information about moisture content and canopy structure. The limitations of optical indices in dense or mixed forests also showed the need to include structural information from radar data to improve the accuracy and reliability of forest attribute estimates.

In addition to optical vegetation indices, several studies showed the value of radar-based features from Sentinel-1 SAR data for estimating forest structure. Unlike optical indices, Sentinel-1 provided backscatter information that was sensitive to canopy geometry, woody biomass and differences between stands. Studies that combined Sentinel-1 with Sentinel-2 showed that VV and VH backscatter, their ratios and seasonal summaries added useful structural information. This was especially important in dense or mature forests, where optical indices could become saturated

(Puliti et al., 2020; Silveira et al., 2023). Suleymanov et al. (2024) also showed that Sentinel-1 features improved model reliability and supported spatial uncertainty assessment in growing stock volume estimation. These findings suggested that combining optical vegetation indices with radar-based structural features gave a more complete description of forest attributes for inventory-focused ML applications.

1.4. Machine Learning for Forest Attribute Modelling

Machine learning techniques have become a key component of forest attribute modelling, particularly in studies that integrate remote sensing data with field-based forest inventories. In this context, supervised learning methods were most commonly applied, as they enable a direct connection between satellite-derived predictors and reference forest attributes from NFI or field plots.

1.4.1. Supervised learning approaches for forest attribute estimation

Supervised learning methods were widely used in the reviewed literature to estimate forest attributes from labelled reference data, most often from NFI or field plot measurements. In these studies, remote sensing predictors were linked to known forest attributes using models trained on reference observations. This allowed both continuous attributes, such as above-ground biomass and growing stock volume, and categorical attributes, such as forest type and species composition, to be estimated (Puliti et al., 2020; Francini et al., 2024).

Many of the reviewed studies used supervised regression methods to estimate continuous forest attributes. Random Forest (RF) regression was the most commonly used model because it could handle many predictor variables and worked well across different forest conditions. Comparative studies showed that RF regression performed better than traditional linear regression models, as it could capture more complex relationships between spectral features, vegetation indices and forest attributes. For example, Fararoda et al. (2021) and Omoniyi and Sims (2024) showed that RF regression achieved higher prediction accuracy than linear regression across several forest types. Similar results were reported by Khan et al. (2024), where ensemble-based regression models produced more reliable estimates than parametric models.

The reviewed literature also showed that RF did not always outperform other supervised regression models. Studies comparing multiple algorithms indicated that results depended on forest structure, predictor choice, and spatial scale. For example, Zhang et al. (2024) and Bui et al. (2024) found that optimised or combined ensemble methods offer more robustness and sometimes

slight performance gains over standard RF. These findings suggest that while RF regression is popular and competitive, no single supervised regression model consistently outperforms others across all forest conditions.

In addition to regression tasks, several studies used supervised classification methods to estimate categorical forest attributes, such as tree species and forest type. In the reviewed literature, RF classifiers and Support Vector Machine (SVM) were among the most commonly used algorithms. This showed their ability to handle high-dimensional spectral-temporal features from Sentinel-2 imagery. Comparative studies showed that these algorithms produced stable results across different forest conditions. For example, Wessel et al. (2018) showed that RF and SVM classifiers achieved consistent classification accuracy across several forest types, while Nasiri et al. (2023) confirmed their usefulness in structurally complex mixed forests. At larger spatial scales, national and regional mapping studies, such as Punalekar et al. (2021) and Grabska-Szwagrzyk et al. (2024), also showed the practical value of supervised classification methods for operational forestry and inventory applications.

Among the reviewed studies, RF models were the most commonly used supervised learning methods for both regression and classification tasks. This showed their reliability and wide use across different forest conditions. RF regression was primarily used to estimate continuous forest attributes, while Gradient Boosting (GB) and Support Vector Regression (SVR) were mainly used in comparative analyses. Most studies classified forest attributes using RF and SVM models, while newer research increasingly experiments with neural-network-based classifiers, although these are not yet the standard approach.

1.4.2. Feature engineering, feature importance, and uncertainty

Feature engineering was recognised as a crucial element in supervised ML methods for estimating forest attributes. The reviewed studies indicated that model performance depended heavily on how input features were derived from Sentinel-2 data, rather than on the choice of algorithm. Several studies have demonstrated that integrating spectral bands, vegetation indices, and temporal summaries improves predictive accuracy more than relying solely on raw spectral data. For example, Khan et al. (2024) and Amuyou et al. (2022) showed that engineered features capturing canopy condition and seasonal variability improved the estimation of biomass and GSV.

Temporal feature engineering emerged as particularly important in studies using multi-date Sentinel-2 imagery. Research by Heyer et al. (2025) on national-scale forest monitoring demonstrated that aggregating spectral information over time reduced sensitivity to noise from cloud contamination or short-term phenological effects. Similarly, studies estimating forest

structure and species composition reported improved robustness when harmonic predictors or multi-temporal metrics were incorporated into supervised models (Francini et al., 2024).

To make model results easier to interpret, many studies used feature importance analysis, especially with tree-based ensemble models. In the reviewed literature, RF-based importance measures were often used to identify which spectral bands, vegetation indices or temporal features had the strongest effect on model predictions. Studies on forest variable estimation often highlighted the importance of red-edge bands, vegetation indices and shortwave infrared features for modelling forest structure (Novo-Fernández et al., 2024). Similar results were reported by Amuyou et al. (2022) and Khan et al. (2024), where vegetation indices and red-edge-related predictors were found to be important for biomass and volume estimation. Feature relevance was also examined in studies using harmonised spectral-temporal predictors for forest mapping, which further supported the interpretability of ensemble-based models (Francini et al., 2024).

Besides optical features from Sentinel-2 imagery, several studies highlighted the value of adding radar-based features from Sentinel-1 Synthetic Aperture Radar (SAR) data. Sentinel-1 backscatter metrics, such as VV and VH polarisation and their combinations, provided additional information about forest structure that optical vegetation indices could not fully capture. Studies combining Sentinel-1 and Sentinel-2 data showed improved model reliability and reduced problems with saturation in dense or mature forests, especially for structural attributes such as biomass and growing stock volume (Puliti et al., 2020; Silveira et al., 2023). Suleymanov et al. (2024) also showed that Sentinel-1 features supported spatial uncertainty assessment and improved the stability of ML estimates for forest attributes.

Uncertainty assessment showed variability in consistency throughout the literature, but its importance in applying ML outputs practically has become more widely recognised. While many studies included standard accuracy metrics, fewer explicitly addressed prediction uncertainty or spatial variation in model performance. Recent research has increasingly concentrated on uncertainty analysis, demonstrating that spatially explicit uncertainty estimates improve the interpretability and dependability of forest attribute maps. This trend was apparent in studies like (Suleymanov et al., 2024) and (Heyer et al., 2025). Validation-centred research also highlighted the need for independent assessment with NFI data to identify systematic errors and promote responsible use of ML models in forest inventory systems, as shown in (Kovárník and Janová, 2025). Similar conclusions about the significance of validation and uncertainty reporting were observed in regional biomass and carbon stock assessments (Zheng et al., 2025).

Taken together, the reviewed studies indicated that careful feature engineering, clear analysis of feature importance, and explicit evaluation of uncertainty are crucial for dependable ML-based

forest attribute estimation. These elements significantly enhance model robustness and interpretability, and make the models more suitable for inventory-focused applications.

1.4.3. Summary of existing studies

The literature review highlighted the extensive use of supervised ML methods to estimate forest attributes using Sentinel-2 data across various forest ecosystems and spatial scales. Consistently, studies indicated that integrating multispectral satellite imagery with ML enabled the estimation of both continuous attributes, such as above-ground biomass and growing stock volume, and categorical attributes, such as forest type and species composition, thereby aiding forest inventory and monitoring efforts.

Most studies focused on biomass and volume variables, using ensemble regression models most commonly. Regional and national research found that RF and hybrid ensemble methods provided reliable results across various forests, especially with NFI data. Besides biomass and volume estimation, numerous studies used ML models to predict forest age, detect disturbances, and map large-scale forest structures, demonstrating the wide applicability of Sentinel-2-based methods for various forest attributes (Chen et al., 2024; Yadegari et al., 2025).

Studies addressing categorical forest attributes further showed that supervised classification models enabled reliable mapping of tree species composition and forest types at regional and national scales. Harmonised Sentinel-2 time series combined with inventory data supported operational forest mapping suitable for management and reporting purposes.

In addition to optical imagery, multiple studies have shown that combining Sentinel-1 SAR data with Sentinel-2 features enhances the estimation of forest structure attributes. Radar-based features offer additional insights into canopy structure and are less affected by saturation in dense or mature forests, leading to more robust inventory models and better uncertainty management (Puliti et al., 2020; Silveira et al., 2023; Suleymanov et al., 2024).

Overall, the reviewed studies indicated that satellite-based ML methods represented a well-developed and actively progressing research area. Although significant improvements in predictive performance and spatial coverage have been achieved, the literature consistently emphasises the need for transparent feature selection, explicit uncertainty assessment, and independent validation to ensure the reliable use of ML outputs in forest inventory frameworks.

1.5. Identified Knowledge Gap and Justification for Chosen Approach

Reliable forest inventory information is essential for sustainable forest management, economic planning, carbon accounting, and policy reporting. Field-based inventory data are expensive to collect and are updated only periodically, which can lead to temporal mismatches and inconsistencies within stands. In operational contexts, outdated or inconsistent inventory attributes can result in poor management choices, suboptimal harvest planning, or skewed carbon stock evaluations.

At the same time, satellite data from Sentinel missions provide continuous spatial coverage and repeated observations of forest conditions. Integrating them with NFI data creates opportunities not only for predicting forest attributes but also for systematically evaluating the consistency of inventory data.

The reviewed literature shows substantial progress in the use of Sentinel-2 imagery and supervised ML methods for forest attribute estimation. Many studies demonstrate that these approaches can predict above-ground biomass, growing stock volume, stand height, species composition, and structural characteristics across different forest ecosystems and spatial scales. The integration of multi-temporal spectral features and red-edge information, and in some cases Sentinel-1 SAR data, has improved model performance and spatial consistency.

Despite these advances, several limitations remain. Most studies primarily aim to optimise predictive accuracy and treat NFI data as fully reliable reference information. In practice, NFI datasets may contain uncertainties related to sampling design, plot configuration, field measurement procedures, and temporal mismatches between inventory and satellite acquisition dates. When such factors are not explicitly considered, model outputs may reproduce inconsistencies present in the reference data.

Furthermore, strong predictive performance at regional or national levels does not automatically ensure robustness across diverse forest conditions. Variability in stand age, species composition, and structural complexity affects the relationship between satellite predictors and inventory attributes. Models calibrated under specific conditions may therefore show reduced reliability in structurally different stands.

The purpose of this master's thesis is to design and evaluate a supervised ML-based procedure that supports forest inventory data quality enrichment. Satellite-derived predictors from Sentinel-2 multispectral imagery, including spectral bands and vegetation indices, are combined with structural metrics derived from Sentinel-1 SAR backscatter and linked to selected NFI attributes. Rather than treating inventory data as unquestionable reference values, the modelling framework is used to examine model behaviour, residual structures, and feature importance

patterns in relation to forest development stages and stand characteristics. Through this approach, prediction deviations are analysed as potential indicators of inconsistencies, outdated values, or structural biases within inventory records.

By integrating remote sensing data with a systematic evaluation of model residuals and uncertainty patterns, the study looks to offer a transparent and reproducible procedure that enhances the interpretability and operational reliability of forest inventory data in practical forest management contexts.

2. RESEARCH METHODOLOGY

This chapter presents the research methodology applied in this study. It first describes the data sources used, including the NFI dataset and Sentinel satellite imagery. The chapter then explains the data preprocessing procedures and feature engineering steps adopted to prepare the predictor variables. It outlines the supervised ML modelling framework used for regression and classification tasks. Finally, the chapter describes the accuracy assessment approach and the software tools applied to ensure transparent, reproducible, and inventory-orientated forest attribute estimation. The overall workflow of the study, from data acquisition to model evaluation, is summarised in Figure 2.1 (a detailed version of the pipeline is provided in Appendix II).

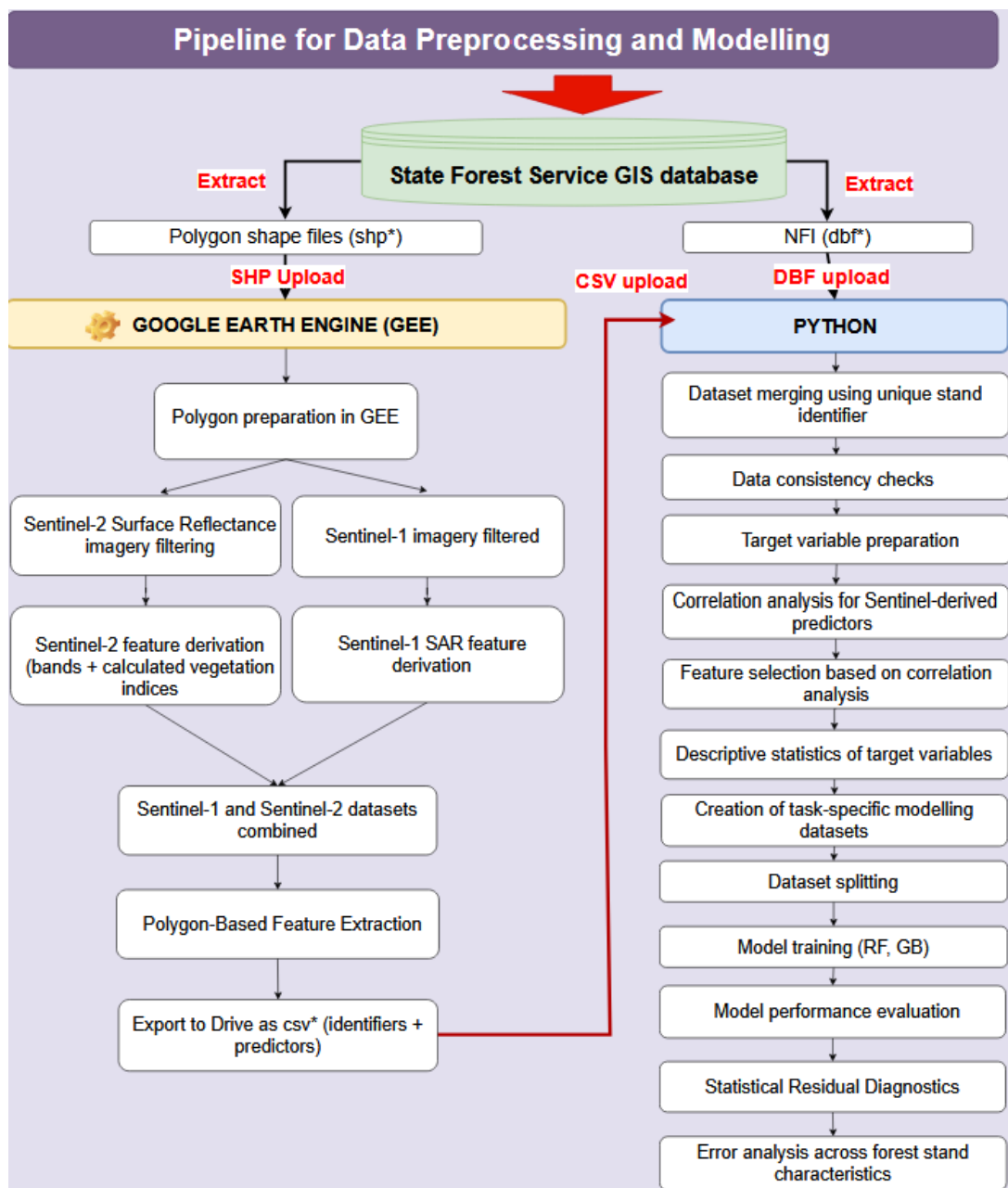


Fig. 2.1. Pipeline for data preprocessing and modelling

2.1. Introduction - Data Sources

Modern forest monitoring increasingly relies on integrating field inventory data and satellite remote sensing. Previous studies showed that Sentinel-2 imagery, combined with forest observations, effectively models attributes such as species, canopy, and biomass (Grabska-Szwagrzyk et al., 2024; Wessel et al., 2018). This method enabled precise ground truth calibration and spatial mapping across extensive regions. Additionally, recent studies have demonstrated that using Sentinel-1 synthetic aperture radar (SAR) data enhances the estimation of forest structural variables. This data offers sensitivity to canopy structure and moisture levels, regardless of cloud cover (Puliti et al., 2020; Silveira et al., 2023).

NFI data provide detailed stand-level information such as species composition, age, stand structure, and productivity indicators. These served as reference data for supervised ML models. The forest inventory data were provided as polygon-based GIS shapefiles with attribute tables, enabling spatial connection between field observations and satellite-derived predictors. Past research indicates that combining NFI plots with satellite data enhances the accuracy of forest attribute estimation at regional and national levels (Francini et al., 2024; Nasiri et al., 2023).

In this study, both forest inventory data and satellite imagery referred to the same forest properties located in Latvia. The study area covered 11,155.14 ha of forestland owned by a private investment company, with the properties' spatial distribution shown in Figure 2.2. Sentinel-2 multispectral imagery and Sentinel-1 SAR data were processed using Google Earth Engine (GEE) to derive spectral, radar, and temporal features. NFI data were used for model training and validation, while satellite data provided spatially continuous information for detecting patterns and inconsistencies in forest attributes. A detailed description of the inventory dataset, satellite imagery, and preprocessing steps is provided in the following subsections.

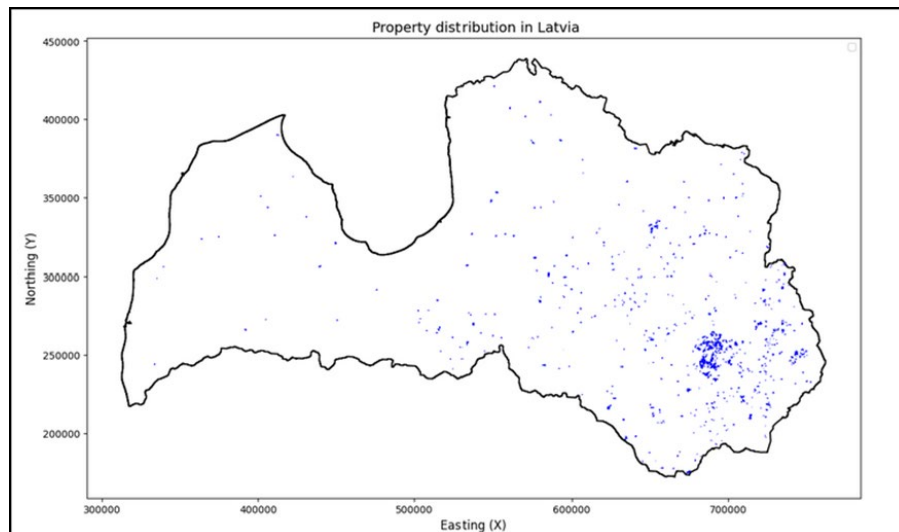


Fig. 2.2. Property distribution in Latvia

2.2. Forest Inventory Dataset Description

The forest inventory data used in this study followed the national regulation on forest inventory and forest state register information circulation (Cabinet Regulation No.384, 2016). This regulation required the collection of essential stand-level variables during each forest inventory cycle, such as spatial location, stand structure, species composition, site characteristics, stand origin, management history, and land-use details. These guidelines helped maintain data consistency and standardisation across all forest regions.

The State Forest Service offers access to these data via its online forest management platform. Once users, such as forest owners, authenticate, they can access and download inventory datasets at the stand level in various formats, including PDF, Excel, and DBF. For the purposes of this research, the DBF format was selected because it contained the full set of 170 machine-readable attributes, followed a strict schema defined by the official specification file, and was optimally suited for integration into geospatial and ML workflows.

To illustrate the traditional format in which forest inventory information was presented, an example of a PDF-based inventory report was included in Appendix III. While this format provided a readable summary of key stand characteristics, it contained only a limited subset of variables and was not suitable for quantitative analysis or automated processing. For this research, the DBF format was selected as the primary data source. The DBF file contained a set of 170 stand-level attributes in a consistent, machine-readable structure defined by the State Forest Service, including detailed information on stand structure, species composition, management history, regeneration, protection categories and metadata. This detailed schema enabled smooth integration with geospatial tools and was perfect for feature engineering and ML workflows. Appendix IV offers a detailed specification of all DBF attributes, including their names, descriptions, units, categories, and data types.

To support the methodological overview, Table 2.1 summarises the core attribute groups included in the NFI dataset and provides examples of their main variables. These groups represent the essential descriptive elements needed to characterise forest stands and to link field-measured conditions with remote sensing data.

table 2.1

The main categories of forest inventory

Category	Core Attributes Included	Short Description
Identification & Geometry	APS, KAD, KV, NOG, PLAT	Stand identifiers, cadastral information, compartment and stand numbers, and total stand area used to locate and reference forest stands.
Site Conditions	ZKAT, MT, BON	Soil type, forest/site type, and fertility class describing ecological growing conditions.
Stand Origin & Regeneration	IZC, MDZG, GI, VGR, GK	Stand origin (natural or planted), birth/planting year, age group, and regeneration year indicating stand development history.
Stand Structure (Main Layer)	G1, K10, S10, A10, H10, D10, G10, N10, V10, B10	Main canopy attributes: species, age, height, diameter, basal area, tree density, growing stock, and canopy density.
Stand Structure (Additional Layers, if present)	G2, S22, A22–A24, H22–H24, D22–D24, V22–V24, B22	Structure of secondary canopy layers, capturing multi-layered forest stands where species or age groups differ vertically.
Species Composition	S10–S14, SKOK1–SKOK3	Dominant and secondary species codes, species shares, and retained tree species used to characterise stand composition.
Management History	P_DARBV, P_DARBG, P_CIRP, P_CIRG, ATZARP, NEP_ATJ, NEP_JKOP	Records of silvicultural operations (thinning, pruning, final felling), years of execution, and required future management actions.
Land-Use Elements	EXPL_MEZS, EXPL_CELI, EXPL_GRAVJI	Proportions of forested area, road area, and ditch area within the stand.
Protection & Special Conditions	VA_KODS1–5, AIZS, VA_UNIK, IPAT1–5	Legal protection categories, protected objects, and special environmental features or restrictions affecting forest management.
Metadata	INV_DATUMS, REG_DATUMS, GTF_NATURE, ORIG_ID, VMD_ID	Administrative information on inventory dates, fieldwork year, and system identifiers required for data traceability.

Although NFI data were collected according to protocols, several studies showed that some attribute groups consistently exhibit higher uncertainty. Puliti et al. in (2020) noted that many structural attributes in forest inventories originated from measurements taken on only a small number of sample trees, which were then expanded to the stand level through allometric equations. This sampling-based workflow naturally introduced uncertainty, particularly in heterogeneous stands or areas with steep terrain. Amuyou et al. (2022) similarly showed that field-derived measurements of stand condition and biomass exhibited substantial plot-level variability and did not always accurately reflect on-ground conditions due to rapid ecological changes, highlighting the challenges of maintaining consistent accuracy in manual forest inventory data. Together, these studies indicated that forest inventory datasets may contain errors due to sampling limitations, observer bias, subjective classification, and temporal mismatch with actual stand conditions.

2.3. Sentinel Satellite Imagery Selection and Acquisition

Satellite remote sensing data were selected to provide spatially continuous and temporally consistent information that complements field-based forest inventories. As demonstrated in prior inventory-focused studies (Grabska et al., 2019; Francini et al., 2024), Sentinel satellite imagery provides an optimal balance of spatial resolution, temporal frequency, and open accessibility. Therefore, this research utilised Sentinel-2 multispectral imagery together with Sentinel-1 synthetic aperture radar (SAR) data.

Sentinel-2 imagery was selected as the primary optical data source because its 10–20 m spatial resolution enables canopy-level analysis while supporting large-scale forest assessments. Grabska et al. (2019) showed that this resolution captures variability in forest structure across managed and semi-natural forests, and Čabaravdić and Balić (2019) found that it supports stand-level modelling in diverse forest landscapes. Additionally, Sentinel-2's multispectral setup offers reflectance data across visible, near-infrared, red-edge, and shortwave infrared bands, which are valuable for estimating forest attributes (Novo-Fernández et al., 2024).

To enhance optical data, Sentinel-1 SAR data were integrated due to their sensitivity to forest structure and their independence of cloud cover and illumination conditions. Puliti et al. (2020) noted that radar backscatter features provide extra insights into vegetation volume and canopy structure, especially in dense forests where optical data may saturate. Similarly, Silveira et al. (2023) indicated that combining Sentinel-1 and Sentinel-2 data improves the accuracy of forest structure mapping at the national level. The dual-polarisation modes (VV and VH) of Sentinel-1 allow the extraction of structural features related to canopy roughness and moisture levels.

All satellite imagery was accessed and processed through GEE, a cloud-based platform that facilitates efficient management of extensive satellite archives and integrates smoothly with vector-based NFI data. Sentinel-1 and Sentinel-2 image collections were spatially filtered to include only imagery overlapping the study area, using the forest inventory polygon boundaries.

Following frequent practice in forest remote sensing studies (Puliti et al., 2020), a seasonal window from June 1 to August 31, 2025, was selected to depict the growing season, when forest canopies are fully developed and spectral responses are most consistent. This selection minimises the effects of snow cover, low solar angles, and phenological changes. Additionally, Sentinel-2 images were filtered for cloud-cover metadata prior to compositing to enhance data quality.

All satellite data acquisition and initial preprocessing steps were completed in GEE, ensuring consistent spatial resolution, coordinate systems, and temporal alignment across datasets. The derived satellite features were subsequently linked to forest inventory polygons for further preprocessing, feature creation, and ML analysis.

2.4. Selection of Forest Inventory Attributes for Modelling

Forest inventory datasets contain numerous attributes related to forest structure, composition, and development. Previous inventory-orientated remote sensing studies have shown that not all variables are equally effective for satellite-based modelling, especially when considering inventory uncertainty and the observability at the canopy level (Puliti et al., 2020; Suleymanov et al., 2024). Consequently, a selective approach was used to choose forest inventory attributes that are both practical for operational use and feasible to predict with Sentinel satellite data.

Attribute selection was based on three criteria. Firstly, attributes needed to be essential for forest inventory and management, thereby supporting planning, reporting, and resource assessment. Secondly, preference was given to attributes known to show uncertainty or inconsistency in NFI data, such as measurement error, timing discrepancies between inventory cycles, or spatial generalisation (Kovárník and Janová, 2025; Silveira et al., 2023). Thirdly, attributes were selected if prior research indicated their sensitivity to spectral or structural data obtained from Sentinel-2 multispectral imagery and Sentinel-1 SAR data (Francini et al., 2024; Puliti et al., 2020).

Based on these criteria, four forest inventory attributes were selected for modelling based on these criteria, encompassing both continuous and categorical data. GSV (m^3/ha) and stand height (m) serve as continuous regression targets because they are crucial indicators of forest structure and productivity, and both are influenced by uncertainty in inventory records. Additionally, dominant tree species and age class function as categorical classification targets, providing vital information for forest classification, management, and reporting. Age class was selected instead of exact age to minimise sensitivity to inventory noise and enhance classification robustness.

Other inventory variables were deliberately excluded from modelling. Attributes that mainly serve administrative purposes, have weak connections to canopy-level properties, or rely heavily on site-specific conditions were deemed unsuitable for satellite-based prediction. This approach ensured the model concentrated on attributes where ML could effectively enhance inventory data accuracy.

Table 2.2 presents a summary of the chosen forest inventory attributes, the types of models used, and the reasons these particular attributes were selected.

table 2.2

Selected forest inventory attributes for modelling

Forest inventory attribute	Variable type	Modelling task	Justification for selection
Growing stock volume (m ³ /ha)	Continuous	Regression	Core forest inventory variable; strongly linked to canopy structure and biomass; known uncertainty in inventory data; widely modelled using Sentinel-1 and Sentinel-2 features.
Stand height (m)	Continuous	Regression	Key structural indicator of forest development; stable inventory attribute; indirectly observable from canopy structure and SAR backscatter; complementary to volume estimation.
Dominant tree species	Categorical	Classification	Essential for forest classification and management; strong spectral sensitivity in multispectral imagery; supported by red-edge information; widely addressed in inventory-oriented studies.
Age class	Categorical	Classification	An operationally robust representation of forest development stage; less sensitive to inventory noise than exact age; commonly used in forest management and reporting.

2.5. Data Preprocessing

Data preprocessing was an essential step to ensure consistency between forest inventory data and satellite-derived predictors before ML modelling. Procedures were applied separately to the forest inventory dataset and satellite imagery, then harmonised and aligned spatially. All steps aimed to minimise noise, reduce systematic bias, and ensure that satellite features accurately reflected the forest stands detailed in the inventory data.

2.5.1. NFI dataset preprocessing

The forest inventory dataset was extracted in DBF format and imported into the Python environment for further processing. Preprocessing focused primarily on attribute selection, consistency checks, and preparation for integration with Sentinel-derived predictors.

First, relevant forest inventory attributes were selected for modelling, including growing stock volume (V_NOG), stand height (H10), dominant tree species (S10), and stand age (A10). These attributes were retained as target variables for subsequent regression and classification tasks.

Second, a composite identifier (merge_ID) was constructed using cadaster number, quartal number, polygon number, and subpolygon number. This identifier enabled consistent merging between the NFI dataset and the polygon-based Sentinel feature table exported from GEE.

Basic data consistency checks were performed to identify duplicate records, missing values, implausible attribute values, and incorrect data types. No extensive manual corrections were

applied, as the objective was to retain the original inventory information and evaluate its consistency through modelling.

For classification purposes, age class categories were derived from stand age using predefined thresholds (0–20, 21–40, 41–80, ≥ 81 years). This transformation enabled categorical modelling while preserving the original age information.

NFI preprocessing aimed to ensure structural consistency and compatibility with satellite-derived predictors while maintaining the integrity of the original inventory records.

2.5.2. Satellite data cloud masking and temporal filtering

Satellite imagery was temporally filtered to focus on the main growing season, during which forest canopies are fully developed and spectral responses remain most stable. A seasonal window from 1 June to 31 August 2025 was chosen to reduce the effects of snow cover, low solar angles, and phenological changes.

For Sentinel-2 imagery, cloud-contaminated observations were removed using quality-assessment information and scene classification layers. Pixels affected by clouds, shadows, or atmospheric artefacts were masked prior to further processing. This step reduced spectral noise and ensured that vegetation indices and reflectance values accurately reflected true canopy conditions, free of transient atmospheric effects.

Sentinel-1 SAR data added useful structural information compared with optical imagery. Still, SAR backscatter can be affected by topography and may also become saturated in dense forest conditions. In areas with complex terrain, differences in slope, surface conditions, and forest structure can change the radar response and reduce prediction accuracy. Similar limitations have also been reported in previous studies (Nian et al., 2025).

2.5.3. Satellite data harmonisation and alignment

After individual preprocessing, the satellite datasets and forest inventory polygons were aligned to verify spatial and temporal consistency. All satellite data were resampled to match a common spatial resolution based on the main Sentinel-2 grid (with a pixel size 10x10m). Temporal harmonisation made sure that both Sentinel-1 and Sentinel-2 features represented comparable seasonal conditions.

Spatial alignment was carried out by limiting satellite imagery to the boundaries of the forest inventory polygons, ensuring only pixels within these areas contributed to feature extraction. All data were processed within a uniform spatial reference system, enabling accurate integration of optical, radar, and inventory data.

2.5.4. Satellite data polygon-based spectral extraction

To connect satellite-derived data with forest inventory information, spectral and radar features were summarised at the polygon level. For each forest stand polygon, satellite pixel values within its boundaries were aggregated using statistical methods.

Mean values primarily reflected average canopy conditions, providing stable predictors for ML models. This aggregation minimised pixel noise and ensured compatibility with the inventory dataset's stand-level structure. As a result, each forest polygon had a corresponding set of satellite-derived predictor values aligned with the inventory attributes used as a reference.

Due to computational limitations and the large number of polygons, the extraction was performed in batches. This approach enabled efficient processing and consistency across the study area. The output was tabular datasets linking satellite features to individual forest inventory polygons, serving as inputs for feature engineering and modelling.

2.6. Sentinel-derived Feature Description

Satellite-derived predictors were extracted from Sentinel-2 multispectral imagery and Sentinel-1 SAR data using the GEE platform. These predictors represented spectral reflectance, vegetation condition, canopy moisture, and structural properties of forest stands. All satellite features were spatially linked to forest inventory polygons and aggregated to the stand level to ensure consistency with the inventory data structure.

Table 2.3 summarises the complete set of Sentinel-derived features used in the modelling framework, including original spectral bands, radar backscatter metrics, and vegetation indices. A detailed explanation and justification of the selected vegetation indices is provided in the following subsection.

Mean values were used to summarise satellite features at the polygon level, as they provide a stable representation of average stand conditions after cloud filtering and seasonal compositing.

In addition to individual VV and VH backscatter values, difference and ratio-based SAR metrics (VV–VH and VV/VH) were included to capture relative changes in scattering mechanisms related to forest structure and canopy complexity, as suggested in previous inventory-orientated studies (Puliti et al., 2020; Becker et al., 2023; Silveira et al., 2023).

table 2.3

Satellite-derived features

Feature group	Feature name (CSV)	Satellite	Aggregation	Description
Spectral band	B2	Sentinel-2	Mean	Blue reflectance (490 nm), sensitive to atmospheric scattering, shadows, and water bodies
Spectral band	B3	Sentinel-2	Mean	Green reflectance (560 nm), related to canopy colour and vegetation vigour
Spectral band	B4	Sentinel-2	Mean	Red reflectance (665 nm), strongly absorbed by chlorophyll, sensitive to photosynthetic activity
Spectral band	B5	Sentinel-2	Mean	Red-edge reflectance (705 nm), sensitive to chlorophyll content and early stress signals
Spectral band	B6	Sentinel-2	Mean	Red-edge reflectance (740 nm), sensitive to canopy structure and biomass variations
Spectral band	B7	Sentinel-2	Mean	Red-edge reflectance (783 nm), sensitive to dense vegetation
Spectral band	B8	Sentinel-2	Mean	Near-infrared (NIR) reflectance (842 nm), strongly related to biomass and canopy density
Spectral band	B8A	Sentinel-2	Mean	Narrow near-infrared reflectance (865 nm), enhanced sensitivity to vegetation structure
Spectral band	B11	Sentinel-2	Mean	Shortwave infrared (SWIR1) reflectance (1610 nm), sensitive to canopy moisture content
Spectral band	B12	Sentinel-2	Mean	Shortwave infrared (SWIR2) reflectance (2190 nm), responsive to vegetation moisture and structural characteristics
Vegetation index	NDVI	Sentinel-2	Mean	Normalised Difference Vegetation Index (NIR–Red), an indicator of canopy greenness and density
Vegetation index	NDRE	Sentinel-2	Mean	Red-Edge Normalised Difference Vegetation Index, improved sensitivity in dense canopies
Vegetation index	NDMI	Sentinel-2	Mean	Normalised Difference Moisture Index, indicator of vegetation water status
Vegetation index	NBR	Sentinel-2	Mean	Normalised Burn Ratio, sensitive to canopy structure, disturbance, and moisture changes
SAR backscatter	VV	Sentinel-1	Mean	Co-polarised C-band radar backscatter, sensitive to surface and double-bounce scattering mechanisms
SAR backscatter	VH	Sentinel-1	Mean	Cross-polarised C-band radar backscatter, sensitive to volume scattering within vegetation canopies
SAR metric	VV_VH_ratio	Sentinel-1	Mean	Ratio of VV to VH backscatter; an indicator of structural variation and scattering dominance
SAR metric	S1_VVminusVH	Sentinel-1	Mean	Difference between VV and VH backscatter, indicating relative dominance of surface vs. volume scattering

2.7. Feature Engineering

Feature engineering was applied to transform the satellite-derived variables into predictors appropriate for ML models. In addition to the original spectral and radar features, a selection of vegetation indices and SAR-based metrics was developed to increase sensitivity to forest canopy

characteristics and mitigate the effects of illumination and background influences. The process of feature engineering concentrated on enhancing the representation of forest structure, moisture levels, and vegetation vitality, while ensuring alignment with forest attributes at the inventory scale.

2.7.1 Vegetation index calculation and selection justification

Vegetation indices were calculated from Sentinel-2 multispectral bands to enhance sensitivity to canopy greenness, chlorophyll content, and moisture conditions. These indices reduce the influence of illumination variability and provide more robust indicators of forest canopy properties than single spectral bands. Based on the reviewed literature and previous inventory-orientated studies, four vegetation indices were selected: NDVI, NDRE, NDMI, and NBR.

The Normalised Difference Vegetation Index (NDVI), defined in Equation (2.1), is widely used as an indicator of vegetation greenness and canopy density (Amuyou et al., 2022). NDVI exploits the contrast between strong reflectance in the near-infrared (NIR) region and chlorophyll absorption in the red region. It has been shown to correlate with aboveground biomass and growing-stock volume, particularly in forests with low to moderate canopy density (Puliti et al., 2020; Amuyou et al., 2022). Where *NIR* corresponds to Sentinel-2 Band 8 (842 nm) and *red* corresponds to Band 4 (665 nm).

$$NDVI = \frac{NIR - Red}{NIR + Red} \quad (2.1)$$

The Red-Edge Normalised Difference Vegetation Index (NDRE) exploits the NIR and red-edge spectral regions, which are particularly sensitive to variations in chlorophyll concentration and canopy structure. Compared with visible bands, the red-edge region can provide additional sensitivity in dense vegetation conditions, where NDVI may be affected by saturation (Yadegari et al., 2025). Therefore, NDRE was included to capture red-edge information relevant to forest structural attributes such as GSV and canopy height. In Equation (2.2), NIR corresponds to Sentinel-2 Band 8 (842 nm), while red-edge corresponds to Band 5 (705 nm). NDRE was calculated using Equation (2.2) (Sentinel Hub, n.d.):

$$NDRE = \frac{NIR - RedEdge}{NIR + RedEdge} \quad (2.2)$$

The Normalised Difference Moisture Index (NDMI), defined in Equation (2.3), includes shortwave infrared (SWIR) information and is sensitive to vegetation water content and canopy

moisture status. NDMI exploits the strong water absorption in the SWIR relative to high reflectance in the NIR, making it sensitive to changes in leaf and canopy moisture. It provides complementary information on forest condition and structural variability and has been shown to improve the robustness of biomass and growing-stock volume estimates when combined with greenness-based indices (Grabska et al., 2019). In Equation (2.3), NIR corresponds to Sentinel-2 Band 8 (842 nm), while SWIR1 corresponds to Sentinel-2 Band 11 (1610 nm). Following Sentinel Hub (n.d.), NDMI was calculated as follows:

$$NDMI = \frac{NIR - SWIR1}{NIR + SWIR1} \quad (2.3)$$

The Normalised Burn Ratio (NBR), defined in Equation (2.4), was originally developed for burnt area and disturbance detection but has also proved useful for characterising forest structure and canopy condition. NBR exploits the contrast between high reflectance in the NIR region and strong absorption in the SWIR region, which is influenced by vegetation moisture content and structural integrity. Consequently, NBR is sensitive to changes in canopy density, disturbance, and shadowing effects and can provide additional information on stand condition that supports forest attribute estimation beyond disturbance analysis (Novo-Fernández et al., 2024). Equation (2.4) follows the formulation provided by Sentinel Hub (n.d.), NIR corresponds to Sentinel-2 Band 8 (842 nm) and SWIR2 corresponds to Band 12 (2190 nm).

$$NBR = \frac{NIR - SWIR2}{NIR + SWIR2} \quad (2.4)$$

All vegetation indices were aggregated as mean values over the growing season to reduce noise from short-term atmospheric effects and intra-seasonal variability.

2.7.2 Correlation analysis and feature selection

Before model development, a correlation analysis was performed to identify and reduce multicollinearity among predictor variables. The initial predictor set included Sentinel-2 spectral bands (B2-B12), vegetation indices (NDVI, NDRE, NDMI, NBR), and Sentinel-1 SAR backscatter metrics (VV, VH, VV/VH ratio, and VV–VH difference).

Pairwise Pearson correlation coefficients were calculated for all predictor variables. A threshold of $|r| \geq 0.9$ was applied to identify highly correlated feature pairs. When this threshold was exceeded, one variable from each correlated pair was removed to reduce redundancy among predictors.

To ensure transparency and reproducibility in the variable selection process, a predefined priority framework was established before variable elimination. Predictor retention was guided by three criteria: theoretical relevance, ecological representativeness, and spectral uniqueness. Red-edge and shortwave infrared bands are known to capture structural and biochemical canopy properties relevant for forest inventory modelling (Silveira et al., 2023), while SAR backscatter metrics provide additional information on forest structure (Suleymanov et al., 2024). Vegetation indices such as NDVI and NDMI describe canopy density and moisture conditions (Bui et al., 2024; Zheng et al., 2025), and SAR-derived variables contribute sensitivity to vertical structure and volume scattering effects (Becker et al., 2023). When several predictors represented similar wavelength regions or derived metrics, a single representative variable was retained.

2.8. Machine Learning Modelling

This section presents the supervised ML framework applied to model selected forest inventory attributes from satellite-derived predictors. Separate regression and classification models were developed and evaluated using independent training and testing datasets. The modelling strategy supports both quantitative assessment of accuracy and interpretation of prediction errors in an inventory-orientated context.

2.8.1. Selected algorithms

Supervised ML was used to model both continuous and categorical NFI attributes with satellite-derived predictors. Tree-based ensemble methods were chosen because they are widely applied in remote sensing inventory studies and demonstrate stable performance when predictor sets include correlated spectral bands and indices.

Random Forest (RF) was selected due to its robustness to noisy reference data, ability to model non-linear predictor response relations, and suitability for mixed spectral structural feature sets. RF has been successfully applied to forest type and attribute modelling using Sentinel data, including comparative algorithm studies and operational mapping contexts (Wessel et al., 2018; Van Geffen et al., 2025). RF-style modelling is also common in multi-source biomass and structure estimation workflows, where redundancy among predictors is expected (Wittke et al., 2019).

Gradient Boosting (GB) was selected as a complementary ensemble method because it builds models sequentially and may improve predictive performance by reducing residual errors, especially in complex regression problems. Ensemble learning has been used in forest age prediction and in biomass/structure modelling, where the relationship between remote sensing predictors and stand variables is non-linear and scale-dependent (Chen et al., 2024; Singh et al.,

2023). Boosting-type approaches are also frequently included in multi-source modelling comparisons, including GEE based frameworks (Zurqani, 2025).

During preliminary experiments, SVM and neural network (MLP) models were also tested because they are used in tree species and forest mapping research, including deep learning approaches (Bolgen et al., 2022; Marconi et al., 2022). In the present dataset, their performance did not clearly exceed that of RF or GB, while requiring more tuning and offering weaker interpretability for inventory-quality analysis. Since the main objective was not only predictive performance but also interpretation through residual patterns and feature relevance assessment, RF and GB were chosen as the final modelling approaches.

2.8.2. Training and validation strategy

For each prediction task, a separate modelling dataset was constructed. Four task-specific datasets were defined: (1) growing stock volume, (2) stand height, (3) dominant tree species, and (4) age class. Each dataset contained the final set of cleaned Sentinel-derived predictors and one corresponding forest inventory attribute serving as the response variable.

The datasets were split into training (75%) and test (25%) sets. The training subset was used for model fitting, while the testing subset remained unseen during model development and was used exclusively for independent performance evaluation. This approach enables assessment of model generalisation ability and reduces the risk of overfitting.

For regression tasks (GSV and stand height), random sampling was applied during dataset splitting. For classification tasks (dominant tree species and age class), stratified sampling was used to preserve the class proportions in both the training and test subsets. This procedure ensures balanced representation of forest development stages and species categories and prevents biased performance estimates due to class imbalance.

To ensure reproducibility of the modelling process, a fixed random seed (`random_state = 42`) was specified during dataset partitioning and model initialisation. This guarantees consistent data splits and identical model results across repeated runs.

Hyperparameter settings for RF and GB models were selected based on configurations commonly reported in forest remote sensing studies and preliminary experimentation conducted within this research. The objective was to obtain stable and interpretable models suitable for inventory-orientated analysis rather than to perform exhaustive hyperparameter optimisation.

All models were trained exclusively on the training data and evaluated on the independent testing data to ensure objective performance assessment.

2.8.3. Reproducibility considerations

Reproducibility was an important methodological principle in this research. All preprocessing, modelling, and evaluation procedures were implemented using scripted workflows to ensure transparency and consistency.

Satellite-derived predictors were generated in GEE using explicitly defined filtering criteria, temporal windows, and feature derivation formulas. This ensures that identical preprocessing steps can be repeated using the same data and parameters. Polygon-based feature extraction was performed using scripted functions, avoiding manual intervention.

Hyperparameter values were defined prior to final model fitting and were not optimised extensively. The objective was to ensure stable and interpretable modelling behaviour rather than to maximise predictive performance through complex tuning procedures. The main preprocessing settings and model hyperparameters used in the final modelling workflow are summarised in Table 2.4.

table 2.4

Hyperparameter settings used for machine learning models

Model	Main hyperparameters used
Random Forest Regressor (RFR)	n_estimators = 500, min_samples_leaf = 5, random_state = 42, n_jobs = -1
Random Forest Classifier (RFC)	n_estimators = 500, min_samples_leaf = 5, random_state = 42, n_jobs = -1
Gradient Boosting Regressor (GBR)	n_estimators = 300, learning_rate = 0.05, max_depth = 3, random_state = 42
Gradient Boosting Classifier (GBC)	n_estimators = 300, learning_rate = 0.05, max_depth = 3, random_state = 42

In the modelling stage, all data processing and analysis were conducted in the Python (Google Colab) environment using open-source libraries, including pandas, numpy, scikit-learn, and matplotlib. Predictor selection followed a predefined correlation-based screening procedure, and the final predictor list was explicitly defined prior to model training.

To ensure consistent dataset partitioning and model results, a fixed random seed (random_state = 42) was specified during train-test splitting and model initialisation. This guarantees that the same training and testing subsets, as well as identical model configurations, are produced across repeated executions.

Hyperparameter values were defined prior to final model fitting and were not optimised extensively. The objective was to ensure stable and interpretable modelling behaviour rather than to maximise predictive performance through complex tuning procedures.

Together, these measures support methodological transparency and allow the modelling workflow to be replicated or adapted in other forest inventory contexts.

2.9. Accuracy Assessment and Evaluation Metrics

Model performance was evaluated separately for regression and classification tasks. Table 2.5 summarises the evaluation metrics applied in this study.

table 2.5

Evaluation metrics used for regression and classification tasks

Task Type	Metric	Description	Interpretation
Regression	R ²	Coefficient of determination	Proportion of variance explained by the model
	RMSE	Root Mean Squared Error	Penalises larger prediction errors
	MAE	Mean Absolute Error	Average magnitude of prediction error
	Bias	Mean prediction error	Indicates systematic over- or underestimation
Classification	Overall Accuracy	Proportion of correctly classified instances	General model performance
	Cohen's Kappa	Chance-corrected agreement	Reliability beyond random classification
	Precision	True positives / predicted positives	Class-level reliability
	Recall	True positives / actual positives	Sensitivity per class
	F1-score	Harmonic mean of precision and recall	Balanced class-level performance
	Confusion Matrix	Cross-tabulation of predicted vs observed classes	Identifies misclassification patterns

In addition to global performance metrics, residual analysis was conducted for regression tasks to examine systematic error patterns across forest development stages and stand characteristics.

2.10. Ethical and Legal Considerations

This master's thesis is based on NFI data, satellite-derived information, and corporate forest management data. These datasets do not include personal or sensitive individual information, but they may contain commercially sensitive information related to forest resources and asset characteristics.

To ensure responsible data use, cadastral identifiers were anonymised and replaced with non-identifiable stand-level IDs. Spatial information remains visible. The research focuses exclusively

on forest structural attributes and does not involve ownership, financial transactions, or identifiable stakeholders.

The use of satellite data (Sentinel-1 and Sentinel-2) complies with open-access data policies under the Copernicus programme. All data were used only for research purposes and in accordance with their intended use.

Furthermore, the results are presented in an aggregated and non-sensitive manner to avoid potential misuse or identification of specific locations or assets. The study does not aim to evaluate individual properties or business entities but rather to analyse general patterns in forest inventory data quality.

The research follows principles of responsible and ethical data handling, ensuring appropriate use of both public and corporate datasets.

3. RESULTS

This chapter presents the results of the data analysis and ML modelling conducted in this master's thesis. The section begins with an overview of the dataset characteristics, including descriptive statistics of the forest inventory attributes used as target variables, and evaluates the performance of the developed ML models for both regression and classification tasks. The model interpretation and residual analyses are presented to identify patterns in prediction errors and assess the reliability of the modelling results.

3.1. Dataset Characteristics

The key characteristics of the dataset used in the modelling workflow are summarised through descriptive statistics of the forest inventory target variables, including GSV, stand height, dominant tree species, and stand age class. These statistics characterise variability within the dataset and support the interpretation of subsequent modelling results.

3.1.1. Forest inventory attribute statistics

Descriptive statistics are presented for the continuous target variables, growing stock volume (GSV) and stand height, while the distributions of categorical variables (dominant species and age class) are analysed separately using frequency-based summaries. The dataset comprises 11,497 forest stands, representing a wide range of structural conditions and species compositions.

Table 3.1 summarises the distribution of GSV and stand height. GSV shows substantial variability, ranging from 0 to 887 m³/ha, with a mean of 136.29 m³/ha and a standard deviation of 117.91 m³/ha, indicating a highly heterogeneous distribution across the dataset. In contrast, stand height shows a narrower range (0-34 m), with a mean of 12.96 m and a standard deviation of 9.18 m, reflecting lower variability than GSV.

table 3.1

Descriptive statistics of GSV and Height variables

	count	mean	std	min	25%	50%	75%	max
GSV	11497	136,29	117,91	0	23	127	218	887
Height	11497	12,96	9,18	0	4	14	21	34

The distribution of GSV is presented in Figure 3.1, showing a strongly right-skewed distribution, with a high concentration of stands at lower GSV values and a gradual decrease in frequency as volume increases. The mean GSV is approximately 136.3 m³/ha, while the median is 127.0 m³/ha, indicating a slight positive skew caused by a limited number of stands with very high

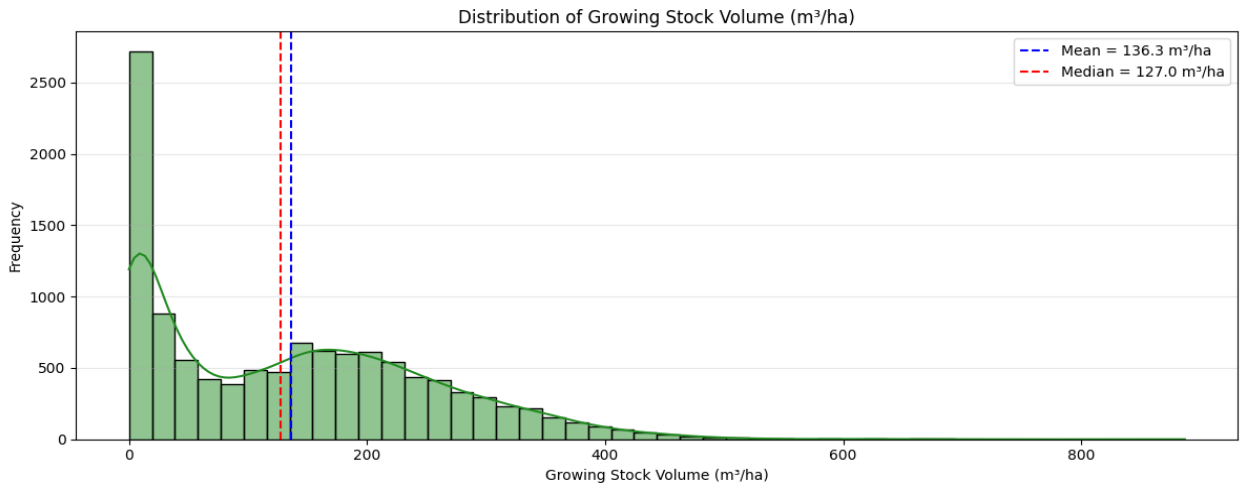


Fig. 3.1 Distribution of Growing Stock Volume m³/ha

volumes. Most forest stands fall within the range of approximately 50 - 250 m³/ha, representing typical productive stands, whereas only a small number of stands exceed 400 m³/ha. The distinct peak at very low GSV values, including zero, indicates the presence of young, recently regenerated, or sparsely stocked stands, where tree biomass is minimal or has not yet built up to measurable levels. Overall, the distribution reflects the heterogeneous structure of the forest landscape and the different growth stages of forest stands.

The distribution of stand height among the analyzed forest stands can be observed on the basis of Figure 3.2. The distribution of stand height indicated average variability with an average height of 12.96 m and median height of 14 m. A slightly lower average than median shows the

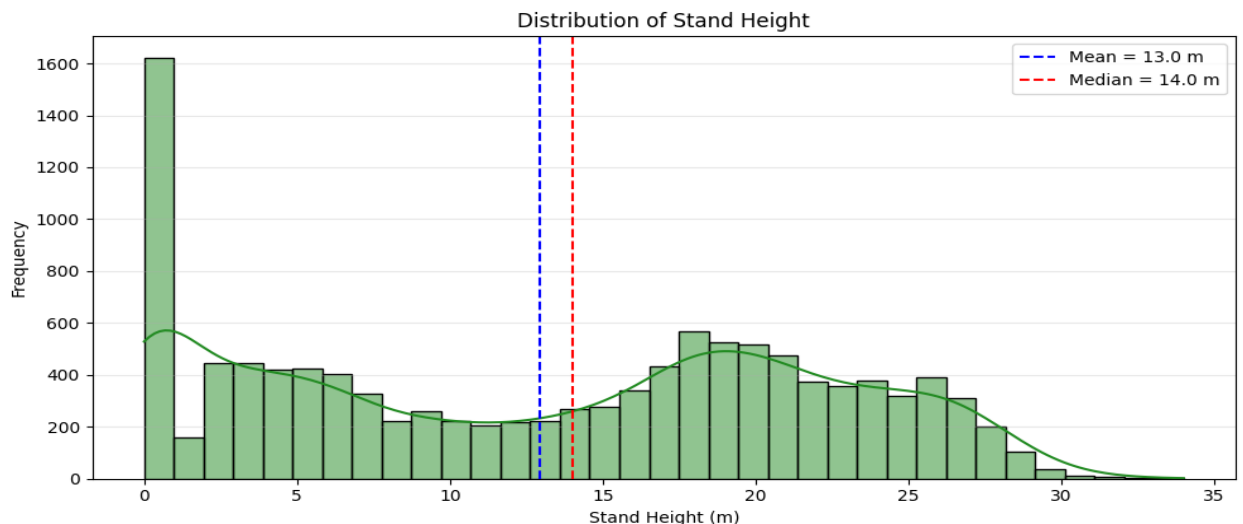


Fig. 3.2 Distribution of stand height

presence of slight negative skew due to the presence of stands with low heights. The majority of stands belonged to middle height category, while extremely low or high heights were not so common. Stands that attained a height up to 25-30 m were associated with older and more developed forest.

To assess potential outliers and overall variability in the structural variables, boxplots were used. Figure 3.3 presents boxplots of growing stock volume and stand height, illustrating variability and the presence of extreme values in the dataset. The boxplot for GSV confirms the strongly right-skewed distribution observed in Figure 3.1, with several high-value observations exceeding 500 m³/ha and a few extreme cases approaching 887m³/ha. These values represent stands with exceptionally high biomass accumulation and are consistent with mature or highly

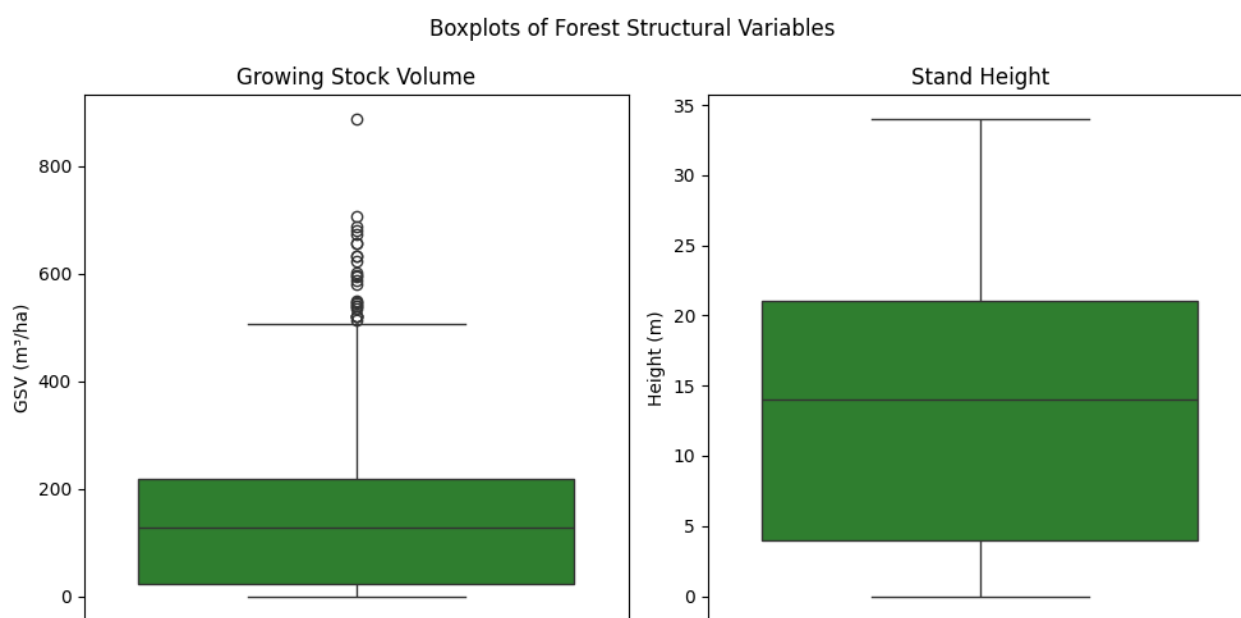


Fig. 3.3 Outlier detection of forest structural variables

productive forest conditions. Similarly, the stand height boxplot shows a broad distribution ranging from near-zero values to approximately 34 m, reflecting the presence of both very young and mature forest stands.

Several observations appeared to be statistical outliers, but they were not excluded from the dataset because they likely reflect genuine ecological variability rather than measurement error. The NFI datasets typically contain stands at various growth stages, from recently regenerated areas to mature forests and highly productive stands. Removing these observations might distort the natural variation of forest structural features and could bias the ML models.

The distribution of forest stands across age classes is presented in Figure 3.4. The majority of stands (42%) fell within the 0-20 year age class (4801 stands), indicating a large proportion of

young or recently regenerated forests. The 41-80 year class (3315 stands) represented the second-largest group (29%), followed by the 21–40-year class (2698 stands). The age group of 81 years and older had the fewest stands (683). Overall, the distribution showed that younger forest stands were more common, possibly due to recent forest management, harvesting cycles, or natural regeneration in the area.



Fig. 3.4 Distribution of forest age class

Figure 3.5 shows the distribution of dominant tree species across various forest stands. Each stand is classified according to its dominant tree species. Birch was the most common species, occurring in 4044 stands, indicating its strong presence within the study area. Spruce was the

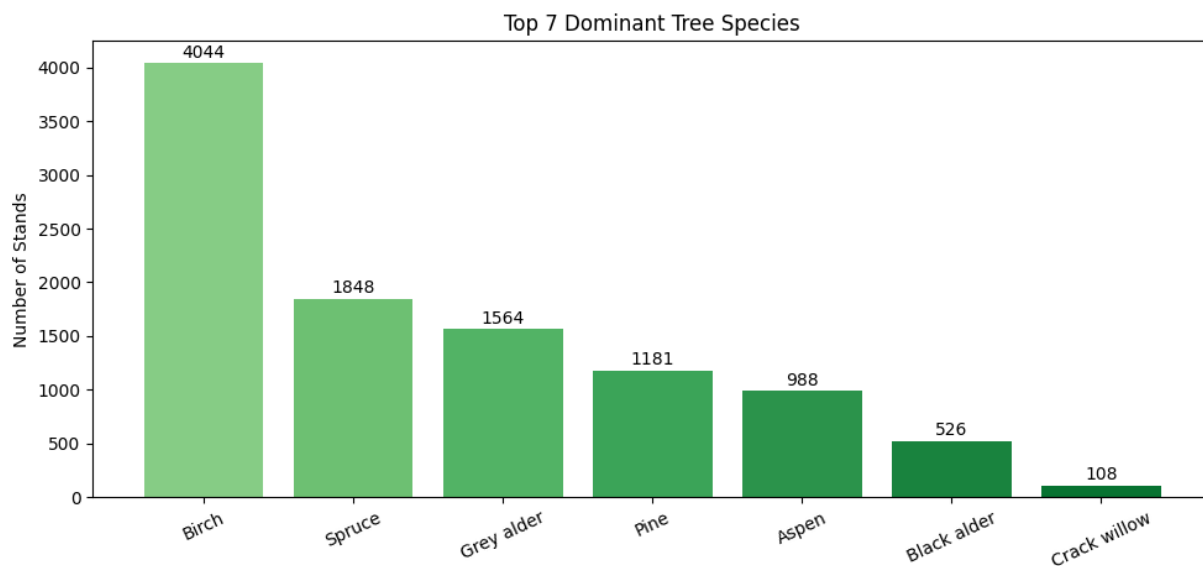


Fig. 3.5 Top 7 dominant tree species across forest stands

second most frequent species (1848 stands), followed by grey alder (1564 stands) and pine (1181 stands). Aspen appeared in 988 stands, while black alder (526) and crack willow (108) were less common among the seven most dominant species. Overall, the results indicate that deciduous

species, particularly birch and grey alder, form a substantial proportion of the dominant species composition in the analysed forest stands.

To sum up, the descriptive statistics indicate substantial variability in forest structural attributes across the analysed stands, reflecting differences in stand development, species composition, and biomass accumulation within the study area. This variability provides a suitable basis for evaluating the performance of ML models in predicting forest attributes.

3.1.2. Correlation analysis of Sentinel-derived predictors and feature selection

The correlation analysis highlighted strong relationships among several predictors derived from Sentinel-2 and Sentinel-1 data (Figure 3.6). Particularly high correlations were observed among spectral bands in the red-edge and near-infrared regions. Bands B6, B7, B8, and B8A showed correlation coefficients of approximately $r \approx 0.99$. Strong correlations were also observed between the SWIR bands B11 and B12 ($r \approx 0.95$); the same applies to the vegetation indices NDMI and NBR ($r \approx 0.97$); and to the Sentinel-1 backscatter polarisations VV and VH ($r \approx 0.92$).

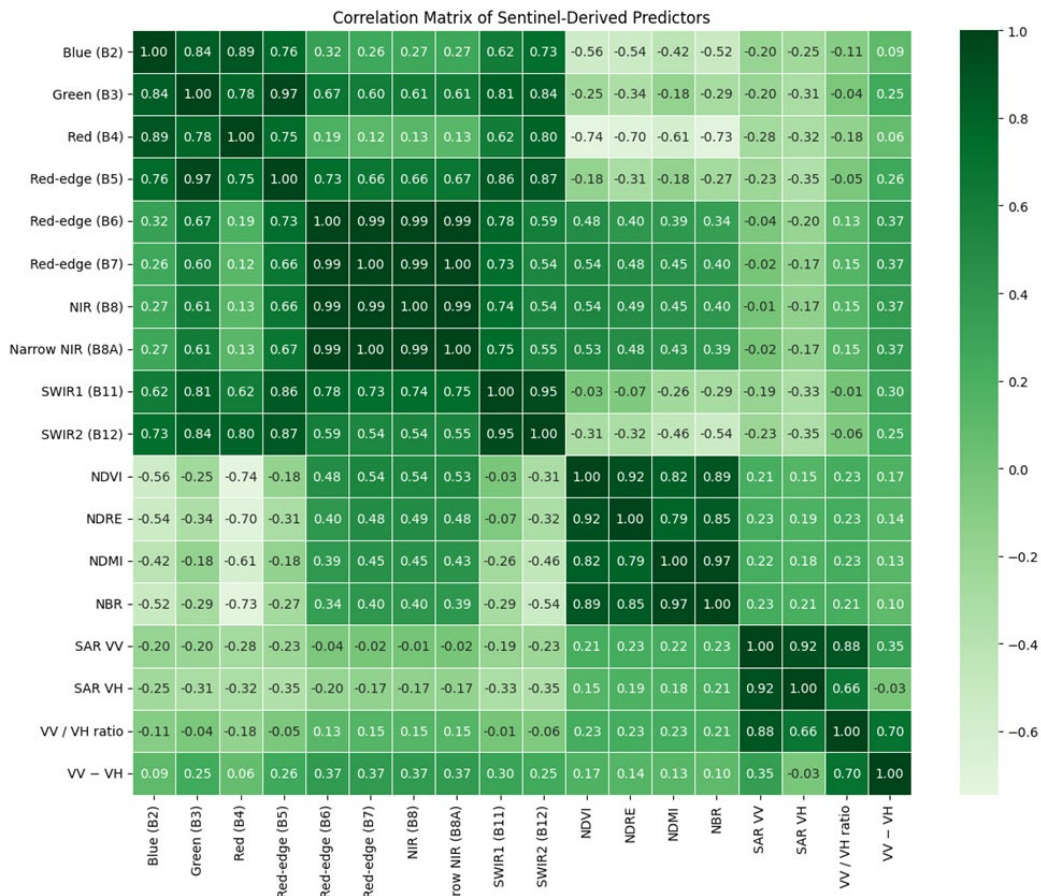


Fig. 3.6 Correlation matrix of Sentinel-derived predictors

In order to avoid redundancy among the predictors, only one predictor out of the highly correlated ones was included for modelling. Therefore, the following predictors: B5, B7, B8, B8A,

B12, NDRE, NBR and S1_VH have been removed from the list. The removal was based on the high correlation between the predictors belonging to the same spectral range or derived index. The priority was given to predictors that had better ecological meaning or broader representation of the spectrum. For example, the red-edge band B6, SWIR band B11 and vegetation indices NDVI and NDMI were preferred. Even though NDRE may be applicable to the dense canopy vegetation, it shows high correlation with NDVI in the current data.

This procedure reduced multicollinearity among predictors while retaining variables that capture key information on canopy spectral properties, vegetation condition and forest structural characteristics.

3.1.3. Predictor variable characteristics

This section summarises the main characteristics of the remote sensing predictors derived from Sentinel-2 optical imagery and Sentinel-1 SAR data. Descriptive statistics of the predictor variables are presented in Table 3.2. The Sentinel-2 spectral bands exhibit reflectance values within expected ranges for forest vegetation, with mean values ranging from approximately 0.03 to 0.28 across the analysed bands. Vegetation indices show relatively stable distributions, with NDVI displaying high values (mean ≈ 0.83) typical of dense vegetation cover, while NDMI values (mean ≈ 0.34) reflect moderate canopy moisture conditions.

table 3.2

Final predictor variable characteristics

Predictor variable	count	mean	std	min	25%	50%	75%	max
Blue (B2)	11497.0	0.034	0.005	0.019	0.030	0.033	0.037	0.104
Green (B3)	11497.0	0.056	0.009	0.032	0.049	0.054	0.061	0.141
Red (B4)	11497.0	0.031	0.008	0.017	0.026	0.029	0.034	0.147
Red-edge (B6)	11497.0	0.275	0.048	0.081	0.242	0.272	0.307	0.489
SWIR1 (B11)	11497.0	0.166	0.028	0.057	0.147	0.164	0.183	0.307
NDVI	11497.0	0.830	0.046	0.403	0.813	0.839	0.859	0.917
NDMI	11497.0	0.345	0.057	-0.149	0.319	0.349	0.378	0.555
SAR VV	11497.0	-9.116	1.048	-15.736	-9.726	-9.023	-8.441	-4.799
VV / VH ratio	11497.0	1.701	0.108	1.364	1.632	1.695	1.760	2.513
VV – VH	11497.0	6.142	0.399	4.376	5.920	6.141	6.367	10.101

Sentinel-1 SAR backscatter variables exhibit negative values for VV polarisation (mean ≈ -9.1 dB), which is characteristic of radar backscatter signals from forest canopies due to volume

scattering effects within vegetation structures. Derived SAR metrics, including the VH/VV ratio and VV–VH difference, show moderate variability across the dataset and provide complementary structural information related to canopy density and vegetation architecture. Overall, the predictor variables demonstrate sufficient variability and realistic ranges of values, supporting their suitability for subsequent ML modelling of forest structural attributes.

3.2. Model Performance

Model performance was evaluated separately for regression and classification tasks using appropriate accuracy metrics, with results presented in the following sections.

3.2.1. Regression results

Regression models were used to estimate GSV and stand height from Sentinel-1- and Sentinel-2-derived predictors. RF and GB algorithms were evaluated using R^2 , RMSE, MAE, and bias metrics (Table 3.3).

table 3.3

Regression model performance results

Variable	Model	R2	RMSE	MAE	Bias
Growing Stock Volume	Random Forest	0.432	89.552	66.732	-0.905
	Gradient Boosting	0.429	89.734	67.882	-1.535
Height	Random Forest	0.453	6.814	5.308	-0.117
	Gradient Boosting	0.442	6.882	5.438	-0.093

For GSV prediction, both models showed moderate performance, with R^2 values of 0.432 for RF and 0.429 for GB. The corresponding RMSE values were 89.55 m³/ha (RF) and 89.73 m³/ha (GB), while MAE values were 66.73 m³/ha and 67.88 m³/ha. Both models exhibited a slight negative bias, indicating a tendency to underestimate higher GSV values.

When considering the GSV distribution described in Section 3.1.1, these error values are relatively high but align with the variable's high variability. GSV varies from 0 to 887 m³/ha, with a mean around 136 m³/ha and a standard deviation of 118 m³/ha. RMSE indicates a massive portion of the mean, highlighting the challenge of modelling highly heterogeneous forest structures.

Stand height was predicted with slightly higher accuracy. The models achieved R^2 values of 0.453 (RF) and 0.442 (GB), with RMSE values of 6.81 m and 6.88 m, respectively. Considering that stand height ranges from 0 to 34 m, with a mean of approximately 13 m, these errors are proportionally smaller, indicating more stable model performance.

Figures 3.7 and 3.8 present the relationship between observed and predicted values for GSV and stand height, respectively. In both cases, the models capture the overall positive trend, with predictions generally following the 1:1 reference line.

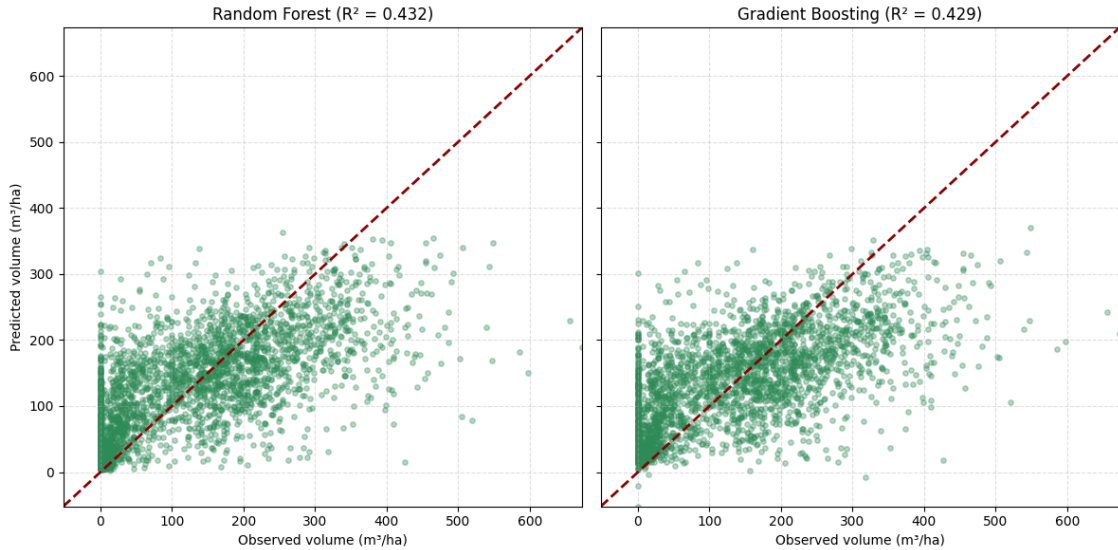


Fig. 3.7 Forest Growing Stock prediction results

For GSV (Figure 3.7), a noticeable dispersion of points around the reference line is observed, particularly at higher values. Both models, RF and GB, tend to underestimate stands with high growing stock volume, as indicated by the concentration of points below the 1:1 line in the upper range. This pattern is consistent with the previously discussed high variability and skewness of GSV.

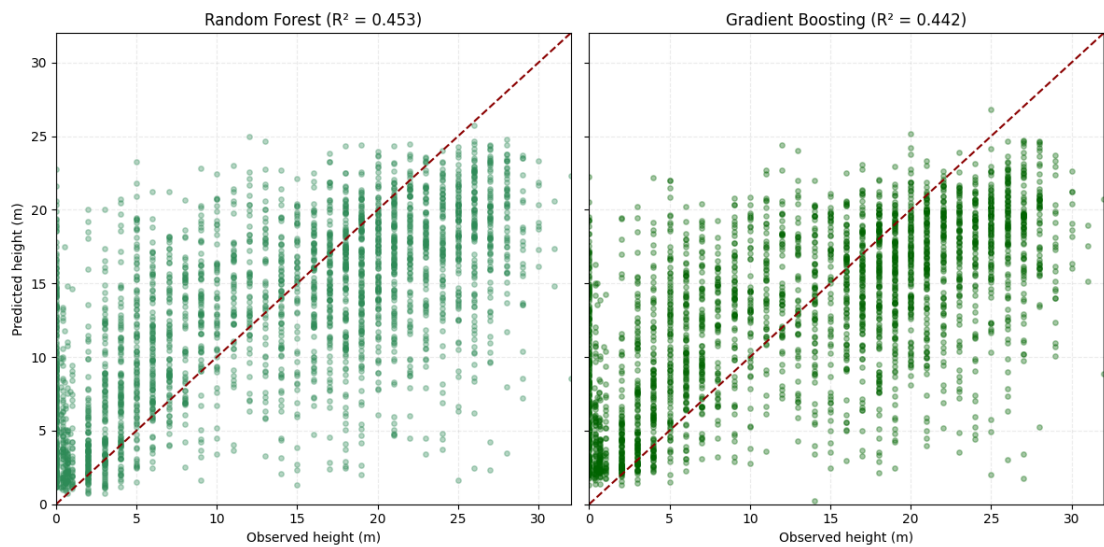


Fig. 3.8 Forest height prediction result

Regarding stand height (Figure 3.8), the observed and predicted values show a stronger relationship, with data points more closely aligned along the 1:1 line. Some variability remains,

particularly for mid-range heights, indicating residual prediction errors. The vertical grouping of points reflects the discrete nature of the reference height measurements. In addition, several cases with observed heights close to zero correspond to non-zero predictions, which may be related to measurement limitations, data inconsistencies, or difficulties in modelling extremely low vegetation heights.

Overall, the visual assessment confirms that both models reproduce general trends but show reduced accuracy for extreme values, especially for GSV.

3.2.2. Classification results

Classification models were developed to predict stand age class and dominant tree species. Model performance was evaluated using overall accuracy, Cohen's kappa, and F1-score, as presented in Table 3.4.

table 3.4

Classification model performance results

Variable	Model	Accuracy	Kappa	F1_score
Age class	Random Forest	0.611	0.416	0.594
	Gradient Boosting	0.601	0.404	0.587
Dominant Species	Random Forest	0.517	0.356	0.471
	Gradient Boosting	0.506	0.348	0.471

For age class prediction, both models achieved moderate performance. RF slightly outperformed GB, with an accuracy of 0.611 compared to 0.601 and a kappa coefficient of 0.416 versus 0.404. The F1-scores were 0.594 and 0.587, respectively, indicating a balanced performance across classes. These results suggest that age class can be reasonably well captured by satellite-derived variables, likely due to its relationship with structural forest characteristics such as canopy height and density.

For dominant tree species classification, model performance was lower. RF achieved an accuracy of 0.517, while GB reached 0.506. Kappa values (0.356 and 0.348) indicate only fair agreement beyond chance, and identical F1-scores (0.471) suggest similar classification behaviour for both models. This lower performance reflects the higher complexity of species discrimination, as spectral and structural signals often overlap between species.

The age-class confusion matrix comparison (Figure 3.9) indicates that both models achieved the highest accuracy for the youngest class (0–20 years) and the mature class (41–80 years). In the RF model, 952 stands from the 0–20 class were correctly classified, while 545 stands were

correctly assigned to the 41–80 class. A similar pattern was observed for GB, with 938 and 536 correct predictions for these classes, respectively.

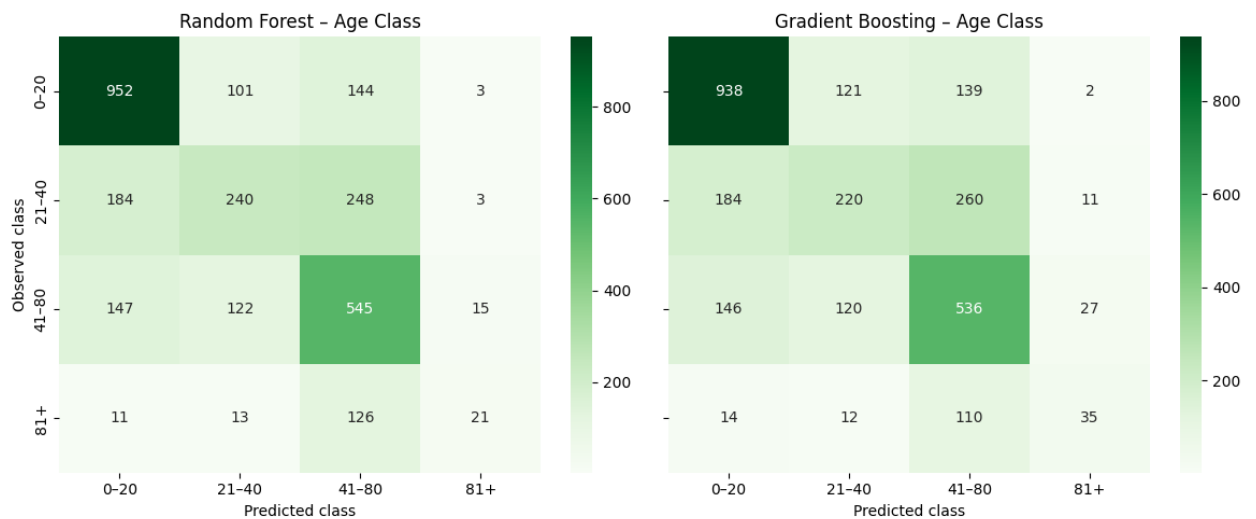


Fig. 3.9 Comparison of confusion matrices for age class classification.

The main source of error occurred between neighbouring age groups. The 21–40 years class was frequently confused with the 41–80 years class and, to a lesser extent, with the 0–20 years class. This suggests that transitional stand development stages are more difficult to separate using satellite-derived predictors. The oldest class (81+) showed the weakest classification performance in both models. Most of these stands were predicted as 41–80 years, indicating that the spectral and structural differences between older and mature stands are not sufficiently distinct in the selected predictor set.

This ecologically plausible pattern shows that stand age isn't directly observed but is inferred from structural and spectral traits such as canopy density, moisture, and radar response. Similar canopy features among nearby age classes reduce separability. The oldest class is under-represented, lowering its classification accuracy.

Overall, the confusion matrix shows that the models capture general age-related patterns, but their performance decreases when class differences are less distinct.

The confusion matrices for dominant tree species classification (Figure 3.10) indicate that performance varies substantially between species. Both models achieved the highest accuracy for the most frequent classes, such as birch and spruce, which show strong diagonal dominance. In the RF model, 798 birch and 262 spruce stands were correctly classified, while GB correctly identified 763 and 255 stands, respectively.

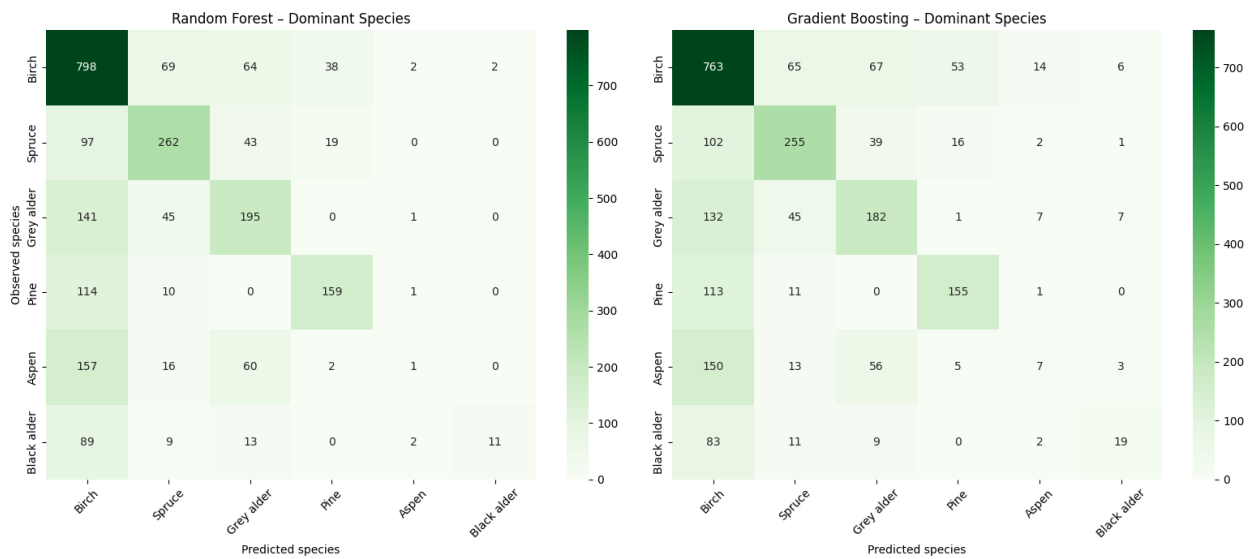


Fig. 3.10 Comparison of confusion matrices for dominant tree species classification.

In contrast, classification performance was considerably lower for less frequent and less distinct species. Aspen showed the weakest performance, with only a small number of correctly classified instances and a large proportion of observations misclassified as birch and grey alder. Similarly, black alder shows low classification accuracy, with only a small number of correctly classified stands (11 in RF and 19 in GB).

Several consistent patterns of confusion are observed across both models. Grey alder and aspen are frequently misclassified as birch, while pine is also occasionally confused with birch and spruce. These patterns suggest that classification is influenced by both class imbalance and spectral overlap between species.

This effect is likely reinforced using Sentinel-2 imagery acquired during the summer leaf-on period, when canopy reflectance is dominated by foliage and spectral differences between deciduous species are reduced. As a result, species such as birch, aspen, and grey alder become difficult to distinguish. This may be explained by the use of summer satellite imagery, where dense foliage causes deciduous species to appear more similar in their spectral response. As a result, species such as birch, aspen, and grey alder are more difficult to distinguish.

Overall, the confusion matrix indicates that while the models can identify dominant species with moderate success, discrimination among less represented and spectrally similar species remains limited.

3.2.3. Model comparison

Across all modelling tasks, RF consistently outperformed GB, although the differences between models were relatively small. This pattern was observed for both regression (GSV and height) and classification (age class and dominant species).

The similar performance of both models suggests that the predictive capacity is primarily constrained by the input data rather than the choice of algorithm. In particular, limitations related to variable variability, class imbalance, and spectral similarity between species appear to play a more significant role than model selection.

Given the marginal performance differences and the robustness of RF, it was selected as the primary model for further analysis and interpretation.

3.3. Feature Importance and Model Interpretation

Feature importance analysis was conducted to identify the most influential predictors contributing to model performance and to support the interpretation of relationships between satellite-derived variables and forest inventory attributes. The analysis focuses on the RF model, which demonstrated slightly better performance and provides robust and interpretable importance measures. The results are presented separately for regression and classification tasks in the following sections.

3.3.1. Regression models

Feature importance for regression models predicting GSV and stand height is presented in Figures 3.11 and 3.12.

For GSV prediction, the most influential predictor is the green band (B3), followed by the red-edge band (B6), blue band (B2), and vegetation indices such as NDMI. The dominance of the green band is notable, as previous studies have often highlighted the importance of red-edge and shortwave infrared (SWIR) bands for estimating forest structural attributes such as biomass and volume (e.g., Puliti et al., 2020; Silveira et al., 2023). These bands are typically associated with canopy structure, chlorophyll content, and moisture conditions (e.g., Zheng et al., 2025)..

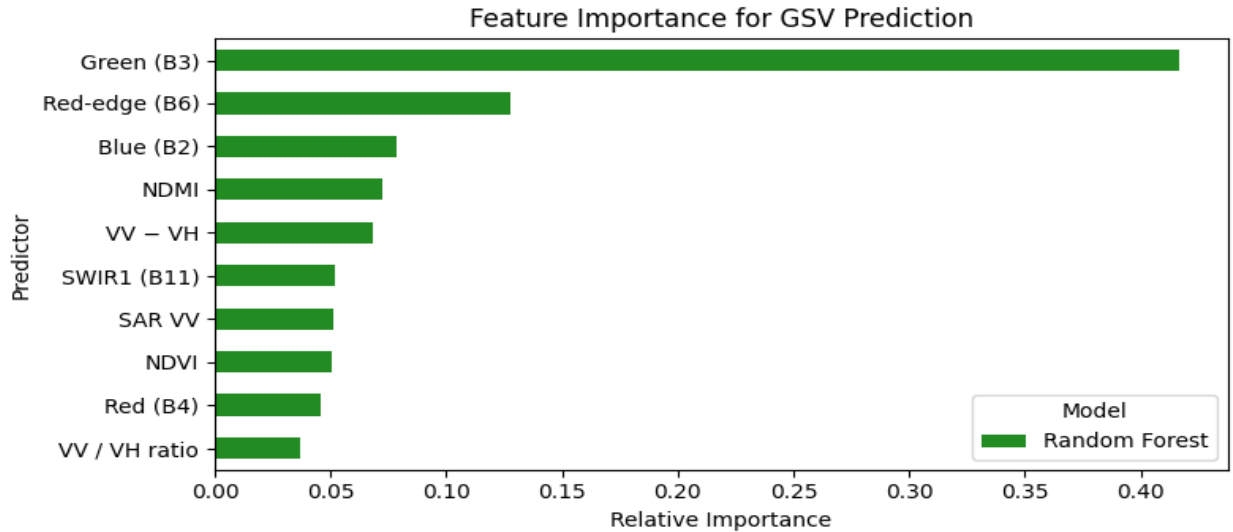


Fig.3.11 Feature importance for GSV prediction

The high importance of the green band observed in this study may be related to several factors. First, the green band captures variations in canopy reflectance that are sensitive to leaf structure and illumination conditions, which may indirectly relate to stand density and productivity. Second, the analysis is based on stand-level aggregated data, where spectral averaging may enhance the contribution of broader reflectance signals. Third, the use of summer (leaf-on) imagery may reduce the relative importance of red-edge and SWIR bands, as canopy reflectance becomes more homogeneous across stands.

The red-edge band (B6) remains one of the most important predictors, confirming its relevance in capturing vegetation's structural and biochemical properties. Similarly, NDMI contributes to the model, indicating that canopy moisture and density help explain variability in GSV. SAR-derived variables, such as the VV–VH difference and VV backscatter, showed moderate importance, suggesting that structural information from radar data complements optical predictors but does not dominate the model.

For stand height prediction (Figure 3.12), a similar pattern is observed, with the green band (B3) again emerging as the most important predictor. However, the relative contributions of other variables are more balanced than those of GSV. NDMI, blue (B2), red-edge (B6), and SAR-derived metrics contribute to the model, reflecting the importance of both spectral and structural information in capturing vertical forest structure.

The strong influence of optical variables, particularly visible and red-edge bands, suggests that stand height is primarily inferred from canopy reflectance patterns rather than direct structural measurements. The relatively lower importance of SAR variables may indicate limitations in capturing vertical structure at the spatial resolution used in this study.

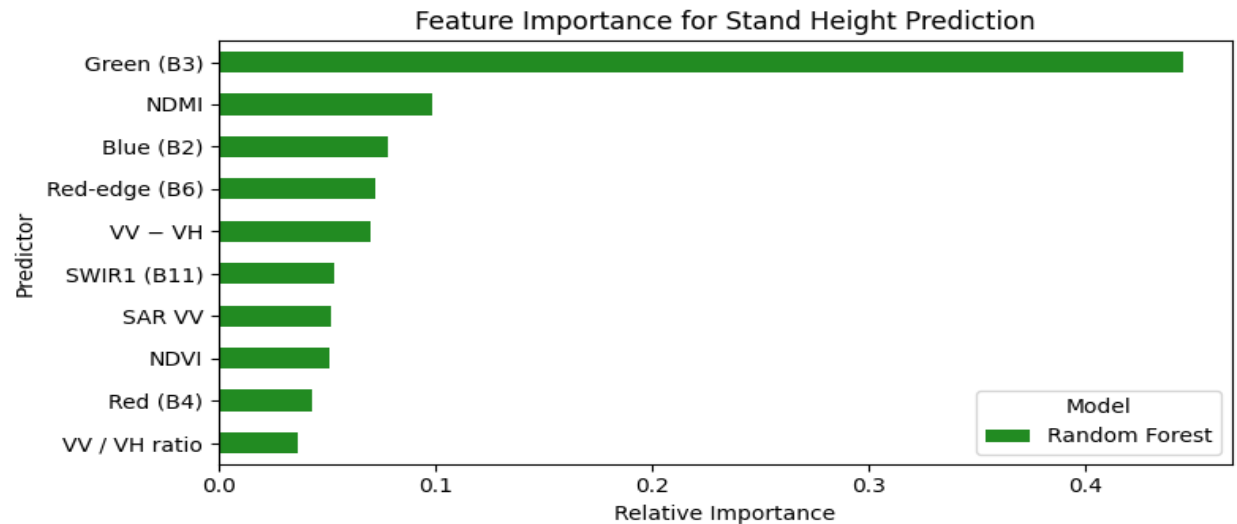


Fig. 3.12 Feature importance for stand height prediction

Overall, the feature importance analysis shows that regression models depend on a mix of spectral and structural predictors, with optical data playing a leading role. The surprising prominence of the green band emphasises the impact of data characteristics and preprocessing choices and indicates that the relationships between spectral signals and forest attributes may vary depending on spatial scale and seasonal conditions.

3.3.2. Classification models

Feature importance for age class and dominant species classification is presented in Figures 3.13–3.14. For age class classification (Figure 3.13), the most influential predictors are the red-edge band (B6), the green band (B3), and SWIR1 (B11), followed by NDMI. The strong importance of the red-edge band reflects its sensitivity to chlorophyll content, canopy condition, and structural variation, all of which change with stand development. The green band also contributes substantially, indicating that differences in canopy reflectance remain relevant for distinguishing between age classes. The high importance of SWIR1 and NDMI suggests that canopy moisture and structural properties are key to distinguishing younger, mature, and older stands.

Visible bands such as red (B4) and blue (B2), as well as NDVI, show moderate importance, indicating that overall vegetation greenness and reflectance patterns also support age-class prediction. SAR-derived variables, including the VV–VH difference, SAR VV, and the VV/VH ratio, contribute less than the main optical predictors but still provide supplementary structural information related to canopy density and vertical complexity.

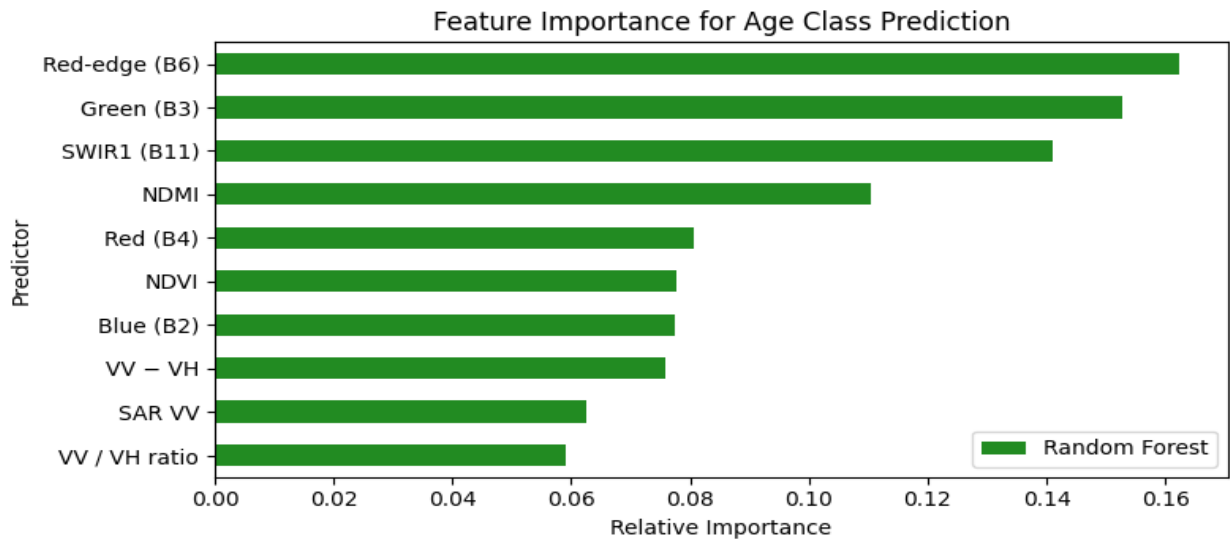


Fig. 3.13 Feature importance for age class prediction

Overall, the feature importance pattern indicates that age class prediction relies primarily on optical variables associated with canopy development, biochemical condition, and moisture status. This is consistent with the ecological meaning of stand age, which is indirectly reflected in spectral and structural canopy properties rather than measured directly by the sensor.

In contrast, variables such as the blue band (B2), the red band (B4), and the VV/VH ratio show lower importance, indicating limited relevance for distinguishing between age classes. These predictors capture more general reflectance properties and are less sensitive to structural differences associated with stand development.

For dominant species classification (Figure 3.14), the most influential predictor is the red-edge band (B6), followed by the SAR-derived VV–VH difference and the green band (B3). The dominance of the red-edge band highlights its sensitivity to chlorophyll content and vegetation structure, which are key factors for distinguishing between tree species. The green band also contributes substantially, reflecting differences in canopy reflectance among species, particularly under leaf-on conditions.

SWIR1 (B11) and vegetation indices such as NDMI and NDVI are relatively important, indicating that moisture-related and general vegetation properties play a significant role in species discrimination. These variables likely capture differences in canopy water content and biochemical composition, which vary across species.

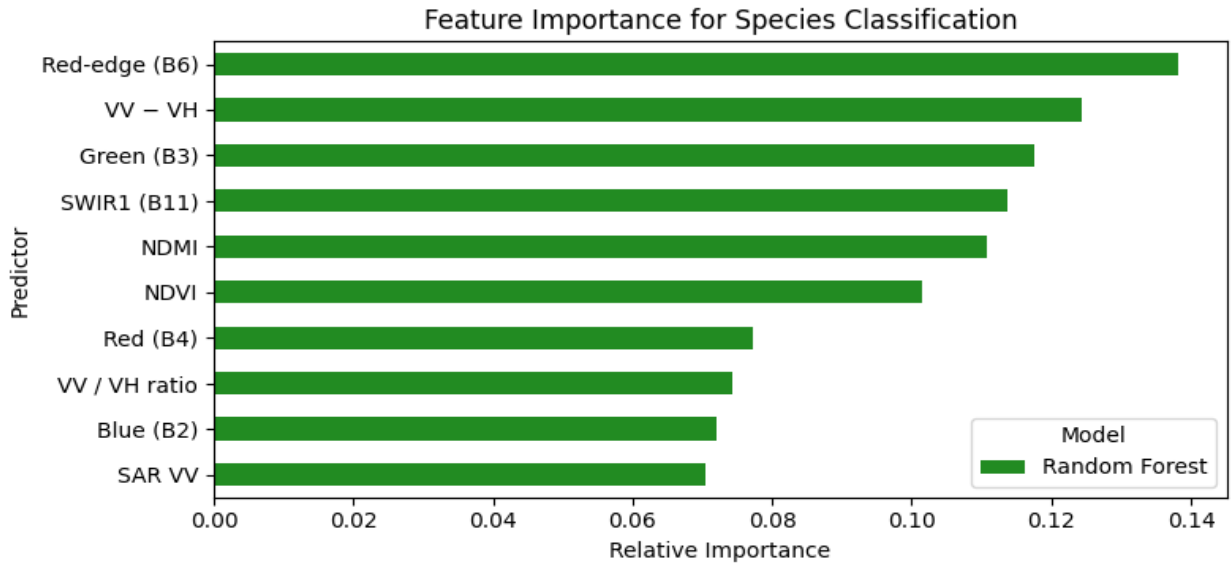


Fig. 3.14 feature importance for dominant species classification

SAR-derived variables, especially the VV–VH difference, rank among the top predictors, suggesting that structural information related to canopy density and volume scattering contributes meaningfully to species classification. Other SAR metrics, such as SAR VV and the VV/VH ratio, are less important, indicating a more limited contribution compared to key optical variables.

Visible bands such as red (B4) and blue (B2) show moderate to low importance, suggesting they provide more general reflectance information and are less effective at separating species classes.

Overall, the feature importance results indicate that dominant species classification relies on a combination of spectral and structural predictors, with a strong contribution from red-edge, green, and moisture-sensitive bands. The importance pattern reflects the complexity of species discrimination using optical satellite data, particularly under leaf-on conditions, where spectral differences between deciduous species are reduced.

3.4. Regression Model Residual Diagnostics and Pattern Analysis

To further evaluate model performance, a detailed residual analysis was conducted. This section focuses on the examination of residuals and prediction discrepancies to assess the reliability of model outputs and to identify systematic patterns in prediction errors. Residual analysis is used to explore how errors vary across stand characteristics and spatial units, enabling the detection of potential inconsistencies and high-deviation observations within the forest inventory dataset.

3.4.1. Regression residual diagnostics

Residual analysis was conducted to evaluate the distribution and behaviour of prediction errors for the regression models estimating GSV and stand height. Residuals, defined as the difference between observed and predicted values, provide insight into model bias, variability, and systematic prediction errors.

The distribution of GSV residuals (Figure 3.15) showed a bell-shaped pattern centred near zero, indicating that the model does not show strong overall bias. The distribution is slightly right-skewed, with a longer tail extending towards positive residual values. This pattern suggests that the model tends to underestimate higher GSV values, as larger positive residuals occur more frequently. Additionally, the widespread occurrence of residuals, ranging from -304 to over 655 m^3/ha , showed significant variability in GSV within the dataset and indicates increased prediction uncertainty, particularly for stands with extreme values.

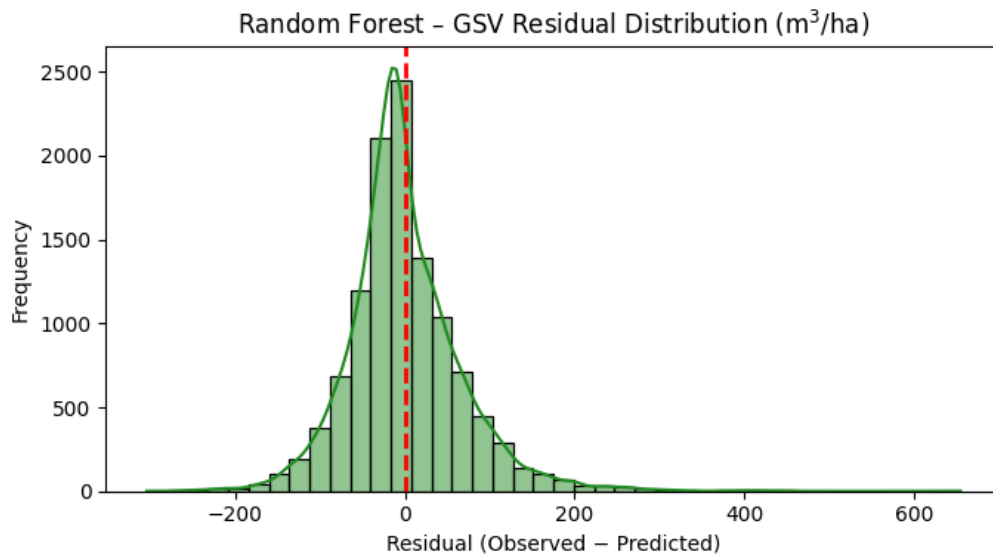


Fig. 3.15 RF- GSV residual (m^3/ha) distribution

In contrast, the residual distribution for stand height (Figure 3.16) is more symmetric and narrowly concentrated around zero. The majority of residuals fall within a relatively limited range (approximately -20 to 20 m), indicating more stable and consistent model performance. The near-symmetric shape of the distribution suggests minimal systematic bias, while the smaller spread compared to GSV reflects lower variability in the target variable and improved predictability.

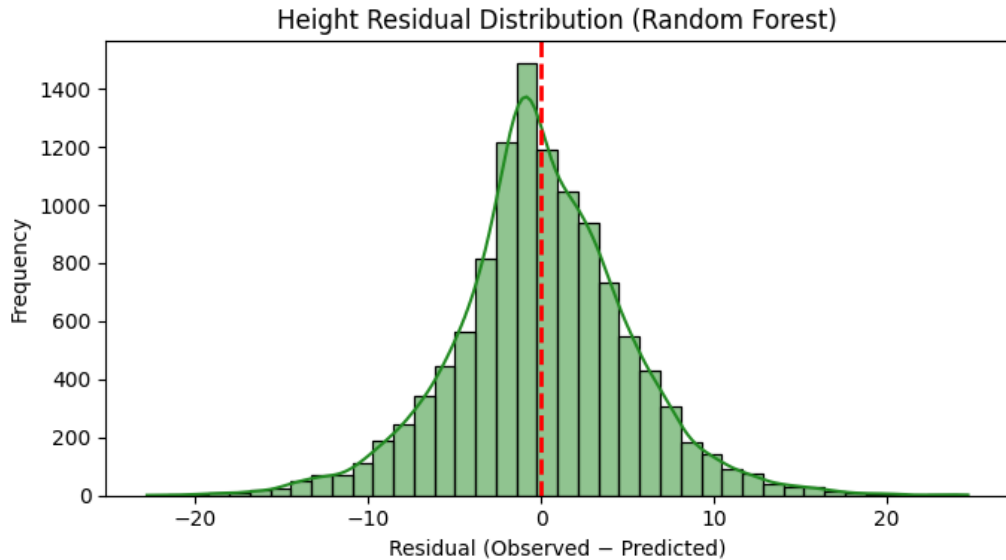


Fig. 3.16 RF – Height residual (m) distribution

The results indicated that growing stock volume is more challenging to predict than stand height. This is primarily due to the higher variability and complexity of GSV, which depends on multiple structural attributes including tree density, diameter, and species composition. In contrast, stand height is more directly related to canopy structure and exhibits lower variability, resulting in more stable and accurate predictions. The wider distribution and skewness of GSV residuals further confirmed greater prediction uncertainty than for height.

3.4.2. Residual patterns across stand characteristics

To further investigate model performance, residual patterns were analysed across selected stand characteristics, including inventory year and stand area (PLAT), for both GSV and stand height. This analysis aimed to assess how prediction errors varied under different conditions and to identify potential sources of uncertainty in the dataset.

The distribution of mean absolute residuals across inventory years revealed clear temporal variability in model performance (Figure 3.17). For GSV, higher error levels were observed in earlier inventory years, particularly between 2000 and 2004, where mean residuals exceeded 100 m³/ha and reached values above 170 m³/ha. In contrast, more recent inventory years showed lower and more stable error levels, generally close to or below the overall mean of approximately 48 m³/ha.

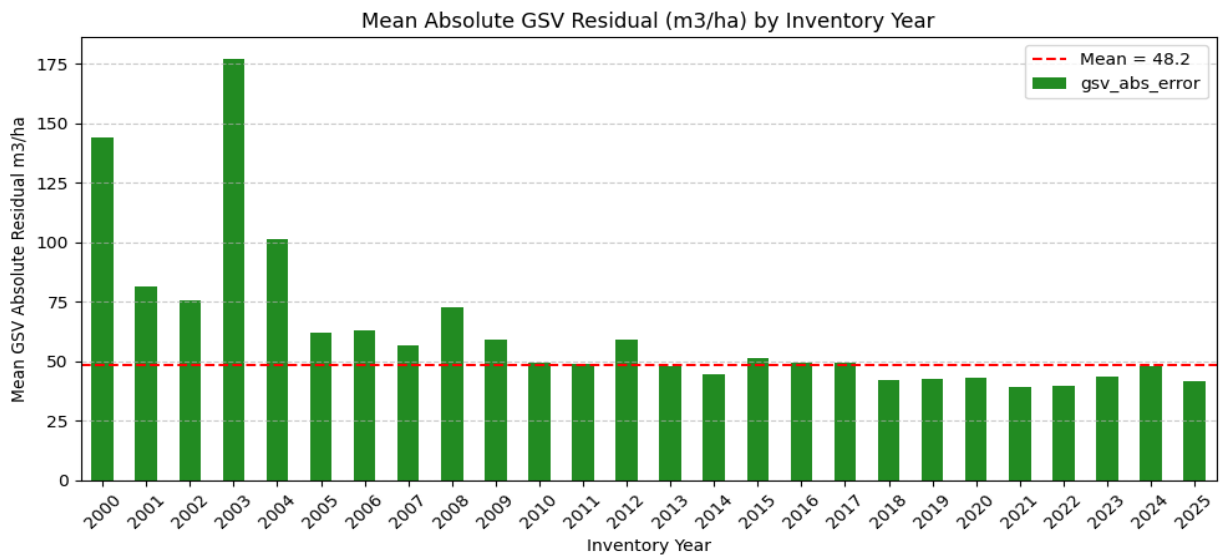


Fig. 3.17 Mean absolute GSV residual (m³/ha) by inventory year

A similar, though less pronounced, pattern was observed for stand height (Figure 3.18). Earlier inventory years exhibited slightly higher mean residuals, exceeding 6 m, while later years showed more stable performance, with residuals typically remaining near the overall mean of approximately 4.2 m. These results suggest that differences in data quality, measurement consistency, or alignment between field observations and satellite data across inventory periods may have influenced model accuracy.

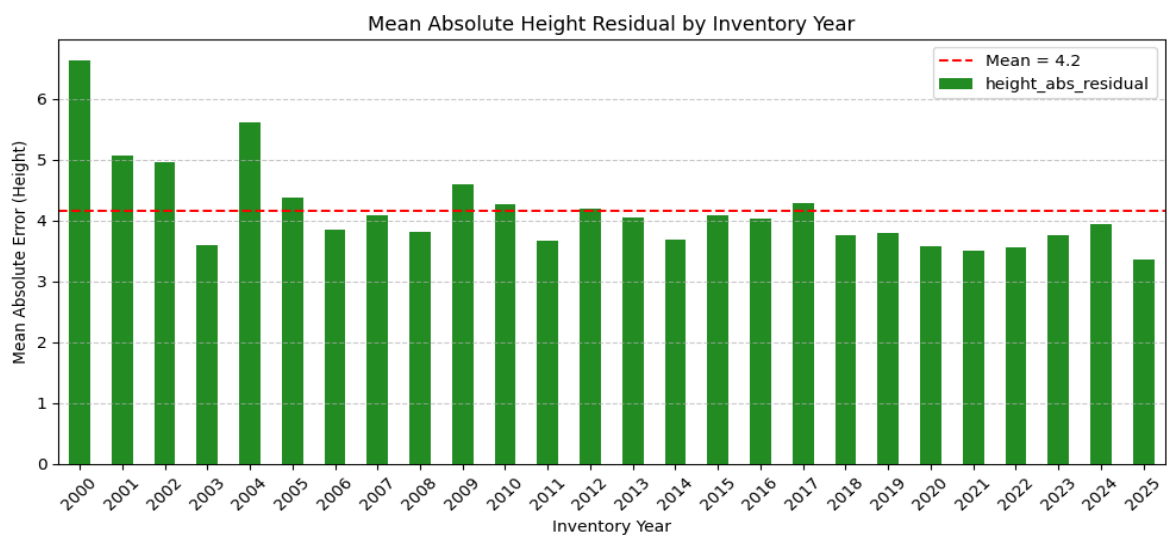


Fig. 3.18 Mean absolute height residual (m) by inventory year

The relationship between residuals and stand polygon area (ha) also highlighted differences in prediction behaviour (Figure 3.19). For GSV, smaller stands reflect a wide range of residual

values, including numerous extreme residuals exceeding 200–600 m³/ha. In contrast, larger stands showed lower and more stable residuals, with most values falling below the overall mean. This pattern indicated that prediction accuracy improved with increasing stand area, likely due to the averaging effect within larger spatial units, which reduced the influence of local heterogeneity and noise.

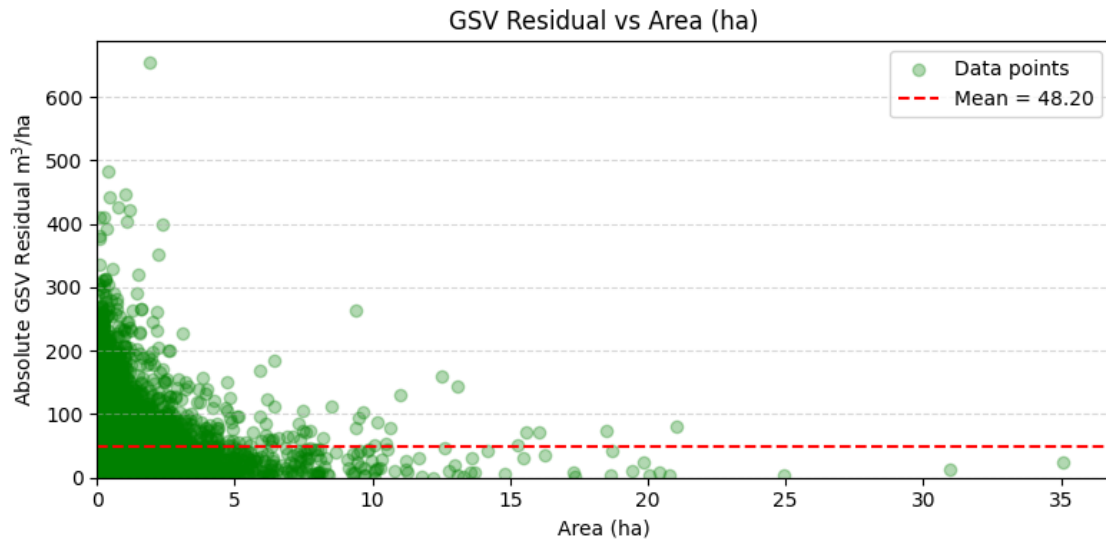


Fig. 3.19 GSV residuals (m³/ha) vs polygon area (ha)

Regarding stand height, a similar but more constrained pattern was observed (Figure 3.20). Smaller stands demonstrated greater residual variability. Despite the lower magnitude of residuals relative to GSV, they remained significant in relation to the overall height range. Larger stands consistently have lower residual values, generally falling below the average error level. This demonstrates that height predictions are less influenced by stand size and spatial heterogeneity than GSV predictions, but prediction errors remain notable in absolute terms.

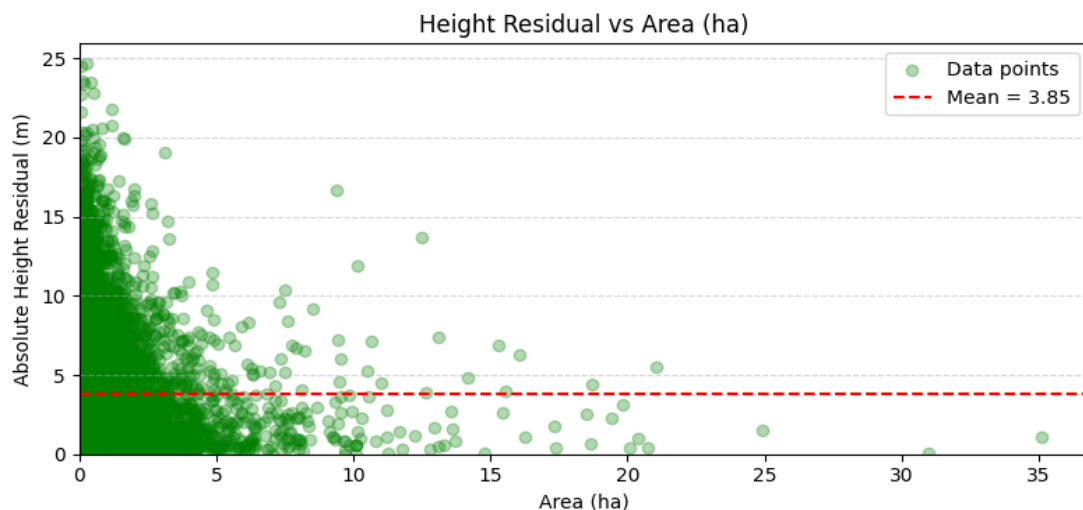


Fig. 3.20 Height residual (m) vs polygon area (ha)

Overall, the analysis demonstrated that prediction residuals were not randomly distributed but were influenced by both temporal and spatial factors. Higher uncertainty was associated with earlier inventory years and smaller stand areas, particularly for GSV, whereas stand height predictions exhibited comparatively lower variability across different conditions. These findings indicated that GSV was more sensitive to data variability and structural heterogeneity, while stand height could be predicted more consistently, although prediction errors remained substantial in absolute terms.

3.4.3. Identification of high-deviation stands

To better understand model uncertainties and identify inconsistent observations, stands with high prediction deviations were selected based on a threshold derived from RF model error metrics, choosing stands where the absolute residual was more than twice the root mean square error ($\text{RMSE} \times 2$) for GSV or stand height.

Using this criterion, 277 stands were identified as high-deviation cases, representing observations with substantially larger discrepancies between observed and predicted values. These stands were further examined to identify potential causes of prediction errors and to assess whether deviations were related to model limitations, spatial characteristics, or inconsistencies in the inventory data.

Three examples are presented in Table 3.5 to illustrate different types of stand behaviour. Polygon 100000000529-1-22-0 (Example 2) was included as a reference case because the observed and predicted GSV values were relatively similar, with an observed GSV of 425 m³/ha and a predicted GSV of 387 m³/ha. This example provides a visual reference for how a high-volume stand may appear in satellite imagery and allows comparison with stands showing much larger deviations.

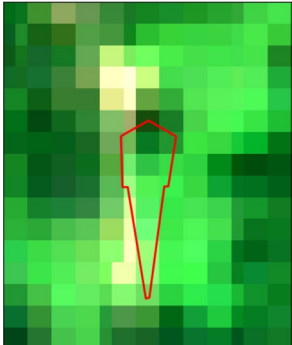
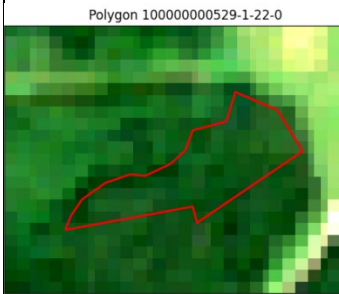
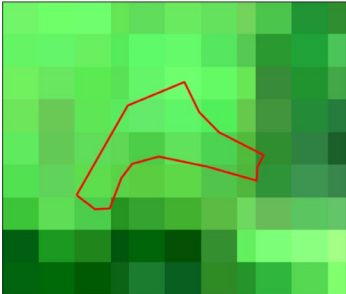
Polygon 100000000867-6-2-0 (Example 1) represents a clear high-deviation case. The observed GSV was 426 m³/ha, while the model predicted only 15 m³/ha. The observed stand height was only 2 m, and the model also predicted a very low height. In addition, the dominant species was correctly predicted as spruce. Therefore, the main inconsistency appears to be related to the recorded GSV value, which is unusually high for a stand with such low height. This may indicate a data-entry error or another inconsistency in the inventory record.

A similar pattern was observed for polygon 100000000711-1-12-0 (Example 3). In this case, the reported GSV was 410 m³/ha, while the model predicted only 73 m³/ha. Although the reported GSV was comparable to that of the reference high-volume stand (Example 2), the spectral characteristics of this stand appear noticeably different. The satellite image suggests a less dense

canopy structure, which does not clearly support the reported high GSV value. This discrepancy may be attributed to limitations in the satellite signal. It may also indicate an inconsistency in the inventory data, where the reported increase in stock volume is not fully supported by the observed spectral response.

table 3.5

Visual comparison of selected high-deviation stands and a reference stand

Example	1. (100000000867-6-2-0)	2. (100000000529-1-22-0)	3. (100000000711-1-12-0)
Satellite Image	 Polygon 100000000867-6-2-0	 Polygon 100000000529-1-22-0	 Polygon 100000000711-1-12-0
Observed GSV:	426 (m ³ /ha)	425 (m ³ /ha)	410 (m ³ /ha)
Predicted GSV:	15 (m ³ /ha)	387 (m ³ /ha)	73 (m ³ /ha)
Observed Height:	2 (m)	27 (m)	26 (m)
Predicted Height:	2 (m)	25 (m)	8 (m)
Observed Age group	1 (0-20 years)	3 (41-80 years)	4 (80+ years)
Predicted Age group	1 (0-20 years)	3 (41-80 years)	1 (0-20 years)
Observed Dominant species	Spruce	Pine	Birch
Predicted Dominant species	Spruce	Pine	Birch

Comparison of these examples showed that stands with similar reported GSV values of approximately 400 m³/ha may represent substantially different spectral patterns. When the spectral response does not correspond to conditions typically associated with high biomass, large prediction errors may occur. Such errors may arise from limitations in satellite data, including signal saturation, but may also indicate inconsistencies in the inventory records.

An additional example was included in Table 3.6, representing the stand with the highest GSV residual identified in the dataset. For polygon 100000000226-1-1-0, the observed GSV was 887 m³/ha, whereas the predicted GSV was only 232 m³/ha. At the same time, the observed and predicted stand height, age group, and dominant species were relatively consistent. This suggested that the extreme deviation was primarily associated with GSV estimation rather than with a broader

mismatch across all stand attributes. The example further demonstrated that substantial errors in GSV could occur even when several other stand characteristics were predicted correctly.

table 3.6

Example of the stand with the highest growing stock volume residual

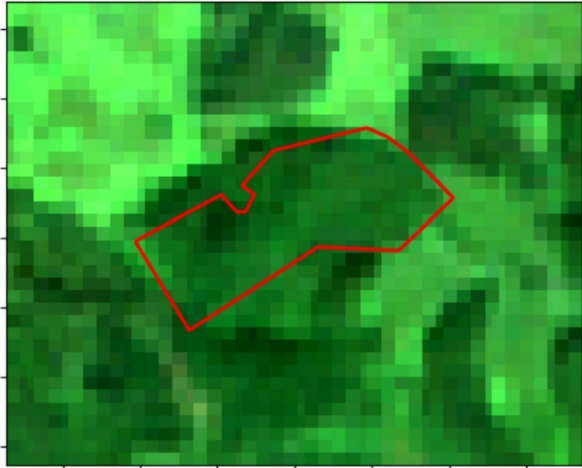
Example	4. (100000000226-1-1-0)
Satellite Image	+5.623e1 Polygon 100000000226-1-1-0 
Observed GSV:	887 (m3/ha)
Predicted GSV:	232 (m3/ha)
Observed Height:	11 (m)
Predicted Height:	10 (m)
Observed Age group	2 (21-40 years)
Predicted Age group	2 (21-40 years)
Observed Dominant species	Grey alder
Predicted Dominant species	Grey alder

Table 3.7 presents a comparison of the original observed values, model predictions, and updated NFI data for property 100000000034. In the original inventory record, the polygon 36800080027-1-3 was described as having a growing stock volume (GSV) of 705 m³/ha, a height of 24 m, belonging to age class 3 (41–80 years), and being dominated by spruce. In contrast, the model predicted substantially lower values for GSV and height, namely 313 m³/ha and 20 m, respectively, and identified birch as the dominant species, while the age class remained the same. After the NFI data update (Appendix V), the recorded GSV decreased to 415 m³/ha, the stand height was revised to 21 m, and the dominant species was again classified as spruce. These updated values were notably closer to the model predictions than the original observations, particularly for GSV and height. This example suggests that the initial NFI record may have overestimated stand volume and height and demonstrates how model-based predictions can support the identification of stands that require further review and correction in the inventory database.

This case is presented as an illustrative example only. A systematic comparison between high-deviation flags and later NFI updates was beyond the scope of this thesis, but it would provide valuable validation in future research.

table 3.7

Comparison of observed, predicted, and updated NFI attributes for property 100000000034-1-3

	GSV	Height	Age class	Dominant Species
Observed	705	24	3 (41-80years)	Spruce
Predicted	313	20	3 (41-80years)	Birch
New NFI 2026 data	415	21	3 (41-80years)	Spruce

3.5. Classification Error Analysis

To better understand the behaviour of the classification models, an additional error analysis was conducted for both age class and dominant species predictions. The analysis focused on differences in error rates across observed class categories and stand area classes in order to identify systematic patterns in model performance.

The results for age class classification showed that prediction errors were not evenly distributed across classes (Fig. 3.21). The highest error rate was observed for the 80+ years class, where misclassification significantly exceeded the overall average. The 21–40 years age group also showed a higher error rate, while the 0–20 years and 41–80 years groups demonstrated significantly lower error rates. This pattern suggested that the model was more effective in distinguishing younger and middle-aged stands than the oldest stands. The high error rate in the 80+ years age class may be related to three factors: a relatively small number of training observations, higher structural variability among mature stands, and limited spectral separability from other older age classes.

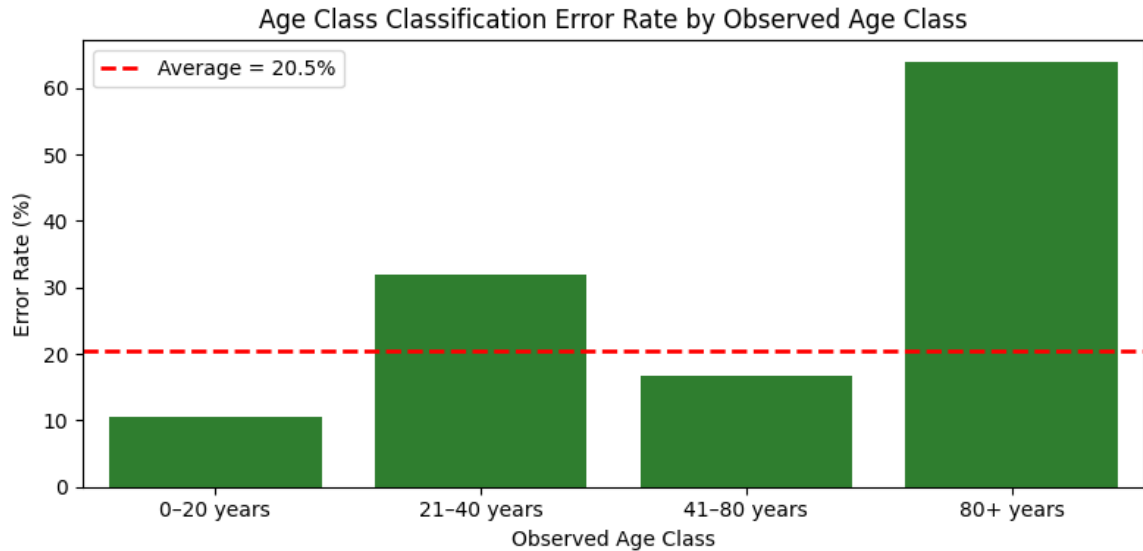


Fig. 3.21 Age class classification error rate by observed age class

A similar class-based pattern was found for the classification of dominant species (Fig. 3.22). The highest error rates were observed for black alder and aspen, both of which were misclassified considerably more often than the overall average. In contrast, birch showed the lowest error rate, while spruce, grey alder, and pine exhibited intermediate performance. These differences indicated that the classification of dominant species was strongly influenced by species-specific separability. The high error rates for black alder and aspen likely reflected both lower class representation and greater spectral similarity to other deciduous species, whereas birch appeared to be more consistently recognised by the model.

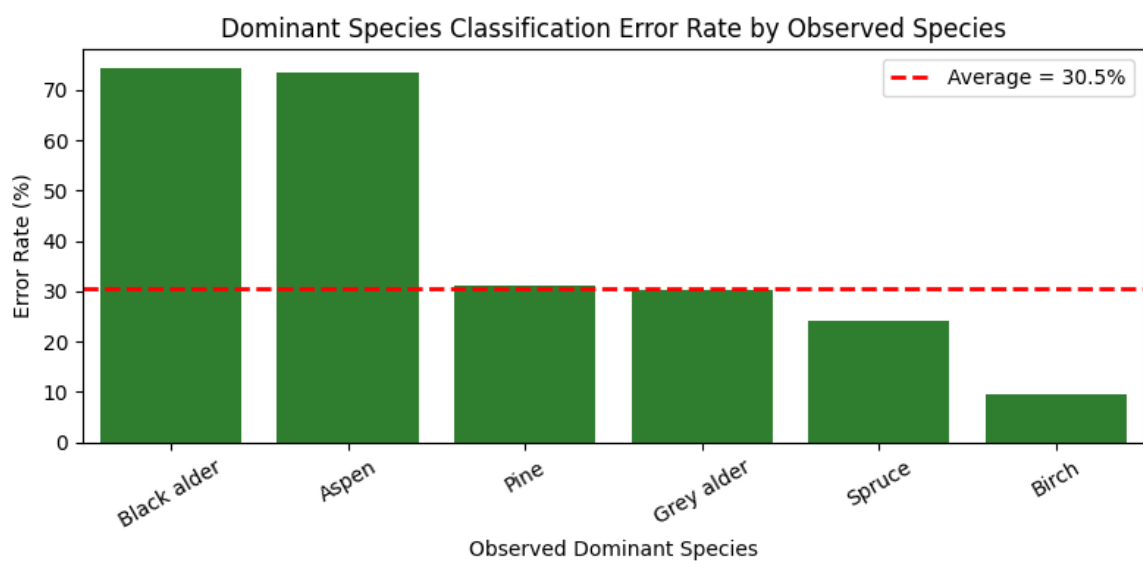


Fig. 3.22 Dominant species classification error rate by observed species

Stand area showed a clear relationship with classification performance for both tasks (Figs. 3.23 and 3.24). For age class classification, the error rate was highest in stands smaller than 0.5 ha (approximately 25%) and lowest in stands larger than 5 ha (around 7%). A similar pattern was observed for dominant species classification, where the error rate was about 36% in the smallest stands and approximately 12% in the largest stands. This consistent pattern indicated that smaller stands were substantially more difficult to classify correctly than larger stands.

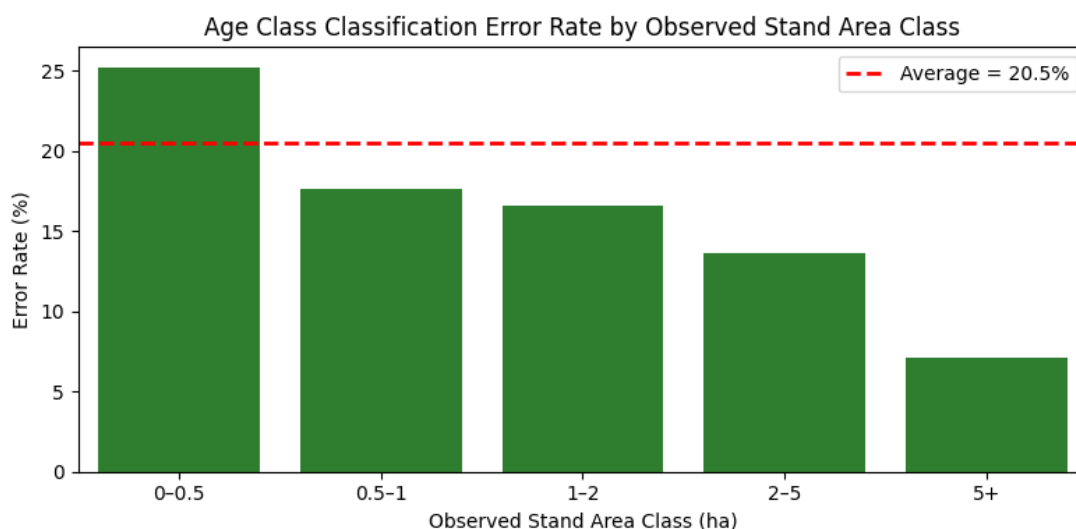


Fig. 3.23 Age class classification error rate by observed stand area class

The observed relationship between stand area and classification error can be explained by the characteristics of medium-resolution satellite data. Small stands are more strongly affected by mixed-pixel effects, boundary uncertainty, and the limited spatial representation of stand conditions within Sentinel imagery. In contrast, larger stands are represented by a greater number of pixels and are generally more spatially homogeneous, which likely improved the stability of extracted predictors and increased classification reliability.

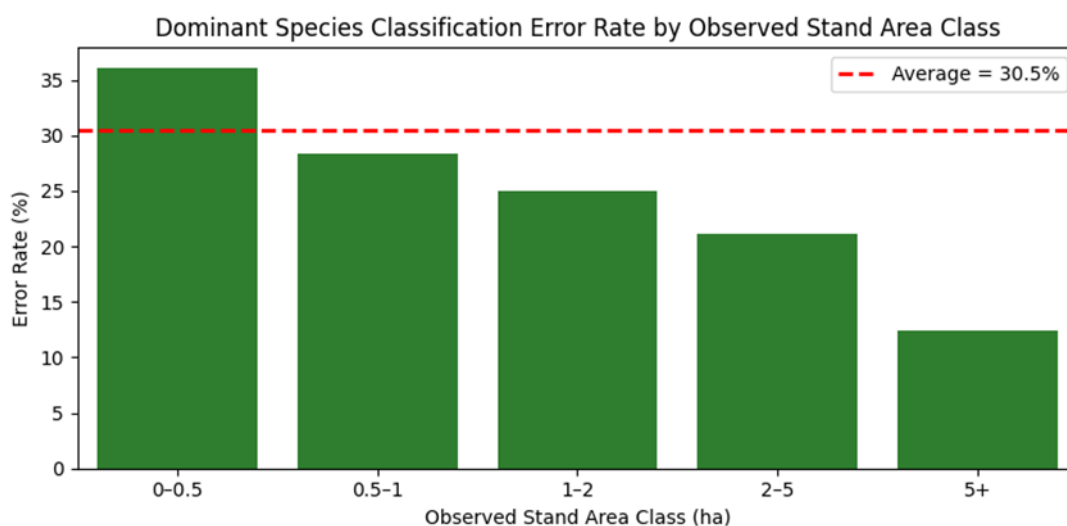


Fig. 3.24 Dominant species classification error rate by observed stand area class

Overall, the classification error analysis demonstrated that prediction performance was influenced not only by class category but also by standard spatial characteristics. Age class errors were particularly high in the oldest stands, while dominant species errors were most pronounced for certain deciduous species. At the same time, the strong decline in error rate with increasing stand area highlighted stand size as one of the most important factors affecting classification reliability. These findings suggested that classification errors were not random but were associated with identifiable data and stand-related properties, which is important for interpreting model limitations and for identifying inventory records that may require additional validation.

4. DISCUSSION

4.1. Interpretation of results

The research findings demonstrated that satellite-derived predictors, combined with ML models, can provide meaningful estimates of forest inventory attributes. Model performance depended on the complexity and variability of the target variables, as well as the characteristics of the input data.

Regression modelling showed moderate predictive performance for both GSV and stand height, with R^2 values of about 0.432 and 0.453. The lower performance for GSV can be explained by its high variability, wide value range (0–887 m³/ha) and the presence of extreme values, which made prediction more difficult. In comparison, stand height had a narrower range (0–34 m) and lower variability, which led to more stable predictions. The results also showed systematic underestimation of high GSV values. This may be linked to signal saturation in optical satellite data, where it becomes difficult to distinguish between high and very high biomass conditions.

The classification results showed moderate performance. Accuracy was higher for age classes (≈ 0.61) than for dominant species (≈ 0.52). Age prediction was more reliable because it is indirectly reflected in canopy structure (height, density, moisture), which changes with stand development. Misclassifications were common between neighbouring age classes, particularly 21–40 and 41–80 years, due to similar structural characteristics.

Dominant species classification was more challenging because of class imbalance and spectral similarity between species. This was partly related to the use of leaf-on, summer satellite data, where canopy reflectance was mainly influenced by foliage. As a result, spectral differences between deciduous species were reduced, making them more difficult to separate.

Residual analysis helped explain how the models behaved under different stand conditions. Prediction errors were not randomly distributed. They were affected by both temporal and spatial factors. Higher residuals were found for older inventory years and smaller stand areas, which suggested that model performance was influenced by data quality, measurement consistency and spatial representation. Stand area had a clear relationship with prediction accuracy, as smaller polygons showed much higher residual variability. This was likely linked to mixed-pixel effects and boundary uncertainty in medium-resolution satellite data.

The analysis of high-deviation stands helped to identify the main sources of prediction errors. Using a threshold of $2 \times \text{RMSE}$, 277 stands were identified as extreme cases. Visual inspection showed that large differences were often linked to small or irregular polygons, mixed canopy structure or possible inconsistencies in the inventory data. For example, stands with very

high reported GSV values and unrealistically low stand height suggested possible errors in the NFI data. At the same time, some high-volume stands were predicted accurately, showing that the model could still capture forest structure when the spectral signal was stable.

4.2. Practical relevance for forest management

The results of this study have important implications for forest management, particularly in operational planning, resource allocation, and economic decision-making.

Forest inventory data are closely linked to estimates of timber volume, forest value and expected financial returns. Therefore, errors or inconsistencies in these data can create financial risk. If GSV is overestimated, forest assets may be valued too highly, harvesting plans may become unrealistic and future revenue may be overestimated. If GSV is underestimated, assets may be valued too low, harvesting opportunities may be missed and investment decisions may be less effective.

The integration of satellite data and ML can provide a more cost efficient and scalable alternative to traditional inventory approaches. Field-based inventories are expensive and time-consuming, especially over large areas. The proposed approach allows forest attributes to be assessed more quickly and across wider areas. This can reduce the need for frequent large field campaigns, lower operational costs and support more timely updates of inventory data for decision-making.

A key economic contribution of this research is the ability to identify stands where inventory data may be unreliable. The results showed that large differences between observed and predicted values may point to outdated or inconsistent inventory records. This means that field checks can be focused on selected stands, rather than carrying out a full re-inventory across the whole area. Such a targeted approach can reduce validation costs and improve efficiency, as resources are directed only to higher-risk areas.

Improved inventory reliability can also support more accurate forest valuation and risk management. In investment-oriented forestry, even moderate errors in GSV estimation can lead to large financial miscalculations at portfolio level. By identifying and reducing these uncertainties, the proposed approach can support more reliable asset valuation, better capital allocation and improved long-term return forecasting.

Satellite data can also support more regular monitoring of forest conditions and improve decision-making in changing environments. Forest growth, harvesting and disturbances can be

monitored more frequently, reducing the risk of decisions being based on outdated information. This supports more responsive forest management and helps to reduce uncertainty in planning.

In addition to direct financial benefits, better data quality can support carbon accounting and sustainability reporting. Accurate estimates of forest biomass and structure are important for carbon credit calculations and environmental reporting, which are becoming increasingly important from an economic point of view.

Integrating remote sensing and ML can improve the efficiency, reliability and scalability of forest inventory management. The proposed approach supports more informed, risk-aware and cost-effective decision-making, especially in forest investment and large-scale forest management.

4.3. Limitations

Despite the promising results, several limitations should be considered.

First, the use of medium-resolution satellite data (Sentinel-1 and Sentinel-2) limits spatial detail. Small and heterogeneous stands are strongly affected by mixed-pixel effects, leading to higher prediction errors, particularly in smaller polygons.

Second, optical data are subject to signal saturation under high-biomass conditions, reducing the ability to distinguish between moderate and very high GSV and leading to systematic underestimation of dense stands.

Third, the models relied on NFI data, which may include inconsistencies related to measurement errors, time differences and subjective interpretation. As shown in the high-deviation analysis, some differences were likely caused by the inventory data rather than by the models themselves.

Fourth, class imbalance affected classification performance, as less represented species were more often misclassified.

Finally, the models were validated using one 75/25 train-test split with a fixed random seed. The dataset was large enough for this approach, but repeated cross-validation was not applied. Therefore, the reported results show model performance for one data split only. Future work should test the models with repeated cross-validation or different random seeds to check how stable the results are.

4.4. Opportunities for future work

First, the integration of higher-resolution data, such as LiDAR or Unmanned Aerial Vehicle (UAV) based imagery, could improve the estimation of forest structure and reduce mixed-pixel effects, particularly in small and heterogeneous stands.

Second, the use of multi-temporal satellite data could enhance species classification by capturing seasonal (phenological) differences, thereby reducing spectral ambiguity.

Third, more advanced modelling approaches, including deep learning methods, could be tested to capture complex relationships between predictors and forest attributes more effectively. Methods for dealing with class imbalance could also be used to improve performance for less represented species.

From an operational point of view, future work should focus on how the proposed approach could be included in forest management workflows. This could include automatic data quality checks and the use of model results in decision-support systems.

Finally, future research should focus on improving the estimation of uncertainty in both satellite data and forest inventory measurements. This would improve the assessment of prediction reliability and facilitate more informed forest management and economic decision-making.

CONCLUSIONS

The goal of this thesis was achieved by developing an approach for improving forest inventory data quality using supervised ML and Sentinel-1/2 data, enabling not only forest attribute estimation but also the identification of potential inconsistencies in inventory records.

The first research question focused on the factors influencing the quality and internal consistency of NFI data. The results, supported by the literature review, showed that NFI data quality was affected by several interacting factors, including sampling design, plot geometry, field measurement uncertainty, temporal organisation of inventory cycles, protocol updates, and limitations in documentation and metadata. These factors influenced key data quality dimensions such as accuracy, consistency, completeness, and reliability. The empirical results showed that higher residuals were associated with older inventory years and smaller stand areas, while several high-deviation cases revealed potentially inconsistent combinations of forest attributes, such as very high reported growing stock volume together with low stand height. These findings indicate that internal inconsistencies may arise not only from measurement errors but also from temporal mismatches, spatial representation issues, and structural heterogeneity within stands.

The second research question examined which satellite-derived features contributed most to the prediction of selected inventory attributes. The feature importance analysis showed that both optical and radar-derived predictors contributed to model performance, although Sentinel-2 variables were generally more impactful than Sentinel-1 SAR metrics. For regression tasks, the green band (B3), red-edge band (B6), and vegetation indices such as NDMI were among the most important predictors. For classification tasks, the red-edge band (B6), green band (B3), and SWIR band (B11) showed the strongest contribution. SAR-derived variables, especially the VV–VH backscatter difference, provided additional structural information, but their overall importance was lower. These results confirmed that combining spectral and structural predictors improved the representation of forest stand characteristics.

The third research question focused on identifying the most accurate and reliable supervised ML methods for estimating forest inventory attributes and supporting the assessment of NFI data quality. The comparison of RF and GB showed similar performance across both regression and classification tasks, although RF demonstrated slightly better overall accuracy and robustness. Regression modelling produced moderate predictive accuracy for GSV and stand height. The classification performance was higher for age classes than for dominant tree species. The relatively small differences between the models suggest that predictive performance was more strongly limited by data quality, class imbalance, and remote sensing constraints than by algorithm

selection. In this Master's thesis, RF was identified as the most reliable and robust method among the tested supervised ML models.

Residual analysis further showed that prediction errors were not randomly distributed but were strongly related to stand characteristics and inventory conditions. Larger deviations were observed in small and heterogeneous stands, as well as in older inventory records. The systematic underestimation of high GSV values indicated limitations related to signal saturation in optical satellite data. These findings show that model residuals and high-deviation stands can serve as practical indicators of potential inconsistencies in forest inventory data and can support targeted field verification instead of full re-inventory.

The proposed approach has practical value because it can improve the efficiency, reliability, and cost-effectiveness of forest inventory management. Instead of repeating expensive and time-consuming field inventories across large areas, the approach allows attention to be focused on stands where the risk of errors is higher. This can support more accurate forest valuation, better investment decisions, improved carbon accounting, and more sustainable forest management.

Overall, the thesis showed that supervised ML combined with Sentinel data can be useful not only for forest attribute estimation, but also for identifying potential inconsistencies and enhancing the quality assessment of NFI data, even when model accuracy is moderate.

To support transparency and reproducibility, the scripts, workflow artefacts, and supplementary materials developed during this thesis are available in the project GitHub repository [\[LINK\]](#). The bibliographic database and literature collection used for the systematic literature review are available through Zotero [\[LINK\]](#).

REFERENCES

1. Amuyou, U. A., Wang, Y., Ebuta, B. F., Iheaturu, C. J., & Antonarakis, A. S. (2022). Quantification of above-ground biomass over the Cross-River State, Nigeria, using Sentinel-2 data. *Remote Sensing*, 14(22), 5741, available at: <https://doi.org/10.3390/rs14225741>
2. Becker, A., Russo, S., Puliti, S., Lang, N., Schindler, K., Wegner, J. D. (2023). Country-wide retrieval of forest structure from optical and SAR satellite imagery with deep ensembles. *ISPRS Journal of Photogrammetry and Remote Sensing*, 195(January), 269–286, available at: <https://doi.org/10.1016/j.isprsjprs.2022.11.011>
3. Bolyn, C., Lejeune, P., Michez, A., Latte, N. (2022). Mapping tree species proportions from satellite imagery using spectral - spatial deep learning. *Remote Sensing of Environment*, 280 (October) 113205, available at: <https://doi.org/10.1016/j.rse.2022.113205>.
4. Breiman, L. (2001) Random forests. *Machine Learning*, 45(1), pp. 5-32, available at: <https://doi.org/10.1023/A:1010933404324>
5. Bui, Q.-T., Pham, Q.-T., Pham, V.-M., Tran, V.-T., Nguyen D.-H., Nguyen Q.-H., Nguyen H.-D., Do, N.T., Vu, V.-M. (2024). Hybrid machine learning models for aboveground biomass estimations. *Ecological Informatics*, 79 (March), 102421, available at: <https://doi.org/10.1016/j.ecoinf.2023.102421>.
6. Chen, J., Du, H., Mao, F., Huang, Z., Chen, C., Hu, M., Li, X. (2024). Improving forest age prediction performance using ensemble learning algorithms base on satellite remote sensing data. *Ecological Indicators*, 166 (September), 112327, available at: <https://doi.org/10.1016/j.ecolind.2024.112327>.
7. Chen, L., Ren, C., Zhang, B., Wang, Z., Wang, Y. (2019). Mapping spatial variations of structure and function parameters for forest condition assessment of the Changbai Mountain National Nature Reserve. *Remote Sensing*, 11(24), 3004, available at: <https://doi.org/10.3390/rs11243004>
8. Chen, M., Dong, W., Yu, H., Woodhouse, I.H., Ryan, C.M., Liu, H., Georgiou, S. and Mitchard, E.T.A. (2025). Multimodal deep learning enables forest height mapping from patchy spaceborne LiDAR using SAR and passive optical satellite data. *International Journal of Applied Earth Observation and Geoinformation*, 143, 104814. doi: 10.1016/j.jag.2025.104814.

9. Čabaravdić, A. and Balić, B. (2019). Modelling stand variables of beech coppice forest using spectral Sentinel-2A data and the machine learning approach. *South-East European Forestry* 10 (2), 274, 137-144, available at:<https://doi.org/10.15177/SEEFOR.19-21>.
10. D'Amico, G., Nilsson, M., Axelsson, A., Chirici, G. (2024). Data homogeneity impact in tree species classification based on Sentinel-2 multitemporal data case study in central Sweden. *International Journal of Remote Sensing* 45 (15), 5050-75, available at: <https://doi.org/10.1080/01431161.2024.2371082>.
11. Elsevier (n.d.) ScienceDirect, available at:<https://www.sciencedirect.com/>, [accessed: 25 September 2025].
12. Elsevier (n.d.) Scopus, available at: <https://www.scopus.com/pages/home>, [accessed: 25 September 2025].
13. Fararoda, R., Reddy, R.S., Rajashekar, G., Chand, T.R.K., Jha, C.S. and Dadhwal, V.K. (2021). Improving forest above-ground biomass estimates over Indian forests using multi-source datasets with machine learning algorithm. *Ecological Informatics*, 65(November), 101392, available at: <https://doi.org/10.1016/j.ecoinf.2021.101392>.
14. Cabinet Regulation No.384 (2016). Forest inventory and forest state register information circulation rules, *Latvijas Vēstnesis*, 28 June 2016, [in Latvian], available at: <https://likumi.lv/ta/id/283091-meza-inventarizacijas-un-meza-valsts-registra-informacijas-aprites-noteikumi> [accessed: 12 December 2025].
15. Francini, S., Schelhaas, M.-J., Vangi, E., Lerink, B., Nabuurs, G.-J., McRoberts, R.E., Chirici, G. (2024). Forest species mapping and area proportion estimation combining Sentinel-2 harmonic predictors and national forest inventory data. *International Journal of Applied Earth Observation and Geoinformation*, 131(July), 103935, available at: <https://doi.org/10.1016/j.jag.2024.103935>.
16. Grabska, E., Hostert, P., Pflugmacher, D., Ostapowicz, K. (2019). Forest stand species mapping using the Sentinel-2 time series. *Remote Sensing*, 11(10), 1197, available at: <https://doi.org/10.3390/rs11101197>.
17. Grabska-Szwagrzyk, E., Tiede, D., Sudmanns, M., Kozak, J. (2024). Map of forest tree species for Poland based on Sentinel-2 data. *Earth System Science Data*, 16(6), 2877–2891, available at: <https://doi.org/10.5194/essd-16-2877-2024>.
18. Heyer, J., Schleeweis, K., Ruefenacht, B., Housman, I., Zhiqiang, Y., Ryerson, D., Reischmann, J., Megown, K., Bogle, M.S. (2025). Annual national tree canopy cover mapping: a novel workflow with temporal transferability and improved uncertainty

- quantification. *Science of Remote Sensing*, 12(October), 100301, available at: <https://doi.org/10.1016/j.srs.2025.100301>.
19. IEEE (n.d.) *IEEE Xplore*, available at: <https://ieeexplore.ieee.org/Xplore/home.jsp>, [accessed: 25 September 2025].
 20. Khan, M. N., Tan, Y., Gul, A. A., Abbas, S., Wang, J. (2024). Forest aboveground biomass estimation and inventory: evaluating remote sensing-based approaches. *Forests*, 15(6), 1055, available at: <https://doi.org/10.3390/f15061055>.
 21. Kovárník, R., and Janová, J. (2025). Validation of Sentinel-2 based machine learning models for Czech national forest inventory. *Ecological Informatics*, 87(July), 103133, available at: <https://doi.org/10.1016/j.ecoinf.2025.103133>.
 22. Laurin, G.V., Puletti, N., Hawthorne, W., Liesenberg, V., Corona, P., Papale, D., Chen, Q., Valentini, R. (2016). Discrimination of tropical forest types, dominant species, and mapping of functional guilds by hyperspectral and simulated multispectral Sentinel-2 data. *Remote Sensing of Environment*, 176(April), 163–176, available at: <https://doi.org/10.1016/j.rse.2016.01.017>.
 23. Litmaps (n.d.) Literature mapping tool, available at: <https://www.litmaps.com/>, [accessed: 27 October 2025].
 24. Marconi, S., Weinstein, B.G., Zou, S., Bohlman, S.A., Zare, A., Singh, A., Stewart, D., Harmon, I., Steinkraus, A., White, E.P. (2022). Continental-scale hyperspectral tree species classification in the United States National Ecological Observatory Network. *Remote Sensing of Environment*, 282 (December), 113264, available at: <https://doi.org/10.1016/j.rse.2022.113264>.
 25. Molnár, T., and Király, G. (2024). Forest disturbance monitoring using cloud-based Sentinel-2 satellite imagery and machine learning. *Journal of Imaging*, 10(1), 14, available at: <https://doi.org/10.3390/jimaging100100144>.
 26. Nasiri, V., Beloiu, M., Darvishsefat, A.A., Griess, V.C., Maftai, C., Waser, L.T. (2023). Mapping tree species composition in a Caspian temperate mixed forest based on spectral-temporal metrics and machine learning. *International Journal of Applied Earth Observation and Geoinformation*, 116 (February), 103154, available at: <https://doi.org/10.1016/j.jag.2022.103154>.
 27. Nian, Y., Chen, S., Chen, J., Che, M., Zhang, W., Ali, J.S., Zhang, H., Wang, X., Liao, B. and Wang, X. (2025). Mapping subalpine forest aboveground biomass in Qilian Mountain National Park using UAV-LiDAR, GEDI, and multisource satellite data. *IEEE Journal of*

- Selected Topics in Applied Earth Observations and Remote Sensing, 18(May), pp.12407–12420, available at: <https://ieeexplore.ieee.org/document/10989585?denied=>
28. Novo-Fernández, A., López-Sánchez, C. A., Cámara-Obregón, A., Barrio-Anta, M., Teijido-Murias, I. (2024). Estimating forest variables for major commercial timber plantations in northern Spain using Sentinel-2 and ancillary data. *Forests*, 15(1), 99, available at: <https://doi.org/10.3390/f15010099>.
 29. Omoniyi, T. O., and Sims, A. (2024). Enhancing the precision of forest growing stock volume in the Estonian national forest inventory with different predictive techniques and remote sensing data. *Remote Sensing*, 16(20), 3794, available at: <https://doi.org/10.3390/rs16203794>.
 30. Puliti, S., Hauglin, M., Breidenbach, J., Montesano, P., Neigh, C.S.R., Rahlf, J., Solberg, S., Klingenberg, T.F. and Astrup, R. (2020). Modelling above-ground biomass stock over Norway using national forest inventory data with ArcticDEM and Sentinel-2 data. *Remote Sensing of Environment*, 236(January), 111501, available at: <https://doi.org/10.1016/j.rse.2019.111501>.
 31. Punalekar, S.M., Planque, C., Lucas, R.M., Evans, D., Correia, V., Owers, C.J., Pośłajko, P., Bunting, P., Chognard, S. (2021). National scale mapping of larch plantations for Wales using the Sentinel-2 data archive. *Forest Ecology and Management*, 501(December), 119679, available at: <https://doi.org/10.1016/j.foreco.2021.119679>.
 32. Sentinel Hub (n.d.). *Sentinel-2 RS indices*. Available at: <https://custom-scripts.sentinel-hub.com/custom-scripts/sentinel-2/indexdb/>, [accessed: 25 February 2026].
 33. Silveira, E.M.O., Radeloff, V.C., Martinuzzi, S., Martínez Pastur, G.J., Bono, J., Politi, N., Lizarraga, L., Rivera, L.O., Ciuffoli, L., Rosas, Y.M., Olah, A.M., Gavier-Pizarro, G.I., Pidgeon, A.M. (2023). Nationwide native forest structure maps for Argentina based on forest inventory data, SAR Sentinel-1 and vegetation metrics from Sentinel-2 imagery. *Remote Sensing of Environment*, 285(February), 113391, available at: <https://doi.org/10.1016/j.rse.2022.113391>.
 34. Singh, B., Verma, A.K., Tiwari, K., Joshi, R. (2024). Above ground tree biomass modeling using machine learning algorithms in western Terai sal forest of Nepal. *Heliyon*, 9(11), e21485, available at: <https://doi.org/10.1016/j.heliyon.2023.e21485>.
 35. Suleymanov, A., Bogdan, E., Gaysin, I., Volkov, A., Tuktarova, I., Belan, L., Shagaliev, R. (2024). Spatial high-resolution modelling and uncertainty assessment of forest growing stock volume based on remote sensing and environmental covariates. *Forest Ecology and*

- Management*, 554(February), 121676, available at:
<https://doi.org/10.1016/j.foreco.2023.121676>.
36. Tolan, J., Yang, H.-I., Nosarzewski, B., Couairon, G., Vo, H.V., Brandt, J., Spore, J., Majumdar, S., Haziza, D., Vamaraju, J., Moutakanni, T., Bojanowski, P., Johns, T., White, B., Tiecke, T., Couprie, C. (2024). Very high-resolution canopy height maps from RGB imagery using self-supervised vision transformer and convolutional decoder trained on aerial lidar. *Remote Sensing of Environment* 300(January), 113888, available at:
<https://doi.org/10.1016/j.rse.2023.113888>.
 37. Van Geffen, F., Hänsch, R., Demir, B., Kruse, S., Herzsuh, U. and Heim, B. (2025). A benchmark dataset for Sentinel-2 based forest type classification in the Siberian summergreen-evergreen forest transition zone. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 18(April), 18509 – 18527, available at:
<https://doi.org/10.1109/JSTARS.2025.3562912>.
 38. Wessel, M., Brandmeier, M., Tiede, D. (2018). Evaluation of different machine learning algorithms for scalable classification of tree types and tree species based on Sentinel-2 data. *Remote Sensing*, 10(9), 1419, available at: <https://doi.org/10.3390/rs10091419>.
 39. Wittke, S., Yu, X., Karjalainen, M., Hyypä, J., Puttonen, E. (2019). Comparison of two-dimensional multitemporal Sentinel-2 data with three-dimensional remote sensing data sources for forest inventory parameter estimation over a boreal forest. *International Journal of Applied Earth Observation and Geoinformation*, 76(April), 167–178, available at:
<https://doi.org/10.1016/j.jag.2018.11.009>.
 40. Yadegari, L.Z., Sohrabi, H., Quirós, E. and Immitzer, M. (2025). Mapping above-ground biomass in old-growth deciduous forests using open-access satellite data, field plots, and machine learning algorithms. *Advances in Space Research*, 76(11), 6566–6584, available at: <https://doi.org/10.1016/j.asr.2025.09.013>.
 41. Yatskiv, I. (2025). Types of data, Data Mining lecture 4, Transport and Telecommunication Institute, available at:
https://e.tsi.lv/pluginfile.php/186648/mod_resource/content/11/DM_4_2026.pdf,
 [accessed: 25 February 2026].
 42. Zhang, X., Shen, H., Huang, T., Wu, Y., Guo, B., Liu, Z., Luo, H., Tang, J., Zhou, H., Wang, L., Xu, W., Ou, G. (2024). Improved random forest algorithms for increasing the accuracy of forest aboveground biomass estimation using Sentinel-2 imagery. *Ecological Indicators*, 159(February), 111752, available at:
<https://doi.org/10.1016/j.ecolind.2024.111752>.

43. Zheng, M., Wen, Q., Xu, F., Wu, D. (2025). Regional forest carbon stock estimation based on multi-source data and machine learning algorithms. *Forests*, 16(3), 420, available at: <https://doi.org/10.3390/f16030420>.
44. Zurqani, H.A. (2025). A multi-source approach combining GEDI LiDAR, satellite data, and machine learning algorithms for estimating forest aboveground biomass on Google Earth Engine platform. *Ecological Informatics*, 86(May), 103052, available at: <https://doi.org/10.1016/j.ecoinf.2025.103052>.

APPENDICES

Appendix I

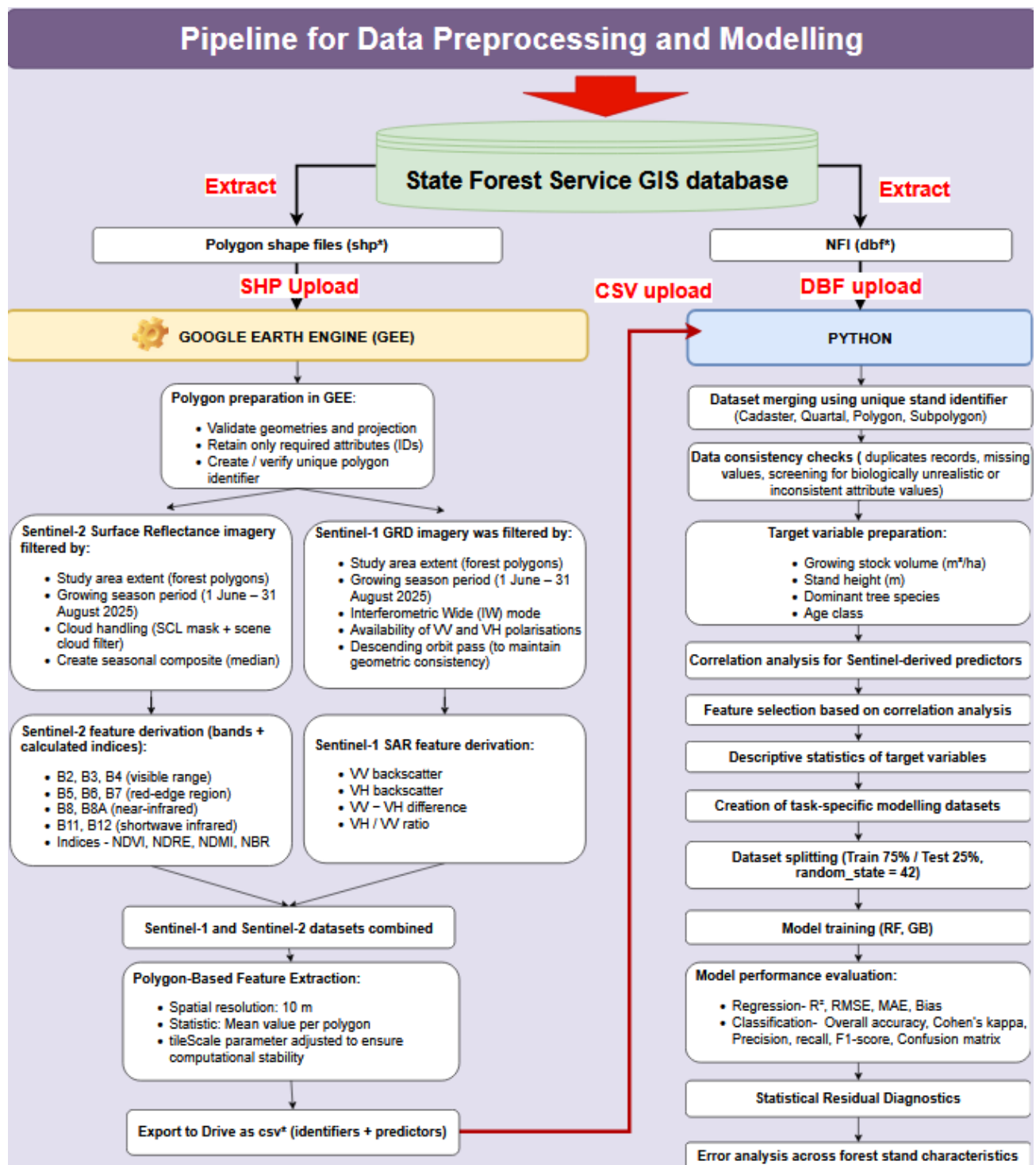
Article citation count

Title	Cross-reference Citations
Very high-resolution canopy height maps from RGB imagery using self-supervised vision transformer and convolutional decoder trained on aerial lidar	160
Discrimination of tropical forest types, dominant species, and mapping of functional guilds by hyperspectral and simulated multispectral Sentinel-2 data	157
Evaluation of different machine learning algorithms for scalable classification of tree types and tree species based on Sentinel-2 data	141
Mapping tree species proportions from satellite imagery using spectral-spatial deep learning	69
Nationwide native forest structure maps for Argentina based on forest inventory data, SAR Sentinel-1 and vegetation metrics from Sentinel-2 imagery	66
Modelling above-ground biomass stock over Norway using national forest inventory data with ArcticDEM and Sentinel-2 data	65
Improving forest above ground biomass estimates over Indian forests using multi source data sets with machine learning algorithm	53
Mapping dominant leaf type based on combined Sentinel-1/-2 data - Challenges for mountainous countries	50
Mapping Forest Aboveground Biomass Using Multisource Remotely Sensed Data	46
Improved random forest algorithms for increasing the accuracy of forest aboveground biomass estimation using Sentinel-2 imagery	45
Country-wide retrieval of forest structure from optical and SAR satellite imagery with deep ensembles	44
Comparison of two-dimensional multitemporal Sentinel-2 data with three-dimensional remote sensing data sources for forest inventory parameter estimation over a boreal forest	39
Wall-to-wall spatial prediction of growing stock volume based on Italian National Forest Inventory plots and remotely sensed data	37
Mapping forest canopy fuel parameters at European scale using spaceborne LiDAR and satellite data	30
Hybrid machine learning models for aboveground biomass estimations	29
Predicting tree species diversity using geodiversity and sentinel-2 multi-seasonal spectral information	26
Utility of hyperspectral compared to multispectral remote sensing data in estimating forest biomass and structure variables in Finnish boreal forest	25
Combination of multi-temporal Sentinel 2 images and aerial image based canopy height models for timber volume modelling	19
Map of forest tree species for Poland based on Sentinel-2 data	16
Forest Aboveground Biomass Estimation and Inventory: Evaluating Remote Sensing-Based Approaches	16
Mapping tree species composition in a Caspian temperate mixed forest based on spectral-temporal metrics and machine learning	15
A multi-source approach combining GEDI LiDAR, satellite data, and machine learning algorithms for estimating forest aboveground biomass on Google Earth Engine platform	12
Improving forest age prediction performance using ensemble learning algorithms base on satellite remote sensing data	11

National scale mapping of larch plantations for Wales using the Sentinel-2 data archive	10
Mapping spatial variations of structure and function parameters for forest condition assessment of the changbai mountain national nature reserve	10
Quantification of Above-Ground Biomass over the Cross-River State, Nigeria, Using Sentinel-2 Data	8
China's larch stock volume estimation using Sentinel-2 and LiDAR data	8
Forest Disturbance Monitoring Using Cloud-Based Sentinel-2 Satellite Imagery and Machine Learning	7
Comparison of multi-source remote sensing data for estimating and mapping above-ground biomass in the West Usambara tropical montane forests	7
Improving country-wide individual tree detection using local maxima methods based on statistically modeled forest structure information	5
Forest species mapping and area proportion estimation combining Sentinel-2 harmonic predictors and national forest inventory data	5
Exploring the potential of SAR and terrestrial and airborne LiDAR in predicting forest floor spectral properties in temperate and boreal forests	4
Above ground tree biomass modeling using machine learning algorithms in western Terai Sal Forest of Nepal	4
Estimating aboveground biomass and biodiversity of a protected area in moist tropical forest ecosystem using remote sensing application	3
Estimating Forest Variables for Major Commercial Timber Plantations in Northern Spain Using Sentinel-2 and Ancillary Data	2
Validation of sentinel 2 based machine learning models for Czech National Forest Inventory	2
Spatial high-resolution modelling and uncertainty assessment of forest growing stock volume based on remote sensing and environmental covariates	2
Regional Forest Carbon Stock Estimation Based on Multi-Source Data and Machine Learning Algorithms	2
Mapping Subalpine Forest Aboveground Biomass in Qilian Mountain National Park Using UAV-LiDAR, GEDI, and Multisource Satellite Data	2
Enhancing the Precision of Forest Growing Stock Volume in the Estonian National Forest Inventory with Different Predictive Techniques and Remote Sensing Data	2
A Spectral-Spatial Approach for the Classification of Tree Cover Density in Mediterranean Biomes Using Sentinel-2 Imagery	2
Integrating multi-source remote sensing data and machine learning for predicting tree density and cover in <i>Argania spinosa</i>	1
A Benchmark Dataset for Sentinel-2 Based Forest Type Classification in the Siberian Summergreen-Evergreen Forest Transition Zone	1
Temporal scalability analysis of spectral remote sensing and Machine Learning-Based models for forest mapping in the tropical region of Nigeria	1
Data homogeneity impact in tree species classification based on Sentinel-2 multitemporal data case study in central Sweden	1
Mapping above-ground biomass in old-growth deciduous forests using open-access satellite data, field plots, and machine learning algorithms	0
Integrating multi-source remote sensing data for mapping boreal forest canopy height and species in interior Alaska in support of radar modelling	0
GEDI and Sentinel data integration for quantifying agroforestry tree height and stocks	0
Enhanced aboveground biomass density estimation in Central Vietnamese forests	0
Using vegetation and chlorophyll indices to model above ground biomass over time in an urban arboretum in subtropical queensland	0
Towards accurate mapping of forest in tropical landscapes: A comparison of datasets on how forest transition matters	0

Sentinel-2 vs. PlanetScope: Comparison and combination for tree species classification in two central European forest ecosystems	0
Multi-sensor modelling of woody vegetation and canopy cover across natural and modified ecosystems	0
Multimodal deep learning enables forest height mapping from patchy spaceborne LiDAR using SAR and passive optical satellite data	0
Modelling stand variables of beech coppice forest using spectral sentinel-2A data and the machine learning approach	0
Continental-scale hyperspectral tree species classification in the United States National Ecological Observatory Network	0
Annual National Tree Canopy Cover Mapping: A Novel Workflow with Temporal Transferability and Improved Uncertainty Quantification	0
Total	1260
Mean	22

Pipeline for Data Preprocessing and Modelling



Nogabalu raksturojošie rādītāji

Kvartāls/ nogabals	Ģeo platība, ha	Meža zemes veids	Meža tips	Nogabala apraksts	Izcelšanās bonitāte	Valdoša koku suga vid. augstums	vid. caurmērs	vecums	biezība	Šķērslauk. m ² /ha	koku sk. gab/ha	krāja m ³ /ha
Zemes vienība 78520060282				inv. veikta 2006. gadā		Īpašums - 78520060282 - Launagi						
				Feimaņu pagasts		Saimniecība: Launagi						
1. kvartāls												
1	0.19	Mežaudze	Dm	10B34	D	Ia	20	15	34	7	18	168
				Pēdējais cirtes izpildes veids un gads: Kopšanas cirte 2021								
				Pēdējais darbības veids un gads: Koku ciršana 2021								
				Platību sadalījums: t.sk. mežs 0.19 ha, t.sk. ceļi 0 ha, t.sk. grāvji 0 ha								
3	0.1	Mežaudze	Ln	9P69 1B59	D	I	24	24	69	7	25	289
				Platību sadalījums: t.sk. mežs 0.1 ha, t.sk. ceļi 0 ha, t.sk. grāvji 0 ha								
4	0.15	Mežaudze	Dm	4P69 4P89 2B74	D	Ia	27	24	69	10	36	463
				Platību sadalījums: t.sk. mežs 0.13 ha, t.sk. ceļi 0.02 ha, t.sk. grāvji 0 ha								
5	0.69	Mežaudze	Ln	7P29 3P34	D	I	13	12	29	7	20	154
				Pēdējais cirtes izpildes veids un gads: Kopšanas cirte 2021								
				Pēdējais darbības veids un gads: Koku ciršana 2021								
				Platību sadalījums: t.sk. mežs 0.66 ha, t.sk. ceļi 0.03 ha, t.sk. grāvji 0 ha								
6	0.1	Mežaudze	Dm	5P69 4E59 1B64	D	Ia	26	24	69	11	40	466
				Platību sadalījums: t.sk. mežs 0.1 ha, t.sk. ceļi 0 ha, t.sk. grāvji 0 ha								
7	0.13	Mežaudze	Dm	10B29	D	II	13	12	29	8	15	101
				Pēdējais cirtes izpildes veids un gads: Kopšanas cirte 2021								
				Pēdējais darbības veids un gads: Koku ciršana 2021								
				Platību sadalījums: t.sk. mežs 0.13 ha, t.sk. ceļi 0 ha, t.sk. grāvji 0 ha								
8	0.11	Mežaudze	Dm	8B1E34 1P23	D	Ia	20	18	34	9	19	157
				Pēdējais cirtes izpildes veids un gads: Kopšanas cirte 2021								
				Pēdējais darbības veids un gads: Koku ciršana 2021								
				Platību sadalījums: t.sk. mežs 0.11 ha, t.sk. ceļi 0 ha, t.sk. grāvji 0 ha								
9	0.08	Mežaudze	Dm	4P121 4P89 2B94	D	I	30	37	121	7	26	350
				Platību sadalījums: t.sk. mežs 0.08 ha, t.sk. ceļi 0 ha, t.sk. grāvji 0 ha								
10	0.26	Mežaudze	Ln	10P27	D	Ia	15	15	27	7	21	172
				Pēdējais cirtes izpildes veids un gads: Kopšanas cirte 2021								
				Pēdējais darbības veids un gads: Koku ciršana 2021								
				Platību sadalījums: t.sk. mežs 0.26 ha, t.sk. ceļi 0 ha, t.sk. grāvji 0 ha								
11	0.49	Mežaudze	Dm	7B29 3B39	D	I	17	11	29	6	16	140
				Pēdējais cirtes izpildes veids un gads: Kopšanas cirte 2021								
				Pēdējais darbības veids un gads: Koku ciršana 2021								
				Platību sadalījums: t.sk. mežs 0.49 ha, t.sk. ceļi 0 ha, t.sk. grāvji 0 ha								
12	1.01	Mežaudze	Dm	7P79 3P89	D	Ia	29	29	79	8	29	390
				2. stāvs: K39								
				Pēdējais cirtes izpildes veids un gads: Kopšanas cirte 2021								
				Pēdējais darbības veids un gads: Koku ciršana 2021								
				Platību sadalījums: t.sk. mežs 0.96 ha, t.sk. ceļi 0.05 ha, t.sk. grāvji 0 ha								
13	5.44	Mežaudze	Ln	7P34 3P29	D	Ia	16	14	34	7	24	195
				Pēdējais cirtes izpildes veids un gads: Kopšanas cirte 2021								
				Pēdējais darbības veids un gads: Koku ciršana 2021								
				Platību sadalījums: t.sk. mežs 5.38 ha, t.sk. ceļi 0.06 ha, t.sk. grāvji 0 ha								
Kopā platība kvartālā (ha):				8.75	Platību sadalījums: t.sk. mežs 8.59 ha, t.sk. ceļi 0.16 ha, t.sk. grāvji 0 ha							
Kopā platība kadastrā (ha):				8.75	Platību sadalījums: t.sk. mežs 8.59 ha, t.sk. ceļi 0.16 ha, t.sk. grāvji 0 ha							
Pavisam kopā (ha):				8.75	Platību sadalījums: t.sk. mežs 8.59 ha, t.sk. ceļi 0.16 ha, t.sk. grāvji 0 ha							

Full DBF attribute table

Attribute	Description	Unit / Type	Category
APS	Manager / inventory file ID	Identification	Identification
KAD	Cadastral unit code	Identification	Identification
KV	Quartal/Block number	Identification	Identification
NOG	Area number	Identification	Identification
ANOG	Sub-stand identifier	Identification	Identification
PLAT	Total geometric area	ha	Geometry
EXPL_MEZS	Exploited forest land	ha	Land use
EXPL_CELI	Road area	ha	Land use
EXPL_GRAVJI	Ditch area	ha	Land use
ZKAT	Soil category	Identifier	Site
MT	Forest / site type	Identifier	Site
BON	Fertility class	Identifier	Site
IZC	Stand origin (natural / planted)	Identifier	Origin
VGR	Age group, code	Identifier	Structure / Development stage
V_NOG	Stand volume	m ³ /ha	Structure
P_DARBV	Last management operation type	Identifier	Management
P_DARBG	Last management operation year	year	Management
P_CIRP	Last cutting type	Identifier	Management
P_CIRG	Last cutting year	year	Management
GK	Actual renewal year	year	Origin/Regeneration
PLANT	“Is the stand a plantation	binary Identifier	Origin / Management
G1	Basal area of the canopy layer 1	m ² /ha	Stand structure – canopy layer 1
K10	Coefficient (dominant species in 1st canopy layer)	Identifier	Stand structure – canopy layer 1
S10	Tree type, canopy layer 1; first dominance	Identifier	Stand structure – canopy layer 1
A10	Age of the first dominating species in the first canopy layer	year	Stand structure – canopy layer 1
H10	Mean height, canopy layer 1, first dominance	m	Stand structure – canopy layer 1
D10	Mean diameter, canopy layer 1, first dominance	cm	Stand structure – canopy layer 1
G10	Basal area, for the first canopy layer; first dominance	m ² /ha	Stand structure – canopy layer 1
N10	Tree count, per species in the first canopy layer; first dominance	pcs/ha	Stand structure – canopy layer 1
BV10	Damage type, per species in the first canopy layer; first dominance	Identifier	Stand structure – canopy layer 1
BA10	Damage severity per species in the first canopy layer; for the first dominant species	Identifier	Stand structure – canopy layer 1
V10	Timber volume, per species in the first canopy layer; first dominance	m ³ /ha	Stand structure – canopy layer 1

B10	Canopy density, canopy layer 1	%	Stand structure – canopy layer 1
K11	Coefficient (2nd species in 1st canopy layer)	coefficient(1-10)	Stand structure – canopy layer 1
S11	Tree type, canopy layer 1; second dominance	Identifier	Stand structure – canopy layer 1
A11	Age of the second dominating species in the first canopy layer	year	Stand structure – canopy layer 1
H11	Mean height, canopy layer 1, second dominance	m	Stand structure – canopy layer 1
D11	Mean diameter, canopy layer 1, second dominance	cm	Stand structure – canopy layer 1
G11	Basal area, for the first canopy layer; second dominance	m ² /ha	Stand structure – canopy layer 1
N11	Tree count, per species in the first canopy layer; second dominance	pcs/ha	Stand structure – canopy layer 1
BV11	Damage type, per species in the first canopy layer; second dominance	Identifier	Stand structure – canopy layer 1
BA11	Damage severity per species in the first canopy layer; for the second dominant species	Identifier	Stand structure – canopy layer 1
V11	Timber volume, per species in the first canopy layer; second dominance	m ³ /ha	Stand structure – canopy layer 1
K12	Coefficient (3rd species in 1st canopy layer)	coefficient(1-10)	Stand structure – canopy layer 1
S12	Tree type, canopy layer 1; third dominance	Identifier	Stand structure – canopy layer 1
A12	Age of the third dominating species in the first canopy layer	year	Stand structure – canopy layer 1
H12	Mean height, canopy layer 1, third dominance	m	Stand structure – canopy layer 1
D12	Mean diameter, canopy layer 1, third dominance	cm	Stand structure – canopy layer 1
G12	Basal area, for the first canopy layer; third dominance	m ² /ha	Stand structure – canopy layer 1
N12	Tree count, per species in the first canopy layer; third dominance	pcs/ha	Stand structure – canopy layer 1
BV12	Damage type, per species in the first canopy layer; third dominance	Identifier	Stand structure – canopy layer 1
BA12	Damage severity per species in the first canopy layer; for the third dominant species	Identifier	Stand structure – canopy layer 1
V12	Timber volume, per species in the first canopy layer; third dominance	m ³ /ha	Stand structure – canopy layer 1
K13	Coefficient (4th species in 1st canopy layer) for the fourth species in the first canopy layer	coefficient(1-10)	Stand structure – canopy layer 1
S13	Tree type, canopy layer 1; fourth dominance	Identifier	Stand structure – canopy layer 1
A13	Age of the fourth dominating species in the first canopy layer	year	Stand structure – canopy layer 1
H13	Mean height, canopy layer 1, fourth dominance	m	Stand structure – canopy layer 1

D13	Mean diameter, canopy layer 1, fourth dominance	cm	Stand structure – canopy layer 1
G13	Basal area, for the first canopy layer; fourth dominance	m ² /ha	Stand structure – canopy layer 1
N13	Tree count, per species in the first canopy layer; fourth dominance	pcs/ha	Stand structure – canopy layer 1
BV13	Damage type, per species in the first canopy layer; fourth dominance	Identifier	Stand structure – canopy layer 1
BA13	Damage severity per species in the first canopy layer; for the fourth dominant species	Identifier	Stand structure – canopy layer 1
V13	Timber volume, per species in the first canopy layer; fourth dominance	m ³ /ha	Stand structure – canopy layer 1
K14	Coefficient (5th species in 1st canopy layer)	coefficient(1-10)	Stand structure – canopy layer 1
S14	Tree type, canopy layer 1; fifth dominance	Identifier	Stand structure – canopy layer 1
A14	Age of the fifth dominating species in the first canopy layer	year	Stand structure – canopy layer 1
H14	Mean height, canopy layer 1, fifth dominance	m	Stand structure – canopy layer 1
D14	Mean diameter, canopy layer 1, fifth dominance	cm	Stand structure – canopy layer 1
G14	Basal area, for the first canopy layer; fifth dominance	m ² /ha	Stand structure – canopy layer 1
N14	Tree count, per species in the first canopy layer; fifth dominance	pcs/ha	Stand structure – canopy layer 1
BV14	Damage type, per species in the first canopy layer; fifth dominance	Identifier	Stand structure – canopy layer 1
BA14	Damage severity per species in the first canopy layer; for the fifth dominant species	Identifier	Stand structure – canopy layer 1
V14	Timber volume, per species in the first canopy layer; fifth dominance	m ³ /ha	Stand structure – canopy layer 1
G2	Basal area of the canopy layer 2	m ² /ha	Stand structure – canopy layer 2
K22	Coefficient (1st species in 2nd canopy layer)	coefficient(1-10)	Stand structure – canopy layer 2
S22	Tree type, canopy layer 2; first dominance	Identifier	Stand structure – canopy layer 2
A22	Age of the first dominating species in the second canopy layer	year	Stand structure – canopy layer 2
H22	Mean height, canopy layer 2, first dominance	m	Stand structure – canopy layer 2
D22	Mean diameter, canopy layer 2, first dominance	cm	Stand structure – canopy layer 2
G22	Basal area, for the second canopy layer; first dominance	m ² /ha	Stand structure – canopy layer 2
N22	Tree count, per species in the second canopy layer; first dominance	pcs/ha	Stand structure – canopy layer 2
BV22	Damage type, per species in the second canopy layer; first dominance	Identifier	Stand structure – canopy layer 2

BA22	Damage severity per species in the second canopy layer; for the first dominant species	Identifier	Stand structure – canopy layer 2
V22	Timber volume, per species in the second canopy layer; first dominance	m ³ /ha	Stand structure – canopy layer 2
B22	Canopy density, canopy layer 2	%	Stand structure – canopy layer 2
K23	Coefficient (2nd species in 2nd canopy layer)	coefficient(1-10)	Stand structure – canopy layer 2
S23	Tree type, canopy layer 2; second dominance	Identifier	Stand structure – canopy layer 2
A23	Age of the second dominating species in the second canopy layer	year	Stand structure – canopy layer 2
H23	Mean height, canopy layer 2, second dominance	m	Stand structure – canopy layer 2
D23	Mean diameter, canopy layer 2, second dominance	cm	Stand structure – canopy layer 2
G23	Basal area, for the second canopy layer; second dominance	m ² /ha	Stand structure – canopy layer 2
N23	Tree count, per species in the second canopy layer; second dominance	pcs/ha	Stand structure – canopy layer 2
BV23	Damage type, per species in the second canopy layer; second dominance	Identifier	Stand structure – canopy layer 2
BA23	Damage severity per species in the second canopy layer; for the second dominant species	Identifier	Stand structure – canopy layer 2
V23	Timber volume, per species in the second canopy layer; second dominance	m ³ /ha	Stand structure – canopy layer 2
K24	Coefficient (3rd species in 2nd canopy layer)	coefficient(1-10)	Stand structure – canopy layer 2
S24	Tree type, canopy layer 2; third dominance	Identifier	Stand structure – canopy layer 2
A24	Age of the third dominating species in the second canopy layer	year	Stand structure – canopy layer 2
H24	Mean height, canopy layer 2, third dominance	m	Stand structure – canopy layer 2
D24	Mean diameter, canopy layer 2, third dominance	cm	Stand structure – canopy layer 2
G24	Basal area, for the second canopy layer; third dominance	m ² /ha	Stand structure – canopy layer 2
N24	Tree count, per species in the second canopy layer; third dominance	pcs/ha	Stand structure – canopy layer 2
BV24	Damage type, per species in the second canopy layer; third dominance	Identifier	Stand structure – canopy layer 2
BA24	Damage severity per species in the second canopy layer; for the third dominant species	Identifier	Stand structure – canopy layer 2
V24	Timber volume, per species in the second canopy layer; third dominance	m ³ /ha	Stand structure – canopy layer 2
SSIZC	Seed origin	Identifier	Origin
ATMIR	Deadwood volume class	Identifier	Structure
GTF	Inventory year	year	Metadata

S51	Tree species (<5% share, first species)	Identifier	Structure – special group 1
A51	Age (<5% share, first species)	year	Structure – special group 1
V51	Volume class (<5% share, first species)	Identifier	Structure – special group 1
S52	Tree species (<5% share, second species)	Identifier	Structure – special group 2
A52	Age (<5% share, second species)	year	Structure – special group 2
V52	Volume class (<5% share, second species)	Identifier	Structure – special group 2
S53	Tree species (<5% share, third species)	Identifier	Structure – special group 3
A53	Age (<5% share, third species)	year	Structure – special group 3
V53	Volume class (<5% share, third species)	Identifier	Structure – special group 3
SKOK1SUG	Species of the first retained tree	Identifier	Species composition
SKOK1SK	Volume of the first retained tree	m3	Species composition
SKOK2SUG	Species of the second retained tree	Identifier	Species composition
SKOK2SK	Volume of the second retained tree	m3	Species composition
SKOK3SUG	Species of the third retained tree	Identifier	Species composition
SKOK3SK	Volume of the third retained tree	m3	Species composition
MRM_AV_NR	MRM source number (stand from which this stand was regenerated)	Identifier	Management / permits
NEP_ATJ	Required regeneration year	year	Management
MIN_SKER	Basal area indicator	M = minimum reached; K = critical threshold reached	Management (thinning thresholds)
APROB	Indicator of strongest management restriction	Identifier	Protection
MRM_PATS	Stand itself as MRM source	Identifier	Management / permits
MRM_SERT1	Certification code 1	Identifier	Certification
MRM_SERT2	Certification code 2	Identifier	Certification
MRM_SERT3	Certification code 3	Identifier	Certification
MRM_SERT4	Certification code 4	Identifier	Certification
MRM_SERT5	Certification code 5	Identifier	Certification
TKS_LIC_N	Surveyor certificate number	Identifier	Permits
ATZARP	Pruning completed	binary Identifier	Management
IEZP	Indicator of fenced area for animal keeping	binary Identifier	Management
NEP_JKOP	Required young stand tending year	year	Management
IPAT1	Feature 1	Identifier	Protection & Special Conditions

IPAT2	Feature 2	Identifier	Protection & Special Conditions
IPAT3	Feature 3	Identifier	Protection & Special Conditions
IPAT4	Feature 4	Identifier	Protection & Special Conditions
IPAT5	Feature 5	Identifier	Protection & Special Conditions
KOMP_KAD	This stand compensates deforestation in another land parcel	cadastral reference	Protection & Special Conditions
KOMP_PLAT	This stand compensates deforestation in another land parcel	compensated area (ha)	Geometry
MDZG	Stand origin year	year	Origin
GTF_NATURE	Actual field inventory year (in nature)	year	Metadata
GI	Planting year	year	Origin & Regeneration
VA_KODS1	Protection category code 1	Identifier	Protection
VA_KODS2	Protection category code 2	Identifier	Protection
VA_KODS3	Protection category code 3	Identifier	Protection
VA_KODS4	Protection category code 4	Identifier	Protection
VA_KODS5	Protection category code 5	Identifier	Protection
AIZS	Protection class	Identifier	Protection
VA_UNIK	Unique protection ID	Identifier	Protection
AIZS_2	Protection class (additional, 2)	Identifier	Protection
VA_UNIK_2	Unique protection ID 2	Identifier	Protection
AIZS_3	Protection class (additional, 3)	Identifier	Protection
VA_UNIK_3	Unique protection ID 3	Identifier	Protection
AIZS_4	Protection class (additional, 4)	Identifier	Protection
VA_UNIK_4	Unique protection ID 4	Identifier	Protection
AIZS_5	Protection class (additional, 5)	Identifier	Protection
VA_UNIK_5	Unique protection ID 5	Identifier	Protection
INV_DATUMS	Date of last inventory received by	date	Metadata
REG_DATUMS	Date of last inventory registered in MVR	date	Metadata
INV_ATT	Was the last submitted inventory rejected in MVR?	Binary Identifier	Metadata
ORIG_ID	External system-assigned stand ID	Identifier	Metadata
IZM_DATUMS	Date of last modification in MVR for this stand	date	Metadata
ATVK	Administrative territorial code	Identifier	Metadata
LAST_UPDATE	Last update timestamp	datetime	Metadata
VMD_ID	VMD stand unique ID	Identifier	Metadata

Updated NFI data for property 36800080027

Nogabalu raksturojošie rādītāji

Species

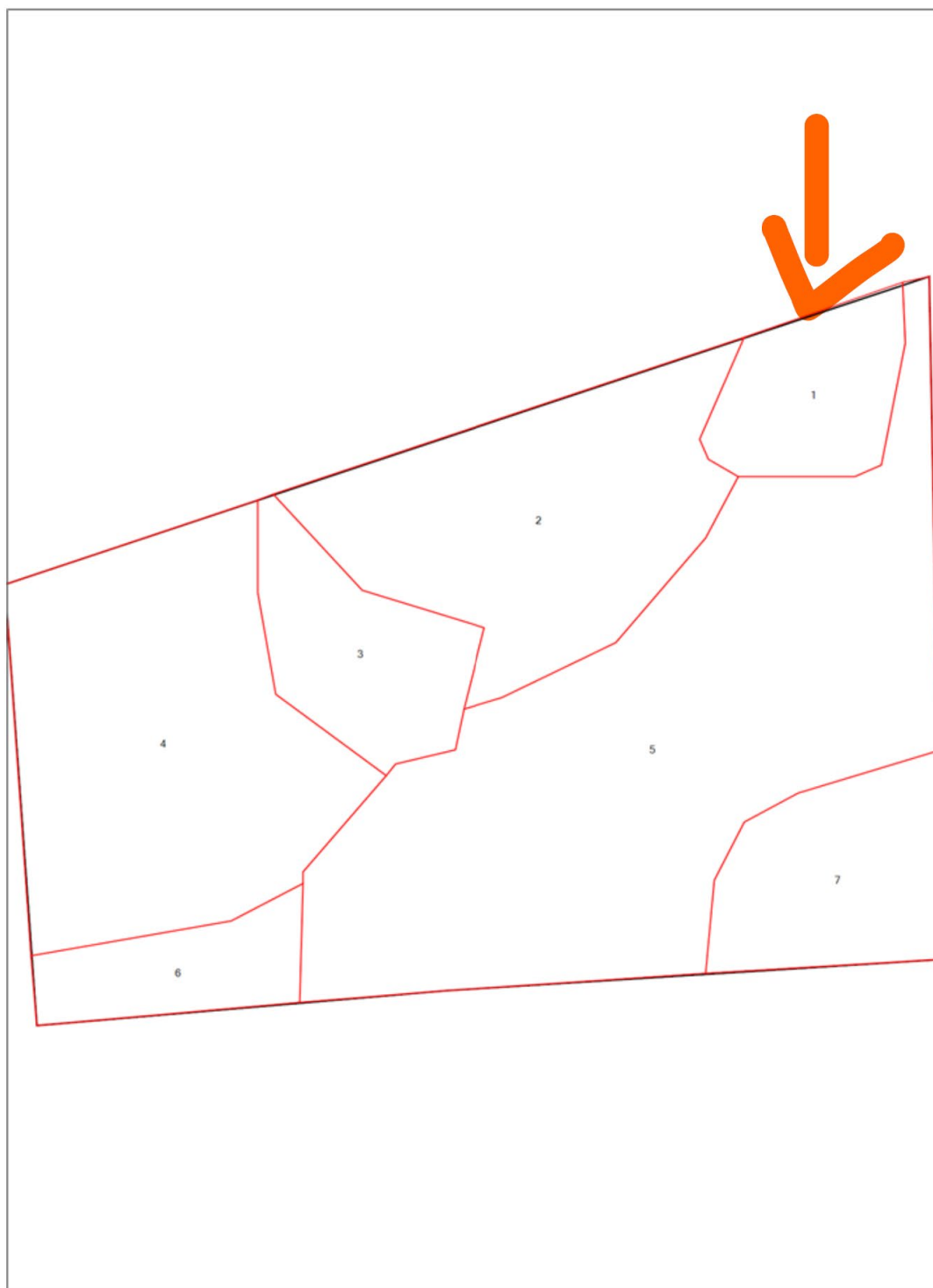
Height

Age

GSV

Īpašuma kadastra apzīmējums- xxxxxxxxxxxxxxxx

Kvartāls/ nogabals	Geo platība, ha	Meža zemes veids	Meža tips	Nogabala apraksts	Izcelšanās	bonitāte	Valdošā koku suga			Mežaudze				krāja m3/ha
							vid. augstums	vid. scaurmērs	vecums	biezība	Šķērslauk. m2/ha	koku sk. gab/ha		
Zemes vienība 100000000034				inv. veikta 2026. gadā Pededzes pagasts			Īpašums - xxxxxxxxxxx - xxxxxxxxxxx Saimniecība: Laimiņu mežs							
1. kvartāls														
1	0.33	Mežaudze	Vr	10E52	D	I	21	23	52	12	42		415	
Platību sadalījums: t.sk. mežs 0.33 ha, t.sk. ceļi 0 ha, t.sk. grāvji 0 ha														
2	0.99	Izcirtums	Vrs	Pēdējais cirtes izpildes veids un gads: Kailcirtē 2024 Pēdējais darbības veids un gads: Koku ciršana 2024 Nepieciešamais atjaunošanas gads: 2029 Platību sadalījums: t.sk. mežs 0.99 ha, t.sk. ceļi 0 ha, t.sk. grāvji 0 ha										
3	0.4	Mežaudze	Vrs	8B2M38	D	II	17	16	38	8	19		161	
Platību sadalījums: t.sk. mežs 0.4 ha, t.sk. ceļi 0 ha, t.sk. grāvji 0 ha														
4	1.29	Mežaudze	Vr	4B3M3Ba38	D	I	20	18	38	9	25		230	
Platību sadalījums: t.sk. mežs 1.29 ha, t.sk. ceļi 0 ha, t.sk. grāvji 0 ha														
5	2.24	Izcirtums	Vr	Pēdējais cirtes izpildes veids un gads: Kailcirtē 2024 Pēdējais darbības veids un gads: Koku ciršana 2024 Nepieciešamais atjaunošanas gads: 2029 Platību sadalījums: t.sk. mežs 2.24 ha, t.sk. ceļi 0 ha, t.sk. grāvji 0 ha										
6	0.26	Izcirtums	Vrs	Pēdējais cirtes izpildes veids un gads: Kailcirtē 2024 Pēdējais darbības veids un gads: Koku ciršana 2024 Nepieciešamais atjaunošanas gads: 2029 Platību sadalījums: t.sk. mežs 0.26 ha, t.sk. ceļi 0 ha, t.sk. grāvji 0 ha										
7	0.47	Izcirtums	Vrs	Pēdējais cirtes izpildes veids un gads: Kailcirtē 2024 Pēdējais darbības veids un gads: Koku ciršana 2024 Nepieciešamais atjaunošanas gads: 2029 Platību sadalījums: t.sk. mežs 0.47 ha, t.sk. ceļi 0 ha, t.sk. grāvji 0 ha										
Kopā platība kvartālā (ha):				5.98	Platību sadalījums: t.sk. mežs 5.98 ha, t.sk. ceļi 0 ha, t.sk. grāvji 0 ha									
Kopā platība kadastrā (ha):				5.98	Platību sadalījums: t.sk. mežs 5.98 ha, t.sk. ceļi 0 ha, t.sk. grāvji 0 ha									
Pavisam kopā (ha):				5.98	Platību sadalījums: t.sk. mežs 5.98 ha, t.sk. ceļi 0 ha, t.sk. grāvji 0 ha									



Zemes vienība: 100000000034 Mērogs: 1:1492

xxxxxxxxxxxxxxxxxx

26.03.2026

lpp 2 no 2

Research methods and tools used

Research stage	Data/method	Tools and software
Literature review	Systematic review of peer-reviewed scientific publications	Scopus, ScienceDirect, IEEE Xplore
Bibliometric analysis	Citation and keyword network analysis	Litmaps, Python (Google Colab)
Reference management	Bibliographic organisation and reference tagging	Zotero [LINK]
Forest inventory data	Polygon-based forest inventory shapefiles and attribute tables in DBF format	GIS database
Spatial data handling	Spatial filtering, polygon handling, and attribute linking	Google Earth Engine
Satellite data acquisition	Sentinel-1 and Sentinel-2 satellite imagery	Google Earth Engine
Satellite data preprocessing	Cloud masking, filtering, compositing, and feature extraction	Google Earth Engine (JavaScript)
Feature engineering	Derivation of spectral and SAR predictors	Google Earth Engine, Python
ML modelling	Supervised regression and classification modelling	Python (Google Colab, scikit-learn)
Model validation	Accuracy evaluation, residual analysis, and uncertainty assessment	Python (scikit-learn)
Code development support	Script structuring and debugging assistance	ChatGPT
Language editing	Grammar and clarity improvement	Grammarly
Visualisation	Map generation and analytical plotting	Google Earth Engine, Python (matplotlib), draw.io
Research documentation and evidence storage	Storage and organisation of code, datasets, GEE scripts, satellite image examples, meeting minutes and thesis-related outputs	GitHub [LINK]